Theoretical Physics I

Lecture notes

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Recommended literature

Essentially any textbook on Classical Mechanics could serve as a companion to the course. A non-exhaustive list of possibilities is:

Goldstein, Safko & Poole, Classical Mechanics: A classic textbook, dating back to the 1950's, which has been updated with chapters on chaos. Perhaps not optimal as companion to this course, as variational techniques are introduced at the very beginning, while we will introduce these towards the end.

Kleppner & Kolenkow, An Introduction to Mechanics: Starts with a brief introduction to the mathematics required. Does not contain advanced topics such as Lagrangian (or Hamiltonian) Mechanics.

Kibble & Berkshire, Classical Mechanics: Begins with an insightful critical discussion of the (often only implicit) assumptions underlying Newtonian Mechanics (similar to what is attempted in Section 6.1).

Morin, Introduction to Classical Mechanics: Very readable modern textbook, containing many exercises (partially with solutions) and an occasional limerick. Begins with statics, as we do here, too. Contrary to this course, only a minimum of mathematical methods is introduced.

Nolting, Theoretical Physics 1. Classical Mechanics: English translation of the first volume of a series on Theoretical Physics popular with German physics students. Begins with a detailed exposition of the required mathematical tools. Does not cover Lagrangian Mechanics, which is dealt with in the second volume of the series.

Scheck, Mechanics: Mathematically rigorous and concise formulation.

Taylor, Classical Mechanics: Very readable modern textbook. In contrast to many other textbooks it covers friction very early on. Contains many exercises.

Tong, Classical Mechanics: A very new textbook based on popular lecture notes which are available on the author's webpage.

1 Introduction

1.1 What is Theoretical Physics?

This is the first of a series of courses on Theoretical Physics. It may thus be appropriate to spend some time to discuss what the role of Theoretical Physics within Physics is. Let us start with Physics. The term "physics" stems from the Latin "physica", which was borrowed from the greek Φυσική. While "physica" broadly meant "study of nature", we nowadays consider physics to be the natural science concerned with natural phenomena[1] which are elementary enough so as not to count as chemistry or biology (of course the boundaries between these subjects are neither sharp nor static). Experimental Physics and Theoretical Physics pursue this study using different and complementary methods: In an experiment, one observes a system under controlled circumstances in a reproducible manner. One role of Theoretical Physics is to explain the phenomena observed in experiments. The basis for this is a *theory*, i.e., a set of abstract concepts and relations between these, often expressed in the form of mathematical equations. An example would be Newtonian mechanics, which involves concepts such as "point particle", "coordinates", "time", "mass", "force", etc., and axioms which relate these. Another example would be Maxwell's Electromagnetism, involving concepts like "charge", "current", "electric" and "magnetic field", and the Maxwell equations relating these.

Typically, a theory cannot directly be used to explain the outcomes of an experiment. For this, a *model* is needed, which provides a mapping from the parameters describing the system under consideration to the concepts provided by the theory (and typically some further input). Depending on how complicated the system is, this may involve a high degree of *idealization*, i.e., disregarding a huge amount of information and concentrating on those aspects that are essential for the understanding of the phenomenon under consideration.[2] As an example consider planetary motion: While the underlying theory is Newtonian mechanics, we model planetary motion by idealizing planets as point particles and (as a further input) using Newton's gravitational force law.[3] A model is considered more successful the less supplementary parameters it has.[4] In the case of planetary motion, there is effectively only one free parameter (entering the potential), namely the product GM of Newton's constant G and the mass M of the sun.

The role of Theoretical Physics is however not limited to the construction of models that fit the outcome of experiments. Typically, to even describe the setup and the outcome of an experiment, some of the abstract concepts of Theoretical Physics are used. Theoretical

[1] "Natural phenomena" of course includes phenomena occurring under circumstances that could also be characterized as highly "artificial", as is typically the case in a lab (clean samples, ultra-high vacuum, extremely low temperatures, ...).

[2] The ability to distinguish the relevant from the irrelevant parameters can be considered as one of the key qualifications of a physicist.

[3] The distinction between a model and a theory is not always clear-cut. For example, in the context of particle physics it has become customary to think of general (quantum) field theory as a theory and particular examples of field theories, such as electromagnetism, as models. On the other hand, Newtonian Mechanics together with Newton's gravitational force law is sometimes considered as a theory, Newtonian Gravity.

[4] In this context, John von Neumann is often quoted as "With four parameters I can fit an elephant, and with five I can make him wiggle its trunk", see arXiv:2407.07909 for a recent contribution to the topic.

Physics also guides Experimental Physics, in the sense that the study of a theory (or a particular model therein) leads to the prediction of new phenomena, which can then be tested experimentally. One example is Electromagnetism: Soon after Maxwell completed the system of equations named after him, it was realized that the equations allow for wave-like solutions. This provided the motivation for Hertz to attempt (and achieve) the production and detection of electromagnetic waves (leading to technology radically transforming economy and society).

Conversely, Experimental Physics influences Theoretical Physics, for example by falsifying predictions based on existing theories or by new, unexpected discoveries. To a negative experimental result, Theoretical Physics usually does not respond by altering or giving up a theory. First, one tries to find a model (based on the existing theories), which better describes the experimental result. After all, the mismatch of experimental results and theoretical predictions may be due to an inappropriate model, for example due to an overidealization. Only when models have to be altered repeatedly and become increasingly "baroque" (containing many free parameters, for example), is a modification of the underlying theory seriously considered. In this way (reconsideration of the existing theories induced by experimental findings that could not be explained) Quantum Mechanics was developed during the first quarter of the 20th century.

But tension with experimental results is not the only driver of theory evolution. Another important one is internal consistency. When Maxwell compiled the hitherto known laws of electromagnetism in the form of differential equations, he noticed that the resulting system of equations was incompatible with the principle of charge conservation. Thus, he added the "displacement current" to one of the equations in order to restore consistency.[5] It was also the incompatibility of Electromagnetism and Newtonian Mechanics (in particular the Galilean relativity principle) that led Einstein to his theory of Special Relativity and the incompatibility of the latter with Newtonian Gravity which led him to General Relativity.

A further mode of theory evolution is via generalization. This often proceeds by first finding equivalent formulations of existing theories, which may then have a broader range of applicability. For example, Newtonian Mechanics can be rephrased in terms of concepts such as action, Lagrangian, Hamiltonian, phase space, but these can be applied also to systems beyond the scope of Newtonian Mechanics (such as constrained systems). Even a Lagrangian or Hamiltonian formulation of General Relativity is possible (and useful).

When new theories are developed, typically new mathematical tools have to be invented or learned by physicists. For example, Newtonian Mechanics was developed in parallel to differential calculus. When Einstein developed General Relativity, he heavily relied on concepts from Differential Geometry, at the time a rather obscure branch of Mathematics. And when Heisenberg made a decisive step towards Quantum Mechanics, it was Born (who had a sounder mathematical education) who realized that the calculational rules used by Heisenberg were nothing but matrix multiplication. On the other hand, new physical theories usually also drive the evolution of Mathematics. For example, Electromagnetism led to the development of vector analysis (or, more generally, differential forms), while Quantum Mechanics gave rise to the theory of Operator Algebras.

The investigation of the structure of existing theories is not only relevant for potentially generalizing them. It may also yield general statements which can be extremely useful for understanding the behaviour of concrete systems. One example would be conservation laws (for energy, momentum, angular momentum...) whose validity can be proven on very general grounds, and which are typically extremely useful when applied to concrete problems.

[5] And this supplementary term gave rise to the existence of wave solutions to the system of equations.

1.2 Newtonian Mechanics

Since ancient times, humans studied the movement both of celestial and of earthbound bodies.[6] That these types of objects seemingly behave quite differently, lead Aristotle to postulate two different domains: Within the *sublunary sphere*, movement is directed (typically downwards) but ultimately comes to an end (due to friction or collision). In contrast, beyond the moon, motion is circular and endless. It took a couple of centuries and the genius of Newton to overcome this misconception and to show that the force that is responsible for the fall of an apple and the motion of the moon (or other celestial bodies), is one and the same, gravity. For this, Newton could build on earlier achievements, such as the mechanical investigations of Galilei (leading to the equivalence principle), the development of Cartesian coordinates (named after their inventor Descartes), or the investigation of planetary motion by Kepler.

Newton formalized his conception of Mechanics in his axioms, and also developed the mathematical concepts needed the formulate them (in particular that of differentiation). By postulating a law for the gravitational force, he was then able to derive Kepler's laws of planetary motion (which were phenomenological statements about the observed planetary motion) from these principles.

1.3 Aim and plan of the course

The aim of the course is to introduce Newtonian Mechanics and (depending on time) also one of its generalizations, Lagrangian Mechanics. These are applied to number of physical systems such as the (damped, driven) harmonic oscillator, or planetary motion (the Kepler problem). As explained in the previous section, availability of the appropriate mathematical concepts and methods is essential. These will be introduced in detail in the accompanying course "Mathematical Methods I". It is thus highly recommended to follow that course in parallel.

The most structured approach to Newtonian Mechanics would be to start with Newton's axioms, deriving some general consequences (such as conservation laws), and then applying the developed concepts and methods to concrete problems. However, this is not the way that we will proceed here. The reason is two-fold: i) We need some mathematical tools in order to even properly formulate Newton's axioms. ii) It might be useful for the understanding to first study a few example before discussing the abstract general framework. Of course in order to study examples, one is already using the Newtonian framework to some extent. However, I assume that students are already familiar with Newton's axioms to some extent (in particular the relation $ma = F$ between acceleration and force), so that a study of concrete examples should be possible before discussion Newton's axioms in detail. After a few preliminaries, we will thus start, in Section 3 with statics. For that, we will introduce basic mathematical concepts (vector spaces, scalar product, vector product) and apply them for the discussion of the balance of forces and torques.

In Chapter 4, we will study motion in one dimension, by considering a general dynamical equation of the form $\ddot{u}(t) = f(u(t), \dot{u}(t), t)$ and special cases thereof. In particular, we will define (when the "force" $f(u, \dot{u}, t) = f(u)$ only depends on the position u, but not on the velocity $\dot{u}$ or the time t) the concept of the *energy*, which turns out to be very useful. We will also consider the oscillator (also in its damped and forced variant), which is an extremely important model, as it exemplifies the possible behaviour (oscillation, exponential growth or

[6] Practical motivations given by the desire to have calendars (for agricultural and/or social reasons) or to build weapons (such as slings and arrows).

decay) near equilibria. Important mathematical methods that will be introduced are curves and vector fields and the Taylor approximation.

In Section 5, we generalize the oscillator to several dimensions and/or particles, and study coupled oscillations. For that, we also introduce basic notions of analysis of functions of several variables (such as partial derivatives and the gradient). In the discussion of motion around equilibria we are relying on results from linear algebra (which should by then be provided though the Mathematical Methods course).

2 Preliminaries

2.1 Mathematical symbols and notations

The latin alphabet is often not sufficient for the needs of a physicist, so frequently greek letters are also employed. The following table, which also includes possible variants, summarizes these.

Lower case	Upper case	Name
α		alpha
β		beta
γ	Γ	gamma
δ	Δ	delta
ϵ, ε		epsilon
ζ		zeta
η		eta
θ, ϑ	Θ	theta
κ		kappa
λ	Λ	lambda
μ		mu
ν		nu
ξ	Ξ	xi
π	Π	pi
ρ, ϱ		rho
σ	Σ	sigma
τ		tau
ϕ, φ	Φ	phi
χ		chi
ψ	Ψ	psi
ω	Ω	omega

It is often useful to derive new symbols from existing ones. For example, as variants of the symbol x, one can introduce x' ("x prime"), x'' ("x double prime"), $\dot{x}$ ("x dot"), $\hat{x}$ ("x hat"), $\tilde{x}$ ("x tilde"). For f a function, f' typically denotes the derivative (or $\dot{f}$ if f is a function of time).

Products of numbers can be denoted by $\cdot$, $\times$, but most often no symbol at all we be used (so that xy denotes the product of x and y). Division of x by y is denoted by x/y or $\frac{x}{y}$.

Apart from latin and greek letters and the symbols of elementary calculus, we will need some further symbols, in particular in order to express conditions or definitions of subsets of sets. To express that a certain element x in contained in a set X, one writes $x \in X$ (for example, $3 \in \mathbb{N}$), while the fact that x is not contained in X is expressed as $x \notin X$ (for example, $\sqrt{2} \notin \mathbb{N}$). The fact that a certain relation holds whenever a certain parameter is contained in a certain set is expressed using the symbol $\forall$ ("for all"), for example $2n \geq n \ \forall n \in \mathbb{N}$.[1] In order to define

[1] The symbols $\exists$ ("exists"), $\nexists$ ("not exists"), or $!\exists$ ("exists unique") also appear in the mathematical literature, but we will try to avoid their use.

subsets of a set, one uses a notation that is perhaps best explained by an example: The set of even numbers can be defined as

$$\{n \in \mathbb{N} \mid n = 2m \text{ for some } m \in \mathbb{N}\}, \tag{2.1}$$

which may be read as "the set of natural numbers n such that $n = 2m$ for some natural number m". The symbol $:=$ is used for defining the l.h.s. in terms of the r.h.s. The symbols $\Longrightarrow$ and $\Longleftarrow$ indicate implications in the directions given by the arrows, whereas $\Longleftrightarrow$ indicate implication in both directions, and is often read as "if and only if" and abbreviated by "iff". Examples would be

$$n \bmod 3 = 0 \quad \Longrightarrow \quad n \bmod 6 = 0 \tag{2.2}$$

or

$$n \bmod 3 = 0 \text{ and } n \bmod 2 = 0 \quad \Longleftrightarrow \quad n \bmod 6 = 0. \tag{2.3}$$

The union of sets is denoted by $\cup$, the intersection by $\cap$, the set subtraction by $\setminus$, and the empty set by $\emptyset$. For example, $[-1,0) \cap (0,1] = \emptyset$, while $[-1,0) \cup (0,1] = [-1,1] \setminus \{0\}$. Here we used the standard notation for intervals, i.e., square brackets denoting closed and round brackets open intervals.

Repeated sums and products are indicated by $\sum_{n=n_0}^{n_1} x_n$ and $\prod_{n=n_0}^{n_1} x_n$, where the summation (or product) variable n takes all numbers between n_0 and n_1. If the summation index is drawn from a set N, we may also write $\sum_{n \in N} x_n$. Examples are

$$\sum_{n=0}^{\infty} x^n = \sum_{n \in \mathbb{N}_0} x^n = x^0 + x^1 + x^2 + \cdots = \frac{1}{1-x}, \tag{2.4}$$

$$\prod_{n=1}^{N} n = \prod_{n \in \{1,2,\ldots,N\}} n = 1 \cdot 2 \cdot \cdots \cdot N = N!. \tag{2.5}$$

As for integration variables, summation (or product) variables can be freely renamed, i.e., $\sum_{n=n_0}^{n_1} x_n = \sum_{m=n_0}^{n_1} x_m$. When it is clear from which set the summation index is drawn, the set will often be omitted. For example, in the context of vectors on $\mathbb{R}^3$, we may write

$$\mathbf{v} \cdot \mathbf{w} = \sum_i v_i w_i = \sum_{i=1}^{3} v_i w_i. \tag{2.6}$$

2.2 Dimensional analysis and units

Many quantities in physics depend on the choice of a scale (unit). Such quantities are called *dimensionful*. Quantities which are independent of the choice of a scale are called *dimensionless*. In Mechanics, the relevant scales are those for time, length, and mass, and the powers of these associated to a quantity are called its *dimension*. For example, the dimension of velocity is length per time. One often uses square brackets around a quantity to denote its dimension and T, L, M as abbreviations for time, length, and mass. For example,

$$[v] = LT^{-1}, \qquad [a] = LT^{-2}, \qquad [F] = MLT^{-2}. \tag{2.7}$$

The units used for time, length, and mass in the SI system are second, meter, and kilogram, but also other units can be useful in some contexts. For a dimensionless quantity, also called a

number, we associate dimension 1 (so that taking the product with a number does not change the dimension).

It is an important principle that while dimensionful quantities can be freely multiplied, only quantities of the same dimension can be added. Furthermore, taking powers or products are the only operations allowed for dimensionful quantities. Functions such as sin, exp, or log can only be applied to numbers, and yield numbers. This often provides useful consistency checks in problem solving.

The role of angles sometimes leads to confusion. After all, two different "units" are of widespread use: Degrees (such that a full circle amounts to 360°) and radian (for which a full circle amounts to 2π). The angle in radian of an arc is defined as its length divided by its radius (which are both of dimension length), so that it is dimensionless, i.e., a number. In the same manner that any number can be converted to a percentage by multiplication with 100, multiplication by $\frac{360}{2\pi}$ converts to degrees. However, such a conversion is quite artificial, so by default, angles will be considered dimensionless (even though it is often convenient to state a problem or its solution in terms of degrees).

In simple problems, one can already learn quite a bit by only considering the dimensions of the involved quantities. Consider the example of a pendulum of length ℓ. As usual in such problems, we consider the rope to be massless (or of mass negligible w.r.t. the mass of the body). In the gravitational field of Earth, the mass is subject to the gravitational acceleration g. Suppose we are asked to determine the period of the pendulum for a given maximum displacement angle θ_0. These three quantities ℓ, g, and θ_0 are the only relevant ones for the problem, and they have dimensions L, LT^{-2}, and one respectively. We are looking for the period τ, with has dimension T. The most general expression of dimension time that we can build out of ℓ, g, θ_0 is

$$\tau = \sqrt{\frac{\ell}{g}} f(\theta_0) \tag{2.8}$$

for an arbitrary function f. To see this, we note that $\sqrt{\ell/g}$ is the only possibility to arrive at a time, and θ_0 is the only dimensionless quantity available. Hence, without solving the problem (i.e., without determining $f(\theta_0)$), we can conclude for example that in order to double the period (for the same maximal angle θ_0) we have to quadruple the length. Or that the same pendulum on the Moon would have a period longer by a factor $\simeq \sqrt{6}$. Furthermore, the numerical factors appearing in a dimensional analysis "calculation" of some quantity are typically in the $0.1 - 10$ ballpark. For example, in the above, $\lim_{\theta_0 \to 0} f(\theta_0) = 2\pi$. Hence, dimensional analysis can be used to get order of magnitude estimates.

We note that, in the example above, our confidence that ℓ, g and θ_0 are the only relevant parameters already requires some understanding of the problem (and often a separation of scales). One can easily come up with more parameters, such as the diameter d of the body, or constants of nature such as the speed of light c. When taking these into account, we can form the dimensionless combinations $\frac{d}{\ell}$ and $\frac{g\ell}{c^2}$, in which case we can only conclude that

$$\tau = \sqrt{\frac{\ell}{g}} f(\theta_0, d/\ell, g\ell/c^2). \tag{2.9}$$

However, for realistic length ℓ, we obviously have $g\ell/c^2 \ll 1$. Furthermore, we can imagine the body to have negligible extension, so that also $d/\ell \ll 1$. There is no reason to expect any irregularity in the limit where these quantities are set to 0, i.e., we can simply define $f(\theta_0)$ as the limit of $f(\theta_0, d/\ell, g\ell/c^2)$ as the last two variables tend to 0.

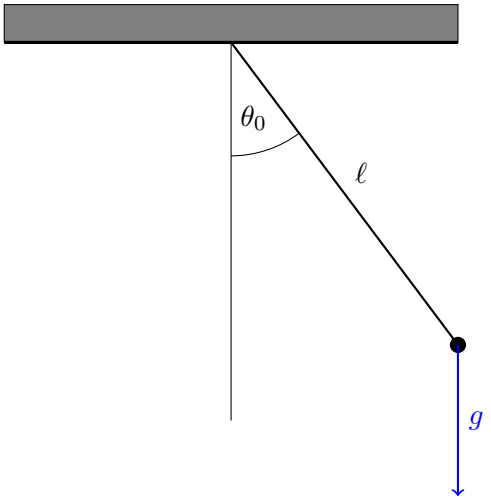


Figure 2.1: Sketch of a pendulum.

2.3 Problem solving strategies

The main aim of this course is to enable you to apply the concepts and methods of Classical Mechanics to solve problems.[2] We will practice this with weekly assignments (and test your problem-solving skills in the final exam). However, a few general guidelines on how to solve a physics problem may be useful:

- Especially in Mechanics, first drawing a diagram is very helpful. In such a diagram you indicate for example the relevant forces, distances, and angles. When appropriate, one can also indicate suitable coordinate axes.
- Especially in complicated problems, it is useful to first write down all the relevant equations relating the knows and unknowns of the problem. It may also be useful to explicitly state which quantities are known (given) and which are unknown (to be determined).
- Work with symbols as long as possible! Even if concrete values for parameters of the problem are given, replace them by symbols and only replace these by their numerical values at the very end!
- When solving differential equations, it is sometimes useful to choose (length and time) units in a convenient manner in order to simplify the equations.
- Do consistency checks! This in particular includes checking dimensions (see above) and considering limiting cases for which the result should be clear.

We will exemplify this in our first few examples, which we will work out in some detail.

[2] A further goal is to enable you to derive general results from basic postulates.

3 Statics: Forces & Torque

Similar to Morin's textbook, we begin by discussing statics. Conceptually, this has the disadvantage that the necessary concepts, force and torque, will only be properly defined later on, in the discussion of Newton's axioms and angular momentum. However, forces and torque are relatively straightforward to introduce without their conceptual background, and having these at our disposal, we can already do some interesting physics. The discussion of forces and torque also serves as a pretext to introduce/recapitulate the underlying mathematical structures, such as vector spaces, scalar products, or the vector product.

3.1 Vectors and the scalar product

Many quantities relevant for Mechanics (forces or velocities, for example) are characterized by a magnitude and a direction. We describe them as elements of $\mathbb{R}^3$. Elements of $\mathbb{R}^3$ are typically denoted by boldface letters, such as $\mathbf{v}$, or by a vector arrow, such as $\vec{v}$, on the blackboard. Depending on the context, we may use different notations for the components of a vector, alternatively

$$\mathbf{v} = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}. \tag{3.1}$$

A defining fact about vectors is that they can be multiplied by real numbers and added, i.e., for $\mathbf{v}$ and $\mathbf{w}$ vectors and λ a real number, we define (c.f. Fig. 3.1 for the interpretation of the addition)

$$\lambda\mathbf{v} = \lambda \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} := \begin{pmatrix} \lambda v_1 \\ \lambda v_2 \\ \lambda v_3 \end{pmatrix}, \qquad \mathbf{v} + \mathbf{w} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} + \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} := \begin{pmatrix} v_1 + w_1 \\ v_2 + w_2 \\ v_3 + w_3 \end{pmatrix}. \tag{3.2}$$

In particular, given two vectors $\mathbf{v}$, $\mathbf{w}$, one can form *linear combinations*

$$\lambda_1\mathbf{v} + \lambda_2\mathbf{w}. \tag{3.3}$$

The numbers λ_1, λ_2 in this equation are called the *coefficients* of $\mathbf{v}$ and $\mathbf{w}$ respectively.

A special vector is the one with vanishing components:

$$\mathbf{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \tag{3.4}$$

It is distinguished by the two properties

$$\mathbf{0} + \mathbf{v} = \mathbf{v} \qquad \forall \mathbf{v}, \tag{3.5}$$

$$0\mathbf{v} = \mathbf{0} \qquad \forall \mathbf{v}. \tag{3.6}$$

Note that on the l.h.s. of the second equation, we perform scalar multiplication by the *number* 0.

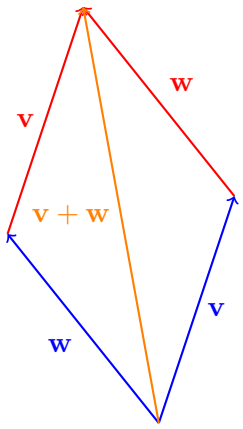


Figure 3.1: The addition of vectors.

Multiplication by a number is sometimes called *scalar multiplication*. For scalar multiplication and addition, one easily checks the rules[1]

$$\lambda_1(\lambda_2\mathbf{v}) = (\lambda_1\lambda_2)\mathbf{v}, \tag{3.7}$$

$$\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w} \qquad \text{associative} \tag{3.8}$$

$$\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v} \qquad \text{commutative} \tag{3.9}$$

$$\lambda(\mathbf{v} + \mathbf{w}) = \lambda\mathbf{v} + \lambda\mathbf{w} \qquad \text{distributive} \tag{3.10}$$

A set allowing for an addition and multiplication by (real/complex) numbers obeying the above rules is called a (real/complex) *vector space*. The above structures can be straightforwardly generalized to $\mathbb{R}^n$, i.e., the set whose elements consist of n real numbers. There are also physically relevant vector spaces different from $\mathbb{R}^n$, for example, the set $C(\mathbb{R})$ of continuous functions.

An important concept in the theory of vector spaces is *linear independence*: We call a collection $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ *linearly independent* if there is a unique linear combination of these that vanishes, namely the one with vanishing coefficients. Mathematically, this can be expressed as

$$\lambda_1\mathbf{v}_1 + \cdots + \lambda_n\mathbf{v}_n = 0 \quad \Longleftrightarrow \quad \lambda_1 = \cdots = \lambda_n = 0. \tag{3.11}$$

Obviously, the zero vector can never be an element of a set of linearly independent vectors. The set $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly dependent, iff there is a least one vector $\mathbf{v}_i$ which can be written as a linear combination of the others.

A *basis* of a vector space is a collection of vectors such that *any* vector can be expressed *in a unique way* as a linear combination of these. In particular, a basis must be linearly independent. One calls the number of vectors in a basis the *dimension* of the vector space. A prime example is the basis of $\mathbb{R}^3$ consisting of the Euclidean basis vectors, for which different notations are in use:[2]

$$\hat{\mathbf{x}} = \mathbf{e}_x (= \mathbf{e}_1) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \hat{\mathbf{y}} = \mathbf{e}_y (= \mathbf{e}_2) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \hat{\mathbf{z}} = \mathbf{e}_z (= \mathbf{e}_3) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \tag{3.12}$$

[1] The commutativity of the addition can be nicely illustrated by a diagram as shown in Fig. 3.1.

[2] The notation $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$ will also be used for a general (orthonormal) basis below. Hence, we will try to avoid its use for the Euclidean basis.

The vector $\mathbf{v}$ as given in (3.1) can expressed as a linear combination of these as

$$\mathbf{v} = v_x\hat{\mathbf{x}} + v_y\hat{\mathbf{y}} + v_z\hat{\mathbf{z}}. \tag{3.13}$$

This is the unique such linear combination in the sense that any other choice of coefficients yields a different vector. This straightforwardly generalizes to $\mathbb{R}^n$, which is n-dimensional. But there are also infinite-dimensional vector spaces, such as $C(\mathbb{R})$.

Given m vectors $\mathbf{v}_1, \ldots, \mathbf{v}_m$ of some vector space V ($\mathbb{R}^n$, for example), their *(linear) span* is the subset of V consisting of all linear combinations of these:

$$\mathrm{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_m) := \{\mathbf{v} \in V \mid \mathbf{v} = \lambda_1\mathbf{v}_1 + \cdots + \lambda_m\mathbf{v}_m \text{ for some } \lambda_1, \ldots \lambda_m \in \mathbb{R}\}. \tag{3.14}$$

If the set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$ is linearly independent, then this span a m-dimensional *subspace* of V, i.e., a subset which is a vector space in its own right. Familiar examples are lines and planes through the origin of $\mathbb{R}^3$, which are one- and two-dimensional submanifolds.

A very important further operation on $\mathbb{R}^3$ (or more generally $\mathbb{R}^n$) is the *scalar product* defined by

$$\mathbf{v} \cdot \mathbf{w} := v_x w_x + v_y w_y + v_z w_z = \sum_{i=1}^{3} v_i w_i. \tag{3.15}$$

An alternative notation that is also sometimes used for the scalar product is $\langle \mathbf{v}, \mathbf{w} \rangle$. Mathematically, the scalar product is a map $\mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}$, with the properties

$$\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v} \qquad \text{symmetry} \tag{3.16}$$

$$\mathbf{v} \cdot (\eta_1\mathbf{w}_1 + \eta_2\mathbf{w}_2) = \eta_1\mathbf{v} \cdot \mathbf{w}_1 + \eta_2\mathbf{v} \cdot \mathbf{w}_2 \qquad \text{(bi)linear} \tag{3.17}$$

$$\mathbf{v} \cdot \mathbf{v} \geq 0, \qquad \mathbf{v} \cdot \mathbf{v} = 0 \iff \mathbf{v} = 0 \qquad \text{positive definite} \tag{3.18}$$

Note that (3.17) only states the linearity in the second factor. But together with the symmetry (3.16), it follows that the scalar product is also linear in the first factor, so that it is in fact bilinear, i.e.,

$$(\lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2) \cdot (\eta_1\mathbf{w}_1 + \eta_2\mathbf{w}_2) = \lambda_1\eta_1\mathbf{v}_1 \cdot \mathbf{w}_1 + \lambda_1\eta_2\mathbf{v}_1 \cdot \mathbf{w}_2 + \lambda_2\eta_1\mathbf{v}_2 \cdot \mathbf{w}_1 + \lambda_2\eta_2\mathbf{v}_2 \cdot \mathbf{w}_2. \tag{3.19}$$

Using the scalar product, one can define the *modulus* (sometimes also called *norm*) of a vector as

$$|\mathbf{v}| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_x^2 + v_y^2 + v_z^2}. \tag{3.20}$$

One often denotes the modulus of a vector by the same letter, but "unboldened", i.e., $v = |\mathbf{v}|$. A vector with modulus one is called a *unit vector*, and unit vectors are often indicated by a hat, as in (3.12).

The *angle* $\theta \in [0, \pi]$ between two non-zero vectors $\mathbf{v}$, $\mathbf{w}$ is defined by

$$\cos\theta := \frac{\mathbf{v} \cdot \mathbf{w}}{vw}. \tag{3.21}$$

This is well-defined, as by the *Cauchy-Schwarz inequality*

$$|\mathbf{v} \cdot \mathbf{w}| \leq vw, \tag{3.22}$$

the r.h.s. of (3.21) is of modulus smaller or equal to 1. Three important special cases are

- $\mathbf{v}$ and $\mathbf{w}$ are *parallel* if $\mathbf{w} = \lambda\mathbf{v}$ with $\lambda > 0$. In this case $\theta = 0$. This is sometimes denoted as $\mathbf{v} \parallel \mathbf{w}$.

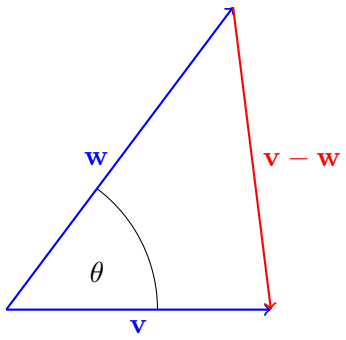


Figure 3.2: Illustration of the law of cosines.

- $\mathbf{v}$ and $\mathbf{w}$ are *anti-parallel* if $\mathbf{w} = \lambda\mathbf{v}$ with $\lambda < 0$. In this case $\theta = \pi$.
- $\mathbf{v}$ and $\mathbf{w}$ are *orthogonal* if $\mathbf{v} \cdot \mathbf{w} = 0$. In this case $\theta = \frac{\pi}{2}$. This is sometimes denoted as $\mathbf{v} \perp \mathbf{w}$.

Using the notations and relations introduced up to now, we arrive at an expression for the norm of the sum $\mathbf{v} + \mathbf{w}$ or difference $\mathbf{v} - \mathbf{w}$ of two vectors:

$$\begin{aligned} |\mathbf{v} \pm \mathbf{w}|^2 &= (\mathbf{v} \pm \mathbf{w}) \cdot (\mathbf{v} \pm \mathbf{w}) \\ &= \mathbf{v} \cdot \mathbf{v} \pm \mathbf{v} \cdot \mathbf{w} \pm \mathbf{w} \cdot \mathbf{v} + \mathbf{w} \cdot \mathbf{w} \\ &= v^2 + w^2 \pm 2vw \cos\theta, \end{aligned} \tag{3.23}$$

with θ the angle between the vectors. For the case of the difference of vectors, this is the *law of cosines*,[3] c.f. Fig. 3.2, a special case of which is the Pythagorean theorem (for orthogonal vectors $\mathbf{v}$, $\mathbf{w}$).

Given a subspace U of a vector space V (think of a line or a plane through the origin in $\mathbb{R}^3$), one defines its *orthogonal complement* as the set of vectors which are orthogonal to all elements of U:

$$U^\perp := \{\mathbf{v} \in V \mid \mathbf{v} \perp \mathbf{u} \, \forall \mathbf{u} \in U\}\,. \tag{3.24}$$

An important result is that if V is n-dimensional and U is m-dimensional, then $U^\perp$ is $n - m$-dimensional. Moreover, every $\mathbf{v} \in V$ can be uniquely decomposed as $\mathbf{v} = \mathbf{v}_\parallel + \mathbf{v}_\perp$ with $\mathbf{v}_\parallel \in U$ and $\mathbf{w}_\perp \in U^\perp$. One then calls $\mathbf{v}_\parallel$ the *projection* of $\mathbf{v}$ onto U. In particular, in $\mathbb{R}^3$ a plane (through the origin) can be alternatively described by two vectors spanning the plane or by a single vector orthogonal to the plane.

A basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ of $\mathbb{R}^3$ (or more generally of $\mathbb{R}^n$) is especially useful if it is *orthonormal* (ONB), i.e., the basis vector are normalized (have unit norm) and are pairwise orthogonal, i.e.,

$$|\mathbf{e}_i| = 1 \qquad \forall i, \tag{3.25}$$

$$\mathbf{e}_i \cdot \mathbf{e}_j = 0 \qquad \forall i \neq j. \tag{3.26}$$

Obviously, the Euclidean basis vectors $\{\hat{x}, \hat{y}, \hat{z}\}$ form an orthonormal basis. The above two equations can be conveniently written as the single equation

$$\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij} \qquad \forall i, j, \tag{3.27}$$

[3] The fact that we recover the law of the cosines can be seen as an a posteriori justification for the definition (3.21) of θ.

using the *Kronecker δ symbol* defined as

$$\delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases} \tag{3.28}$$

Exercise 3.1. *Verify that for any $\theta \in [0, \pi]$, $\phi \in [0, 2\pi)$, the vectors*

$$\hat{\mathbf{r}} = \mathbf{e}_r := \begin{pmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{pmatrix}, \qquad \hat{\boldsymbol{\theta}} = \mathbf{e}_\theta := \begin{pmatrix} \cos\theta\cos\phi \\ \cos\theta\sin\phi \\ -\sin\theta \end{pmatrix}, \qquad \hat{\boldsymbol{\phi}} = \mathbf{e}_\phi := \begin{pmatrix} -\sin\phi \\ \cos\phi \\ 0 \end{pmatrix} \tag{3.29}$$

form an ONB of $\mathbb{R}^3$. Show that the same is true for any $\phi \in [0, 2\pi)$ for the vectors

$$\hat{\boldsymbol{\rho}} = \mathbf{e}_\rho := \begin{pmatrix} \cos\phi \\ \sin\phi \\ 0 \end{pmatrix}, \qquad \hat{\boldsymbol{\phi}} = \mathbf{e}_\phi := \begin{pmatrix} -\sin\phi \\ \cos\phi \\ 0 \end{pmatrix}, \qquad \hat{\mathbf{z}} = \mathbf{e}_z := \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \tag{3.30}$$

When expressing $\mathbf{v}$ in terms of the basis $\{\mathbf{e}_i\}_{i=1,2,3}$ as

$$\mathbf{v} = \sum_i v_i \mathbf{e}_i, \tag{3.31}$$

we can now determine the unique coefficients v_i as follows:

$$\begin{aligned} \mathbf{e}_i \cdot \mathbf{v} &= \mathbf{e}_i \cdot \sum_j v_j \mathbf{e}_j \\ &= \sum_j v_j \mathbf{e}_i \cdot \mathbf{e}_j \\ &= \sum_j v_j \delta_{ij} \\ &= v_i \end{aligned} \tag{3.32}$$

From the first to the second line, we used the linearity (3.17) of the scalar product, in the next step (3.27), and in the last step the following

Kronecker δ rule: To evaluate a sum with summation index j and factor δ_{ij} in the summand, delete the sum and δ_{ij} and replace every occurrence of j in the remainder by i (of course i and j are arbitrary and can be replaced by any other symbol).

The result of the above calculation is thus that the coefficients v_i of a vector $\mathbf{v}$ w.r.t. an ONB $\{\mathbf{e}_i\}$ are given by $v_i = \mathbf{e}_i \cdot \mathbf{v}$. We can summarize this in the identity

$$\mathbf{v} = \sum_i (\mathbf{e}_i \cdot \mathbf{v})\, \mathbf{e}_i. \tag{3.33}$$

Exercise 3.2. *Using (3.27), show that the scalar product of two vectors $\mathbf{v}$, $\mathbf{w}$ can be expressed in terms of their coefficients v_i, w_i w.r.t. a ONB as*

$$\mathbf{v} \cdot \mathbf{w} = \sum_{i,j} v_i w_j \delta_{ij} = \sum_i v_i w_i, \tag{3.34}$$

i.e., the scalar product has the same form in any ONB.

Finally, we discuss how $\mathbf{v} = \mathbf{v}_\parallel + \mathbf{v}_\perp$ splits into a component contained in a subspace U and one orthogonal to it. For this, choose an ONB $\{\mathbf{e}_1, \dots \mathbf{e}_m\}$ of U. In the case of a line in $\mathbb{R}^3$, this would consist of the unit vector pointing along the line (unique up to sign). Then set

$$\mathbf{v}_\parallel := \sum_i (\mathbf{e}_i \cdot \mathbf{v})\, \mathbf{e}_i, \qquad\qquad \mathbf{v}_\perp := \mathbf{v} - \mathbf{v}_\parallel. \tag{3.35}$$

As a linear combination of basis vectors of U, $\mathbf{v}_\parallel$ is contained in U, and obviously $\mathbf{v} = \mathbf{v}_\parallel + \mathbf{v}_\perp$ is fulfilled. It remains to show that $\mathbf{v}_\perp \in U^\perp$. For this, it suffices to prove that $\mathbf{v}_\perp$ is orthogonal to all vectors of a basis of U. And indeed

$$\begin{aligned} \mathbf{e}_j \cdot \mathbf{v}_\perp &= \mathbf{e}_j \cdot \left(\mathbf{v} - \sum_i (\mathbf{e}_i \cdot \mathbf{v})\, \mathbf{e}_i \right) \\ &= \mathbf{e}_j \cdot \mathbf{v} - \sum_i (\mathbf{e}_i \cdot \mathbf{v})\, \delta_{ji} \\ &= 0, \end{aligned} \tag{3.36}$$

where we again used the above Kronecker δ rule.

3.2 Forces

A precise definition of forces will be given in our discussion of Newton's axioms. For the present purposes, we may define a *force* as an influence on a body which can be characterized by a direction (into which the force aims to accelerate the body) and a magnitude (related to the magnitude of the resulting acceleration). Mathematically, we can thus describe it by a vector $\mathbf{F}$. For the "macroscopic" forces that we will be considering, forces are additive, i.e., when several forces $\mathbf{F}_i$ are acting on a body, then the resulting (total) force is the sum of the individual forces, $\mathbf{F}_{\text{tot}} = \sum_i \mathbf{F}_i$. A necessary condition for a static situation is that there is no net force on the body, i.e., $\mathbf{F}_{\text{tot}} = 0$ (a further condition for an extended rigid body to be static will be discussed below). The types of forces which will be relevant to us in the context of statics are:

Gravity: According to Newtons law of gravity, two point-like bodies of masses m_1, m_2 exert an attractive force of magnitude Gm_1m_2/R^2 on each other, with R their distance and G Newton's constant. Expressed in vectorial notation,

$$\mathbf{F}_1 = Gm_1m_2 \frac{\mathbf{r}_2 - \mathbf{r}_1}{|\mathbf{r}_2 - \mathbf{r}_1|^3} = -\mathbf{F}_2, \tag{3.37}$$

with $\mathbf{F}_i$ the force on the mass m_i. As we will later see, if, say, the first body is extended, but spherically symmetric, then the force on the second body is still given by the above expression for $\mathbf{F}_2$, with $\mathbf{r}_1$ the center of the extended body. In particular, the gravitational force exerted on a body of mass m by Earth is $\mathbf{F} = -\frac{GMm}{r^2}\hat{\mathbf{r}}$, with M the mass of Earth, r the distance to the center of Earth and $\hat{\mathbf{r}}$ the unit vector pointing from the center of Earth to the body. Near Earth's surface and on scales $d \ll R$ (with R Earth's radius), both r and $\hat{\mathbf{r}}$ can be considered essentially constant, so that one can approximate $\mathbf{F} = m\mathbf{g}$ with a constant $\mathbf{g}$.

Normal force: The fact that you can effortlessly and statically lie on the ground even though you are subject to gravity is due to the normal force applied by the ground. It is ultimately

due to molecular forces, but for our purposes we only need to consider their net effect. As the name suggests, the normal force is normal, i.e., its acts perpendicularly to the surface on which the body rests. It occurs whenever the sum of the other forces acting on the body have a component perpendicular to the surface, and pointing into the surface. The normal force then precisely cancels this perpendicular component. One may think of the surface as a spring with a huge spring constant, so that it is only negligibly deformed by the body, but precisely by such an amount that there is a balance of the forces in the perpendicular direction.

Friction: The fact that you can stand on a slope even though the normal force only compensates the normal component of the gravitational one, is due to a further tangential force, friction. In the static case, it is of magnitude equal to that of the tangential component of the sum of the other forces, but of opposite direction, c.f. Fig. 3.3. There is a maximal magnitude for the static frictional force, which is typically proportional to the magnitude of the normal force, i.e., there is a number μ (the coefficient of static friction) such that $F_f \leq \mu F_n$. The friction coefficient depends on the composition of body and ground. If the tangential force exceeds the maximal static frictional force, the body will move, but still be subject to a frictional force of magnitude $F_f = \mu_k F_n$, with μ_k the coefficient of kinetic friction.

Tension: When two people pull at the two ends of a rope or chain, it is under tension. This means that each element of the rope is subject to opposing forces from the two neighbouring elements, c.f. Fig 3.4. When the situation is static, then the two forces have to cancel. An exception are the two ends, which both have a neighbouring element only to one side. In this case, the balancing force is provided by the people pulling the rope. As in a static situation the forces exerted on one element from its two neighbours cancels exactly, the magnitude and direction of the forces exerted on each element must be constant, unless there are additional forces. It is convenient to distinguish between forces tangential to the rope and those orthogonal (normal) to the rope. An example for the first case would be the gravitational force acting on each element of a rope hanging vertically, c.f. Fig. 3.5. While the lowest element of the rope is only subject to the gravitational force due to its own weight (and possible some further due to a weight attached to it), the next element is subject to the gravitational force due to its own weight and the force exerted on it from the previous element of the rope. Hence, it is subject to a greater force (tension). Thus, tangential forces on the rope change the magnitude of the force (tension). A further example for a tangential force is friction, as when a rope is wrapped around a pole in order to moor a boat. Often, one idealizes ropes as massless and not subject to friction in order to exclude tangential forces. An example for a normal force is a rope wrapped around a wheel, c.f. Fig 3.6. In that case, the magnitude of the forces exerted by neighbouring elements on each other is constant, but the direction changes. It is important to note that tension always acts tangentially to the rope, at each point.[4] A rigid body, such as a rod or a beam can not only pulled at from both sides, but also pushed at. One calls this *compression* and it is sometimes convenient to treat it as tension with a negative sign.

Exercise 3.3. *Your soles have a friction coefficient of 0.7 on dry concrete. What is the maximum height (above the lowest point) at which you can stand in a half-pipe of 5m diameter? By how much is it reduced on wet concrete (friction coefficient 0.5)?*

[4] Rigid bodies, like rods, can also transmit tension which is normal to the body, as in a balance.

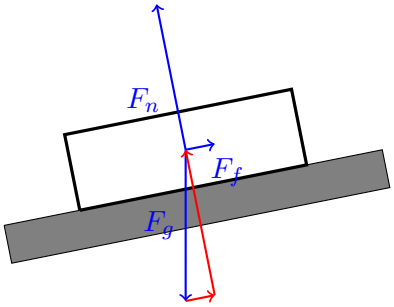


Figure 3.3: Frictional and normal force on a body on a slope.



Figure 3.4: Balance of forces in a rope without additional forces.



Figure 3.5: Balance of forces in a vertically hanging rope, subject to gravity. In red, the gravitational force on the elements, in blue the forces due to the neighbouring elements.



Figure 3.6: Balance of forces in a rope wrapped around a wheel, with normal force indicated in red.

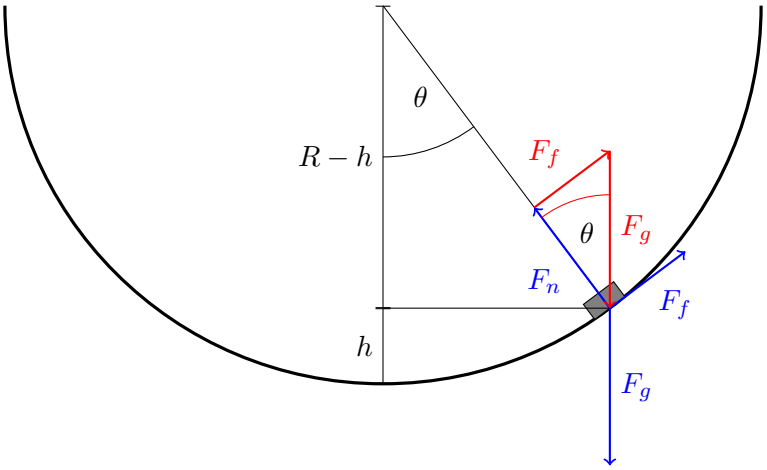


Figure 3.7: Diagram for Exercise 3.3.

We begin by drawing a diagram, c.f. Fig. 3.7. We indicate (in blue) the gravitational force $\mathbf{F}_g$, the normal force $\mathbf{F}_n$, and the friction for $\mathbf{F}_f$, the first one orthogonal to the surface, the latter one tangential. Here we denote vectors by bold face letters and their modulus by normal letters. We also introduced an angle θ that simplifies the discussion considerably. Now what are the relations between the quantities involved? Denoting by μ the friction coefficient, we have $F_f \leq \mu F_n$. We anticipate that equality will correspond to the case of interest, i.e., maximal height, but this will be the result of our considerations. Being static means that the forces cancel out, i.e., $\mathbf{F}_g + \mathbf{F}_n + \mathbf{F}_f = 0$. In the diagram, this equality is indicated by the triangle formed by $\mathbf{F}_n$ and the red arrows for $\mathbf{F}_f$ and $\mathbf{F}_g$. As the normal and the friction force are orthogonal, this is a right triangle, so we have $\tan\theta = \frac{F_f}{F_n} \leq \mu$. As the tangent is a monotonously increasing function, we see that the for a given μ, the maximal angle attainable (corresponding to the maximal height attainable) is $\theta = \arctan\mu$. Consideration of the triangle in black yields $\cos\theta = \frac{R-h}{R}$, so that

$$h = R(1-\cos\theta) = R(1-\cos\arctan\mu) = R\left(1 - \frac{1}{\sqrt{1+\mu^2}}\right). \tag{3.38}$$

Let us subjected it to consistency checks. Dimensionally the expression is fine: As μ is a number, the expression in brackets is well-defined (and a number), and the end result is a length. Simple special cases are $\mu \to 0$ and $\mu \to \infty$, in which cases $h \to 0$ and $h \to R$, as expected. Finally, plugging in numbers, we obtain

$$h_{\text{dry}} = 0.45\text{m}, \qquad h_{\text{wet}} = 0.26\text{m}. \tag{3.39}$$

3.3 The vector product

While the scalar product exists on any (finite-dimensional) vector space, the *vector product* exists only on $\mathbb{R}^3$. It is the map $\mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$ defined by

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \times \begin{pmatrix} w_x \\ w_y \\ w_z \end{pmatrix} = \begin{pmatrix} v_y w_z - v_z w_y \\ v_z w_x - v_x w_z \\ v_x w_y - v_y w_x \end{pmatrix}. \tag{3.40}$$

This map is antisymmetric,

$$\mathbf{w} \times \mathbf{v} = -\mathbf{v} \times \mathbf{w}, \tag{3.41}$$

and bilinear,

$$(\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2) \times (\eta_1 \mathbf{w}_1 + \eta_2 \mathbf{w}_2) = \lambda_1 \eta_1 \mathbf{v}_1 \times \mathbf{w}_1 + \lambda_1 \eta_2 \mathbf{v}_1 \times \mathbf{w}_2 + \lambda_2 \eta_1 \mathbf{v}_2 \times \mathbf{w}_1 + \lambda_2 \eta_2 \mathbf{v}_2 \times \mathbf{w}_2. \tag{3.42}$$

For the Cartesian basis $\{\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}\}$, one easily checks that

$$\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}, \qquad \hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}, \qquad \hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}. \tag{3.43}$$

As we will later see, for any ONB $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ we have

$$\mathbf{e}_1 \times \mathbf{e}_2 = \pm \mathbf{e}_3, \qquad \mathbf{e}_2 \times \mathbf{e}_3 = \pm \mathbf{e}_1, \qquad \mathbf{e}_3 \times \mathbf{e}_1 = \pm \mathbf{e}_2, \tag{3.44}$$

where the sign is the same in all three equations. An ONB in which the above holds with the $+$ ($-$) sign is called a right (left) ONB. Obviously, the Cartesian basis is a right ONB.

Exercise 3.4. *Verify that the bases given in* (3.29) *and* (3.30) *are right ONBs.*

The vector product fulfills a number of relations. These could be derived directly from its explicit form (3.40), but it is more economical to first express it in terms of the *Levi-Civita symbol* and then use properties of the latter to infer properties of the vector product. The Levi-Civita symbol is defined as

$$\varepsilon_{ijk} = \begin{cases} 1 & ijk = 123, 231, 312 \\ -1 & ijk = 213, 132, 321 \\ 0 & \text{otherwise} \end{cases} \tag{3.45}$$

One easily checks that the vector product can now be expressed as

$$(\mathbf{v} \times \mathbf{w})_i = \sum_{jk} \varepsilon_{ijk} v_i w_j, \tag{3.46}$$

where we use $(v_x, v_y, v_y) = (v_1, v_2, v_3)$ (and analogously for $\mathbf{w}$) and on the l.h.s. we consider the ith component of the new vector $\mathbf{v} \times \mathbf{w}$. One easily checks that the Levi-Civita symbol is invariant under cyclic permutations and antisymmetric under the interchange of any two indices:

$$\varepsilon_{ijk} = \varepsilon_{jki} = \varepsilon_{kij}, \qquad \varepsilon_{ijk} = -\varepsilon_{jik} = -\varepsilon_{kji} = -\varepsilon_{ikj}. \tag{3.47}$$

The last property implies the anti-symmetry already noted in (3.41):

$$(\mathbf{w} \times \mathbf{v})_i = \sum_{j,k} \varepsilon_{ijk} w_j v_k = -\sum_{j,k} \varepsilon_{ikj} v_k w_j = -\sum_{j,k} \varepsilon_{ijk} v_j w_k = -(\mathbf{v} \times \mathbf{w})_i. \tag{3.48}$$

In the first step, w used the antisymmetry property of the Levi-Civita symbol and then relabelled summation indices, i.e., interchanged j and k.

Exercise 3.5. *Use the cyclicity property of the Levi-Civita symbol to prove*

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \mathbf{v} \cdot (\mathbf{w} \times \mathbf{u}) = \mathbf{w} \cdot (\mathbf{u} \times \mathbf{v}) \,. \tag{3.49}$$

The product $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ is called the *triple product*. Its sign determines whether the vectors $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ form a right or a left system, and the modulus can be geometrically interpreted as the area of the parallelepiped spanned by the three vectors (this will be justified below).

As a consequence of the antisymmetry of the vector product, we have

$$\mathbf{v} \times \mathbf{v} = 0. \tag{3.50}$$

From this and (3.49), it follows that $\mathbf{v} \times \mathbf{w}$ is orthogonal to $\mathbf{v}$:

$$\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = \mathbf{w} \cdot (\mathbf{v} \times \mathbf{v}) = 0. \tag{3.51}$$

Orthogonality to $\mathbf{w}$ follows analogously.

We will sometimes encounter expressions involving two vector products. To simplify these, the following result is useful:

$$\sum_k \varepsilon_{ijk}\varepsilon_{lmk} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}. \tag{3.52}$$

To see this, one can argue as follows: Consider the term $k = 1$ on the left hand side. It gives contributions only if both i, j and l, m are unequal and drawn from $\{2, 3\}$. Specifically, we get 1 when $i = l$, $j = m$, $i \neq j$, and -1 if $i = m$, $j = l$, $i \neq j$ and i, j are drawn from $\{2, 3\}$. The contributions from $k = 2$ and $k = 3$ are similar, but now i, j are drawn from $\{1, 3\}$ and $\{1, 2\}$. As $\{2, 3\}$, $\{1, 3\}$, and $\{1, 2\}$ exhaust all possible unequal pairs drawn from $\{1, 2, 3\}$, we have shown that

$$\sum_k \varepsilon_{ijk}\varepsilon_{lmk} = \begin{cases} 1 & i = l, j = m, i \neq j, \\ -1 & i = m, j = l, i \neq j, \\ 0 & \text{otherwise.} \end{cases} \tag{3.53}$$

But this coincides with the r.h.s. of (3.52), taking into account that the latter vanishes for $i = j = l = m$.

With (3.52) we can now show

$$\begin{aligned} (\mathbf{a} \times (\mathbf{b} \times \mathbf{c}))_i &= \sum_{j,k} \varepsilon_{ijk} a_j \sum_{l,m} \varepsilon_{klm} b_l c_m \\ &= \sum_{j,k,l,m} \varepsilon_{ijk}\varepsilon_{lmk} a_j b_l c_m \\ &= \sum_{j,l,m} (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jk})\, a_j b_l c_m \\ &= b_i \sum_j a_j c_j - c_i \sum_j a_j b_j. \end{aligned} \tag{3.54}$$

Reexpressing the sum as a scalar product and switching back to vector notation, we thus have the *baccab formula*

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}\,(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}\,(\mathbf{a} \cdot \mathbf{b})\,. \tag{3.55}$$

Exercise 3.6. *Prove the identity*

$$(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \cdot \mathbf{c})\,(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})\,(\mathbf{b} \cdot \mathbf{c})\,. \tag{3.56}$$

A geometrical interpretation of the vector product can be obtained by using (3.56) to compute the norm of $\mathbf{v} \times \mathbf{w}$:

$$|\mathbf{v} \times \mathbf{w}|^2 = v^2 w^2 - (\mathbf{v} \cdot \mathbf{w})^2 = v^2 w^2 \left(1 - \cos^2 \theta\right) = v^2 w^2 \sin^2 \theta \tag{3.57}$$

where θ is the angle between $\mathbf{v}$ and $\mathbf{w}$. Hence, $\mathbf{v} \times \mathbf{w}$ is the vector which is orthogonal to both $\mathbf{v}$ and $\mathbf{w}$, with direction determined by the right hand rule, and of length equal to the area of the parallelogram spanned by $\mathbf{v}$ and $\mathbf{w}$ (the area enclosed in Fig. 3.1). This also justifies the interpretation of the triple product as the volume of the parallelepiped spanned by the three vectors.

Finally, we can now also prove (3.44). Consider the first equation. As $\mathbf{e}_1$ and $\mathbf{e}_2$ are of unit length and orthogonal, their vector product must also be of unit length. As it is both orthogonal to $\mathbf{e}_1$ and $\mathbf{e}_2$, it must be an element of the subspace spanned by $\mathbf{e}_3$, so $\mathbf{e}_1 \times \mathbf{e}_2 = \pm \mathbf{e}_3$. From this, we can get the other equations by taking the vector product with $\mathbf{e}_2$ or $\mathbf{e}_1$ and using the baccab formula (3.55):

$$\mathbf{e}_2 \times \mathbf{e}_3 = \pm \mathbf{e}_2 \times (\mathbf{e}_1 \times \mathbf{e}_2) = \pm \left(\mathbf{e}_1 \left(\mathbf{e}_2 \cdot \mathbf{e}_2\right) - \mathbf{e}_2 \left(\mathbf{e}_2 \cdot \mathbf{e}_1\right)\right) = \pm \mathbf{e}_1. \tag{3.58}$$

Using the Levi-Civita symbol, the three equations in (3.44) can be summarized as

$$\mathbf{e}_i \times \mathbf{e}_j = \pm \sum_k \varepsilon_{ijk} \mathbf{e}_k. \tag{3.59}$$

In particular, if v_i, w_j are the components of $\mathbf{v}$, $\mathbf{w}$ in a, not necessarily Cartesian, right ONB $\{\mathbf{e}_i\}$, then (3.46) also holds (for a left ONB, there would be a $-$ sign on the left hand side).

Exercise 3.7. *Verify that the ONBs* $\{\hat{\rho}, \hat{\phi}, \hat{\mathbf{z}}\}$ *and* $\{\hat{\mathbf{r}}, \hat{\theta}, \hat{\phi}\}$ *given in* (3.30), (3.29) *are right ONBs.*

3.4 Torque

Movement of the center of mass is not the only possible motion for an extended body. Disregarding deformation, i.e., restricting to rigid bodies, a further possibility is rotation. While the balance of forces is necessary of the center of mass to stay at rest, balance of torques is required for an extended body to keep its orientation. Assume a force $\mathbf{F}$ acts at a point $\mathbf{r}$, for example because a rope is attached to that point. The corresponding torque is defined as

$$\boldsymbol{\tau} := \mathbf{r} \times \mathbf{F}. \tag{3.60}$$

When several forces $\mathbf{F}_i$ act on points $\mathbf{r}_i$, then the total torque is the sum of the individual ones, i.e.,

$$\boldsymbol{\tau} = \sum_i \boldsymbol{\tau}_i = \sum_i \mathbf{r}_i \times \mathbf{F}_i. \tag{3.61}$$

This definition seems to depend on the choice of the origin. Shifting it by $\mathbf{r}_0$, i.e., considering $\mathbf{r}' = \mathbf{r} - \mathbf{r}_0$, we obtain (note that the forces in the new frame are not affected, $\mathbf{F}'_i = \mathbf{F}_i$, a result

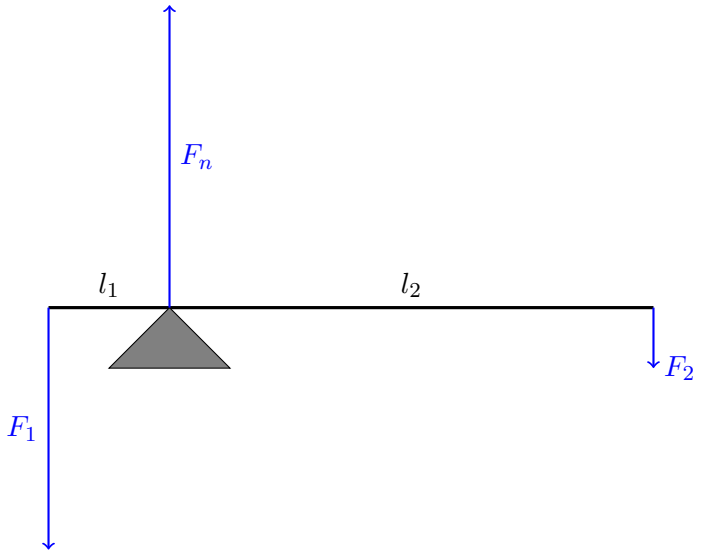


Figure 3.8: The lever.

that we will justify later) for the torque in the new coordinate system

$$\begin{aligned}\boldsymbol{\tau}' &= \sum_i \mathbf{r}_i' \times \mathbf{F}_i \\ &= \sum_i (\mathbf{r}_i - \mathbf{r}_0) \times \mathbf{F}_i \\ &= \sum_i \mathbf{r}_i \times \mathbf{F}_i + \mathbf{r}_0 \times \sum_i \mathbf{F}_i \\ &= \boldsymbol{\tau} + \mathbf{r}_0 \times \mathbf{F}_{\text{tot}}. \end{aligned} \tag{3.62}$$

Hence, if the forces balance, $\mathbf{F}_{\text{tot}} = 0$, the torque in the new frame equals the one in the original frame. In particular, we may choose the origin of coordinates conveniently.

Let us consider the lever as an example, c.f. Fig. 3.8. Forces $\mathbf{F}_1$ and $\mathbf{F}_2$ act on the two ends of a rod, in the same direction and orthogonal to the rod. The rod rest on a tip (fulcrum) which divides the length of the rod in ratio l_1/l_2. To achieve balance of forces, normal force exerted by the tip must be $\mathbf{F}_n = \mathbf{F}_1 + \mathbf{F}_2$. It is now convenient to choose the tip as the origin of coordinates. As the point of attack of the normal force has coordinates $\mathbf{r} = 0$, the normal force does not contribute to the torque. As for the forces on the endpoints, we note that from the tip, the position vectors of the two endpoints are in opposite direction, with the ratio of their lengths $r_1/r_2 = l_1/l_2$. As the forces are orthogonal to the corresponding position vector, their modulus is given by $M_i = l_i F_i$, with opposite directions. Hence, for a static rod, we must have $F_2 = \frac{l_1}{l_2} F_1$, i.e., at the longer end, the force to be exerted is smaller than on the shorter end, by the ratio l_1/l_2. Hence, with an appropriately long lever, one can use a light weight to balance (or lift) a heavy weight. This prompted Archimedes (to whom we owe the oldest known systematic discussion of levers) to pronounce his famous "Give me a lever and I shall move the world".

An important fact about the torque is that (in a homogeneous gravitational field) the total contribution to the torque due to gravitational forces can be obtained by considering only the

gravitational force for the total mass M acting on the *center of mass*

$$\mathbf{R} := \frac{1}{M}\sum_i m_i \mathbf{r}_i \tag{3.63}$$

of the body. To see this, we compute (the subscript g stands for the gravitational contributions only)

$$\begin{aligned}\boldsymbol{\tau}_g &= \sum_i \mathbf{r}_i \times \mathbf{F}_{i,g} \\ &= \sum_i \mathbf{r}_i \times (m_i \mathbf{g}) \\ &= \left(\sum_i m_i \mathbf{r}_i\right) \times \mathbf{g} \\ &= \mathbf{R} \times (M\mathbf{g})\,. \end{aligned} \tag{3.64}$$

Exercise 3.8. *Verify that the center of mass for the lever discussed above is situated at the tip. Why must this be the case?*

As a further example, we consider the leaning ladder, c.f. Fig 3.9 for the setup and notation. We assume that there is negligible friction between the ladder and the wall, while the friction coefficient between ladder and the ground is μ. In the sketch we already used the previous result on the gravitational contribution to the torque. Balance of forces now requires that $F_{1,n} = F_g = mg$ and $F_{2,n} = F_f$. Furthermore, $F_f \leq \mu F_{1,n}$. Balance of torques can most easily be analysed with base point of the ladder as the origin (so that only the gravitational force and the normal force of the wall contribute to the torque). By the properties of the vector product, the two torques are perpendicular to the plane spanned by the ladder and the direction of gravity, the torque due to gravity pointing "out of the paper", the torque due to the normal force of the wall pointing "into the paper". They will thus cancel if their modulus is equal. For the gravitational torque, we have

$$\tau_g = mg\frac{L}{2}\sin(\pi - \theta) = mg\frac{L}{2}\sin\theta, \tag{3.65}$$

while for the "normal one"

$$\tau_{2,n} = F_f L \sin(\tfrac{\pi}{2} + \theta) = F_f L \cos\theta. \tag{3.66}$$

The balance condition thus implies

$$\tan\theta = \frac{2F_f}{mg} \leq 2\mu. \tag{3.67}$$

Hence, the inclination w.r.t. the vertical can not become too large. It may be interesting to note that the slope of the total force $\mathbf{F}_1 = \mathbf{F}_{1,n} + \mathbf{F}_f$ on the base point is twice as large as the slope of the ladder,

$$\frac{F_{1,n}}{F_f} = \frac{2}{\tan\theta}, \tag{3.68}$$

so in particular not along the ladder. This can be understood as follows: Choosing the intersection point of $\mathbf{F}_{2,n}$ and $\mathbf{F}_g$ (starting at their base points) as the origin of coordinates (this point is indicated by the black circle in Fig 3.9), we see that both these forces do not contribute to torque. Hence, the torque due to total force $\mathbf{F}_1$ acting on the base point must vanish, too. But this is only the case if $\mathbf{F}_1$ also points towards the intersection point, i.e., has twice the slope of the ladder.

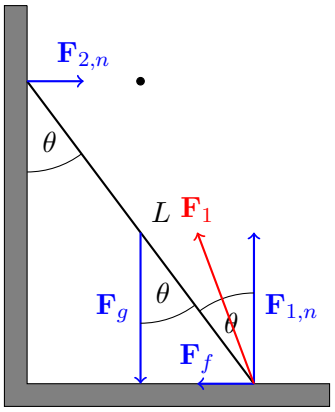


Figure 3.9: The leaning ladder.

4 Motion in one dimension

In the previos chapter, we dealt with static situations. We now turn to dynamical problems, but first confine ourselves to motion in one dimension, i.e., we can describe the evolution of the system by a function $u(t)$, with t the time parameter and u a parameter describing the position of the body. This parameter may or may not be one of the coordinates in Cartesian coordinate system. For example, when describing a pendulum, it is convenient to parameterize it by its angle to the vertical axis.

The general form of the dynamical equation governing such a system is

$$\ddot{u}(t) = f(u(t), \dot{u}(t), t). \tag{4.1}$$

Here the dot stands for a derivative w.r.t. the time parameter t (and a double dot for the second derivative). Even though x may be a general parameter describing the position of the body (not necessarily a coordinate along a straight line), one calls u, $\dot{u}$, and $\ddot{u}$ the position, velocity, and acceleration of the body. The dynamical equation (4.1) equates the acceleration to a "force" f, which may in general depend on the position, the velocity, and the time. It is important to distinguish here the implicit time-dependence through $u(t)$, $\dot{u}(t)$, and an explicit time dependence. The latter occurs for example in the motion of a charged particle in a time-dependent electric field. A dependence on the velocity $\dot{u}$ may be due to friction or drag forces (such as air resistance),[1] but, as we will later see, can also be due to the choice of the position parameter u.

A well-known example of the above is motion along (in the direction of) a uniform gravitational field, such as vertical motion near the surface of Earth. Denoting by h the height, we have the equation

$$\ddot{h}(t) = -g, \tag{4.2}$$

which has the general solution

$$h(t) = -\frac{1}{2}gt^2 + v_0 t + h_0, \tag{4.3}$$

with two free parameters v_0 and h_0. These must be fixed by initial conditions. Evaluating h and $\dot{h}$ at $t = 0$, we obtain

$$h(0) = h_0, \qquad \dot{h}(0) = v_0. \tag{4.4}$$

Hence, the parameters h_0 and v_0 can be identified with the position and velocity at time $t = 0$.

Exercise 4.1. *Show that h_0 and v_0 are also uniquely fixed by initial conditions $h(t) = h_t$, $\dot{h}(t) = v_t$ at some other time t.*

[1] The kinetic friction of bodies sliding or rolling against each other is typically approximately independent on the velocity (Coulomb's law of friction).

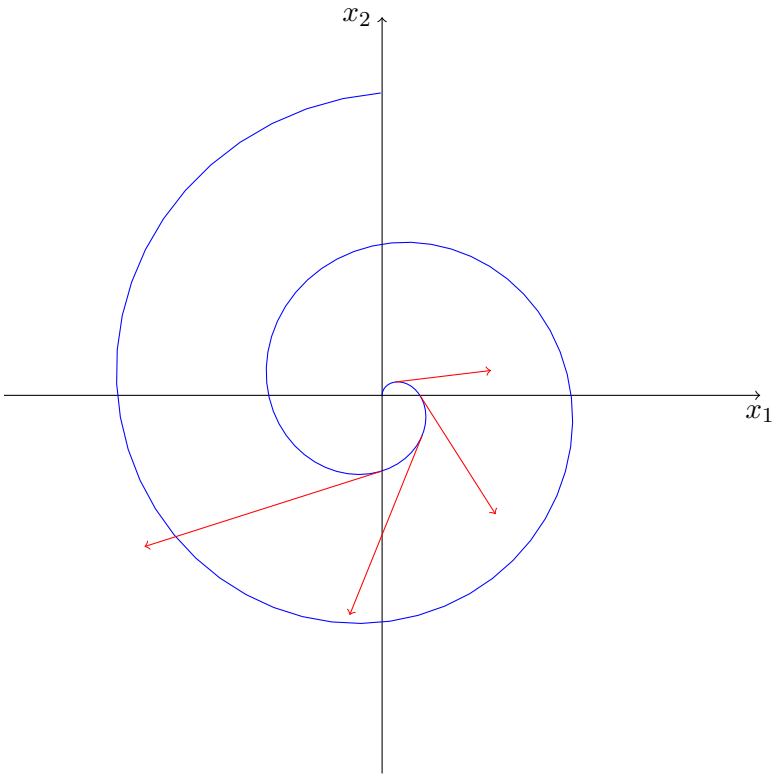


Figure 4.1: The curve $(t\sin(\pi t), t\cos(\pi t))$ on $\mathbb{R}^2$ (in blue) and tangent vectors (in red) at some points. [The arrows are drawn with the base point at the end.]

4.1 Interlude: Curves and vector fields in $\mathbb{R}^n$

Below, we will see that it is convenient to describe the trajectory of a body in one dimension, subject to the dynamical equation (4.1), in terms of a curve in $\mathbb{R}^2$. As we will later also consider curves in higher dimensional spaces, we directly consider curves in $\mathbb{R}^n$. By this, we mean a differentiable function

$$\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix} \tag{4.5}$$

such that all the functions $x_i(t)$ are differentiable. The derivative along the curve, defined by

$$\dot{\mathbf{x}}(t) = \begin{pmatrix} \dot{x}_1(t) \\ \vdots \\ \dot{x}_n(t) \end{pmatrix}, \tag{4.6}$$

specifies, at each point along the curve, a vector $\dot{\mathbf{x}}(t)$ which is tangent to the curve, of length equal to the "velocity" of the curve at the point, c.f. Fig. 4.1.

Another concept that will frequently occur is that of a *vector field* $\mathbf{F}(\mathbf{x})$, i.e., to each point (not just to points along a curve) one associates a vector, c.f. Fig. 4.2. An important related concept is that of an *integral curve* $\mathbf{x}(t)$, which is a curve, such that at each point the tangent

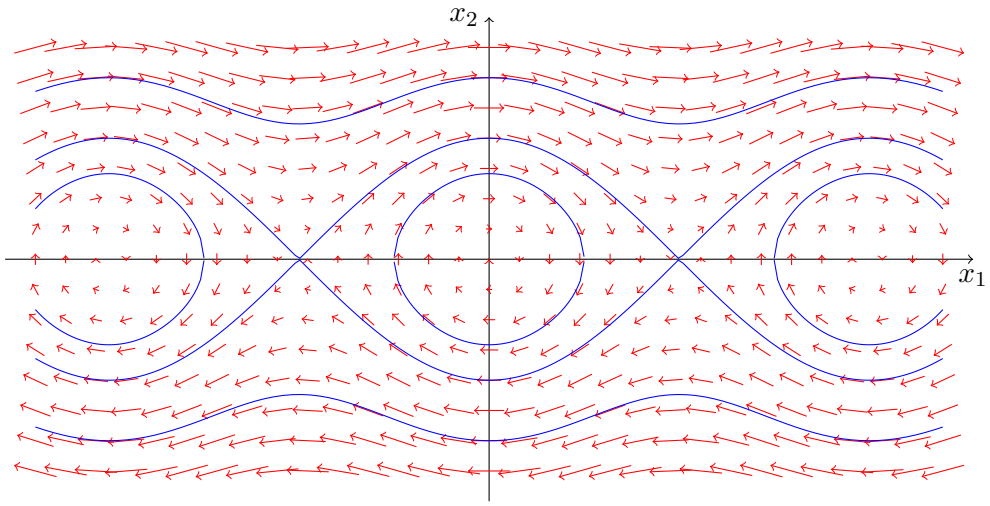


Figure 4.2: The vector field $(x_2, -\sin x_1)$ on $\mathbb{R}^2$ (in red) and some integral curves (in blue). [Note that the arrows are scaled for better visibility. Furthermore, they are drawn with their "base point" in the middle.]

vector coincides with the vector field at the same point, i.e., for all t,

$$\dot{\mathbf{x}}(t) = \mathbf{F}(\mathbf{x}(t)). \tag{4.7}$$

See also Fig. 4.2 for an illustration. An example from physics would be electric field lines, which are integral curves of the electric field.

For later reference, we write out the defining equation (4.7) for integral curves in components:

$$\begin{aligned} \dot{x}_1(t) &= F_1(x_1(t), \dots, x_n(t)), \\ &\vdots \\ \dot{x}_n(t) &= F_n(x_1(t), \dots, x_n(t)). \end{aligned} \tag{4.8}$$

Under some conditions on the functions $F_i(x_1, \dots, x_n)$ to be discussed in more detail below, this system of equations has a unique solution given an initial value $\mathbf{x}_0$ at time $t = 0$. Let us denote by $\mathbf{x}(t, \mathbf{x}_0)$ the curve that starts at $t = 0$ at $\mathbf{x}_0$, i.e., which fulfills

$$\dot{\mathbf{x}}(t, \mathbf{x_0}) = \mathbf{F}(\mathbf{x}(t, \mathbf{x}_0)), \qquad \mathbf{x}(0, \mathbf{x}_0) = \mathbf{x}_0. \tag{4.9}$$

We then have

$$\mathbf{x}(t + s, \mathbf{x}_0) = \mathbf{x}(t, \mathbf{x}(s, \mathbf{x}_0)), \tag{4.10}$$

i.e., starting at $\mathbf{x}_0$ and letting the system evolve for time $t + s$ is equivalent to starting at $\mathbf{x}_0$, letting the system evolve for time s, and use the resulting point as the starting point for another evolution for time t. To prove this, we note that the two functions $t \mapsto \mathbf{x}(t + s, \mathbf{x_0})$ and $t \mapsto \mathbf{x}(t, \mathbf{x}(s, \mathbf{x}_0))$ coincide at $t = 0$, and that both sides fulfill the same differential equation in t, namely:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}(t + s, \mathbf{x}_0) = \dot{\mathbf{x}}(t + s, \mathbf{x_0}) = \mathbf{F}(\mathbf{x}(t + s, \mathbf{x}_0)), \tag{4.11}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}(t, \mathbf{x}(s, \mathbf{x}_0)) = \dot{\mathbf{x}}(t, \mathbf{x}(s, \mathbf{x}_0)) = \mathbf{F}(\mathbf{x}(t, \mathbf{x}(s, \mathbf{x}_0)). \tag{4.12}$$

Hence, by the uniqueness of the solution of such an equation (assuming the necessary conditions for $\mathbf{F}$ are fulfilled), the two functions must coincide, i.e., (4.10) must hold.

An important consequence of the uniqueness of integral curves is that integral curves cannot cross. For assume that we have integral curves $\mathbf{x}_1(t)$, $\mathbf{x}_2(t)$ such that $\mathbf{x}_1(t_1) = \mathbf{x}_2(t_2)$ for some t_1, t_2. But then the curves $t \mapsto \mathbf{x}_1(t_1 + t)$ and $t \mapsto \mathbf{x}_2(t_2 + t)$ coincide at $t = 0$ (by assumption) and both fulfill the differential equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}_i(t_i + t) = \dot{\mathbf{x}}(t_i + t) = \mathbf{F}(\mathbf{x}(t_i + t)), \qquad i = 1, 2, \tag{4.13}$$

so they coincide. Hence up to a possible shift in the parameter t, the two curves coincide.

This result seems to contradict the apparent intersections of curves shown in Fig. 4.2. These take place at $x_2 = 0$ for $x_1 = \pm\pi$ (or more generally for odd multiples of π). At these points, the vector field $\mathbf{F}(\mathbf{x}) = (x_2, -\sin x_1)$ vanishes. Hence, the constant (static) curves $\mathbf{x}(t) = (\pm\pi, 0)$ are integral curves. But the two curves apparently intersecting at this point actually never reach it: The curve $\mathbf{x}(t)$ approaching $(\pi, 0)$ from the "north-west" has the intersection point as the limit point, $\lim_{t\to\infty} \mathbf{x}(t) = (\pi, 0)$, but does not reach the point for any finite time. The same is true for the curve approaching the intersection point from the "south-east". The curves emanating from the intersection point to "north-east" and "south-west" have the intersection point as limit $t \to -\infty$. In physical terms, the zeros of $\mathbf{F}$ are *equilibria*, and those at $(n\pi, 0)$ for even n are stable, while those for odd n are unstable. A trajectory connecting two different unstable equilibria (as in the example sketched in Fig. 4.2) is called a *heteroclinic orbit*, while a trajectory connecting the same unstable equilibrium is called *homoclinic*.

4.2 General properties of one-dimensional motion

Before discussing further concrete cases, we study general properties of one-dimensional motion. We start with terminology. An equation of the form (4.1) is called an ordinary second order differential equation. Here "ordinary" refers to the fact that only derivatives w.r.t. a single variable (here t) occur,[2] and "second order" refers to the fact that the equation depends on the second but no higher derivatives of u. It is sometimes convenient to rewrite the second order equation as a system of two first order equations. For this one defines $v(t) = \dot{u}(t)$ (but variations of this are possible), so that $\dot{v} = \ddot{u}$. Then (4.1) can be equivalently rewritten as

$$\dot{u}(t) = v(t), \tag{4.14a}$$
$$\dot{v}(t) = f(u(t), v(t), t). \tag{4.14b}$$

This system of equations is *coupled*, as it does not separate into two equations in which one only involves u, $\dot{u}$, and t and another equation only involving v, $\dot{v}$, and t.

One advantage of writing the dynamical equation in first order form is that one can apply the *Picard-Lindelöf theorem*, which, under certain conditions, implies the existence of unique solutions $(u(t), v(t))$, given initial conditions $(u(t_0), v(t_0))$ at some time t_0. For a proof of the theorem and the precise technical conditions, we refer to the Mathematics (or Mathematical Methods) modules. The technical conditions refer to the regularity of the function $f(u, v, t)$. They are fulfilled for example when it depends continuously on t, is differentiable w.r.t. u and v, and both these derivatives are bounded as functions of u, v, t. In physical terms, this implies that given an initial position and an initial velocity, the future and past trajectory of the body

[2] If derivatives w.r.t. several variables occur, one speaks of a partial differential equation.

is uniquely determined. However, this does not mean that $u(t)$ (and thus also $v(t)$) can always be determined analytically, i.e., expressed in terms of elementary functions such as polynomials or exponentials.

We now consider some special cases. A very important one occurs when f does not explicitly depend on t, i.e., can be expressed as $f(u, v)$. One calls this an *autonomous* differential equation (or a system of equations). Considering u and v as coordinates on $\mathbb{R}^2$, we see that the system (4.14) is now of the form of the defining equation (4.8) for integral curves of a vector field. Hence, solving (4.14), which is equivalent to solving the original dynamical equation (4.1), is equivalent to finding the integral curves for the vector field

$$\mathbf{F}(\mathbf{x}) = \begin{pmatrix} x_2 \\ f(x_1, x_2) \end{pmatrix}, \tag{4.15}$$

where we identified u with x_1 and v with x_2. By plotting this vector field, one can often learn a lot about qualitative features of solutions to the dynamical equations. For example, for the vector field indicated in Fig. 4.2, one can distinguish two qualitatively different classes of solutions: Periodic bounded solutions, and unbounded solutions (i.e., solutions that leave any finite region in the (u, v) plane in finite time). The boundary between the two regions consists of trajectories that asymptote different instable equilibiria, called *heteroclinic orbits*.

When f does only depend on u (and not on t and v), then a huge simplification takes place, as there is then a *conserved quantity*. Let $V(u)$ be an integral of $-f(u)$, i.e., a function such that $V'(u) = -f(u)$. This function is unique up to an additive constant, i.e., the difference of any two such functions is a constant. Then consider the *energy*

$$E = \frac{1}{2}v^2 + V(u). \tag{4.16}$$

The first term on the r.h.s. is called the *kinetic energy*, while the second one is referred to as the *potential energy*. Differentiating the energy w.r.t. t (using the chain rule), we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}E = v\dot{v} + V'(u)\dot{u} = v\left(\dot{v} - f(u)\right) = 0, \tag{4.17}$$

where we used the chain rule, the definition of V, and $v = \dot{u}$ (with the dot denoting the derivative w.r.t. t). In the last step, we used the dynamical equation (4.1), or equivalently (4.14). This means that under the given dynamics, the energy E does not change, i.e., is conserved.

A conserved quantity such as E is extremely useful for the study of solutions of the dynamical equation. For example, we can solve for v as a function of the position u as

$$v = \pm\sqrt{2(E - V(u))}. \tag{4.18}$$

Hence, given a value for E (provided by initial conditions for u, v), we can determine the velocity v at any given position u. In particular, this provides us with the integral curves of the vector field $\mathbf{F}(u, v)$ given by (4.15), and this is also how the integral curves in (4.2) were computed.

The equation (4.18) provides us with important qualitative information. For example, we know that only those values of u are allowed for which $V(u) \geq E$. Hence, given a value for E, we might learn that the motion is confined to certain regions, c.f. Fig. 4.3. The boundaries of these regions, where $V(u) = E$, are in general *turning points* of the motion, as, by (4.18), the velocity vanishes there. More precisely, for a given energy E, we call u a turning point if $V(u) = E$ and $V'(u) = -f(u) \neq 0$. The latter condition ensures that the motion cannot stop at the turning point but has to turn back. Hence, the sign of v changes at turning point.

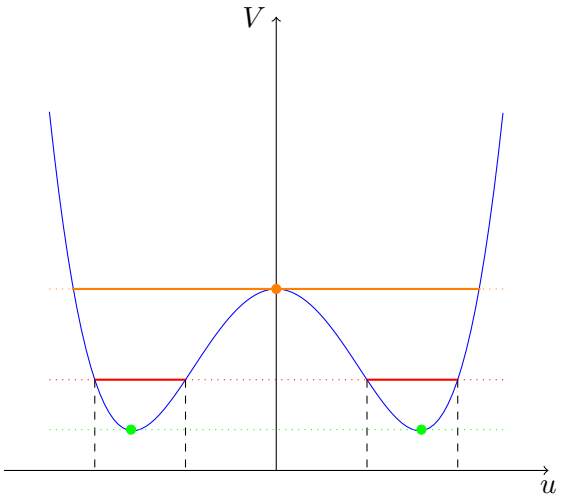


Figure 4.3: A potential in blue and a possible choice of the energy in red. The dotted red line indicates the regions which are forbidden, for the given energy, while the solid red line indicates the allowed regions. The turning points are the boundaries of these regions. The leftmost turning point can only be approached from the right, i.e., with $\dot{u} = v < 0$. It can only be left to the right, i.e., with $\dot{u} = v > 0$. The velocity v changes its sign at the turning point. Stable and unstable equilibria are indicated by green and orange dots and the corresponding energies by lines.

If the constant function $u(t) = u_0$ is a solution to the dynamical equation, then we call u_0 an *equilibrium* configuration. This is the case iff $f(u_0) = 0$ (if f is also a function of the velocity $\dot{u}$, then the condition is $f(u_0, 0) = 0$). By $V'(u) = -f(u)$, the equilibria can also be characterized as the extrema of the potential (in the sense that $V'(u_0) = 0$). In Fig. 4.3, the equilibria are indicated by green and orange dots. Depending on whether an equilibrium is local minimum or maximum of the potential, the behaviour near such an equilibrium is qualitatively very different.

- A local minimum u_0 of the potential $V(u)$ is called a *stable* equilibrium. In Fig. 4.3, these are indicated by green dots. If $E_0 := V(u_0)$ is the energy of the equilibrium configuration (for which the velocity and thus the kinetic energy vanishes), then a stable equilibrium can also be characterized by the fact that it is an isolated point in the allowed region for energy E_0. In Fig. 4.3, the energy E_0 is indicated by the green dotted line. Raising the energy slightly above E_0, the allowed region grows, but will still be bounded (at least the component containing u_0). Hence, for small perturbations of a stable equilibrium, one has bounded periodic motion.

- A local maximum u_0 of the potential $V(u)$ is called an *unstable* equilibrium. In Fig. 4.3, this is indicated by an orange dot. An unstable equilibrium can also be characterized by being an element of an *extended* allowed region for the energy E_0 (indicated by the solid orange line in Fig. 4.3). As we will see, if $V'(u_0) < 0$, then small deviations from the equilibrium will approximately grow exponentially in time.

- If the equilibrium u_0 is a saddle point of $V(u)$, then it is again an element of an *extended* allowed region for the energy E_0 (extending only to one side from u_0). Hence, it is again unstable.

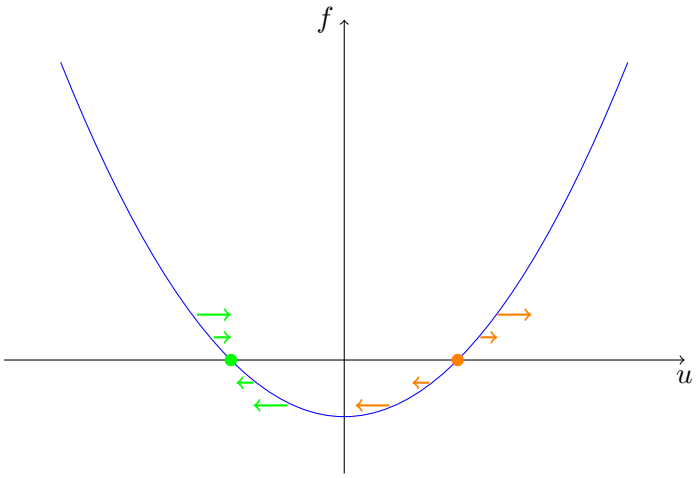


Figure 4.4: Behaviour of the force near a stable (green) and an unstable equilibrium (orange).

The different behaviour near stable and unstable equilibria can also be discussed on the level of the force $f(u)$, c.f. Fig. 4.4: Near the stable equilibrium (green), where $f' = -V''$ is negative, the force tends to drive the system back to the equilibrium. Near the unstable equilibrium (orange), where $f' = -V''$ is positive, the force tends to drive the system away from the equilibrium.

Apart from qualitative information on the motion, (4.18) can in principle also be used to obtain the solution $u(t)$ of the dynamical equation. For this, we write $v = \frac{\mathrm{d}u}{\mathrm{d}t}$ and bring $\mathrm{d}t$ in (4.18) to the other side, obtaining

$$\pm \frac{1}{\sqrt{2(E - V(u))}} \mathrm{d}u = \mathrm{d}t. \tag{4.19}$$

Given E and the initial condition $u(t_0) = u_0$, we can now determine t as a function of u by

$$t - t_0 = \int_{t_0}^{t} \mathrm{d}t' = \pm \int_{u_0}^{u} \frac{1}{\sqrt{2(E - V(u'))}} \mathrm{d}u'. \tag{4.20}$$

The choice of sign corresponds to the choice of the sign of the initial velocity $v(t_0) = v_0$. Once $t(u)$ is determined either by solving the integral analytically or numerically, the inverse function $u(t)$ gives us the desired trajectory. It is important to note that the integral in (4.20) can only be performed up to a *turning point* u_{tp} (beyond the turning point, the argument of the square root in the integrand becomes negative). Thus, when a turning point is reached, one should in principle use it as a new starting point, and use the other sign in (4.20) (as the direction of motion changed). However, as the integral itself is still the same, there is not really a new integral to solve. Concretely, if $u(t)$ is the trajectory which reaches the turning point u_{tp} at t_{tp}, then one can continue the trajectory as $u(t_{\mathrm{tp}} + t) = u(t_{\mathrm{tp}} - t)$ for $t \geq 0$, i.e., by "inverting" the time evolution.

Exercise 4.2. *We use the above general methods for the motion in a uniform gravitational field, i.e., the dynamical equation* $\ddot{u} = -g$.

a) Show that the dynamical equation can be written in first order form as the system

$$\dot{u} = v, \tag{4.21}$$
$$\dot{v} = -g. \tag{4.22}$$

b) Defining the vector

$$\mathbf{q} = \begin{pmatrix} u \\ v \end{pmatrix} \tag{4.23}$$

show that the above can be expressed as

$$\dot{\mathbf{q}} = \mathbf{F}(\mathbf{q}), \tag{4.24}$$

for an appropriately chosen vector field $\mathbf{F}$*.*

c) Sketch the vector field $\mathbf{F}$ *(you may use plotting tools).*

d) Defining $E = \frac{1}{2}v^2 + gu$*, show that* E *is a constant of motion.*

e) For a fixed E*, determine* v *as a function of* u *and plot this function for different values of* E *in your sketch from c). Do these represent integral curves of the vector field* $\mathbf{F}$*? How do you see that?*

f) Show that the above entails

$$\mathrm{d}t = \pm \frac{1}{\sqrt{2(E - gu)}} \mathrm{d}u. \tag{4.25}$$

What is the physical significance of the choice of the sign?

g) Integrate the above with the initial condition $u(t = 0) = u_0$ *to obtain* $t(u)$*.*

h) Invert the function $t(u)$ *to obtain* $u(t)$*. Express* E *in terms of the initial velocity* $v(t = 0) = v_0$ *and compare your result to the well-known solution* $u(t) = u_0 + v_0 t - \frac{1}{2}gt^2$*.*

4.3 The pendulum

We apply the methods developed above to the pendulum in a plane (sometimes also called the *mathematical pendulum*). It consists of a rod of length L, idealized as massless, one end of which is fixed to a mass m, the other to an axis around which it can rotate in a plane, c.f. Fig. 4.5. We denote by θ the angle by which the pendulum is displaced from its equilibrium position, as indicated in the figure. To conveniently describe the motion of the mass, we introduce two orthogonal unit vectors

$$\hat{\mathbf{r}}(\theta) = \begin{pmatrix} \sin\theta \\ -\cos\theta \end{pmatrix}, \qquad \hat{\boldsymbol{\theta}}(\theta) = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \tag{4.26}$$

which are also indicated in the figure. While $\hat{\mathbf{r}}$ points along the rod, $\hat{\boldsymbol{\theta}}$ is orthogonal to the rod (and hence both depend on the angle θ, as explicitly indicated). The position of the mass (when placing the axis of rotation at the origin) can then be described in terms of θ as $\mathbf{r}(\theta) = L\hat{\mathbf{r}}(\theta)$. Using that

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\hat{\mathbf{r}} = \hat{\boldsymbol{\theta}}, \qquad \frac{\mathrm{d}}{\mathrm{d}\theta}\hat{\boldsymbol{\theta}} = -\hat{\boldsymbol{r}}, \tag{4.27}$$

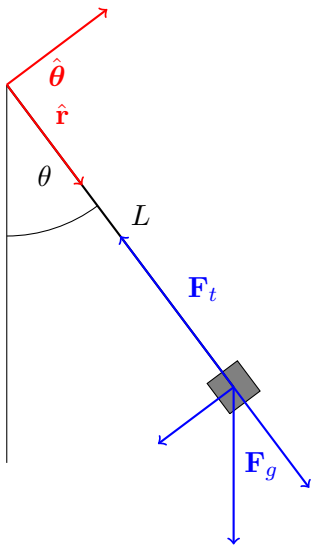


Figure 4.5: The pendulum.

we compute for the velocity and acceleration (first and second derivative w.r.t. t) of the mass

$$\dot{\mathbf{r}} = L\dot{\theta}\hat{\boldsymbol{\theta}}, \qquad \ddot{\mathbf{r}} = L\ddot{\theta}\hat{\boldsymbol{\theta}} - L(\dot{\theta})^2\hat{\mathbf{r}}, \tag{4.28}$$

where we used the chain rule $\frac{\mathrm{d}}{\mathrm{d}t}f(\theta(t)) = \frac{\mathrm{d}}{\mathrm{d}\theta}f(\theta(t))\frac{\mathrm{d}}{\mathrm{d}t}\theta(t)$.

We now anticipate Newton's equation for the motion of a mass point (to be discussed in more detail in Chapter 6), namely

$$m\ddot{\mathbf{r}} = \mathbf{F} \tag{4.29}$$

with $\mathbf{F}$ the total force acting on the body. In the present case, the forces acting on the mass are the gravitational force

$$\mathbf{F}_g = mg\begin{pmatrix} 0 \\ -1 \end{pmatrix} = F_{g,r}\hat{\mathbf{r}} + F_{g,\theta}\hat{\boldsymbol{\theta}}, \tag{4.30}$$

which we decompose as indicated in the figure into a components pointing in directions $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$, and the rod tension

$$\mathbf{F}_t = -F_t\hat{\mathbf{r}}, \tag{4.31}$$

which only acts along the direction of the rod. According to what we learned in Section 3.1, we can determine the coefficients $F_{g,r}$ and $F_{g,\theta}$ by taking the scalar product with $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$:

$$F_{g,r} = \hat{r}\cdot\mathbf{F}_g = mg\cos\theta, \qquad F_{g,\theta} = \hat{\theta}\cdot\mathbf{F}_g = -mg\sin\theta. \tag{4.32}$$

Decomposing (4.29) into components w.r.t. $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$, we obtain

$$-mL\dot{\theta}^2 = mg\cos\theta - F_t, \qquad mL\ddot{\theta} = -mg\sin\theta. \tag{4.33}$$

From the first equation, we can determine the tension in the rod as

$$F_t = m\left(L\dot{\theta}^2 + g\cos\theta\right). \tag{4.34}$$

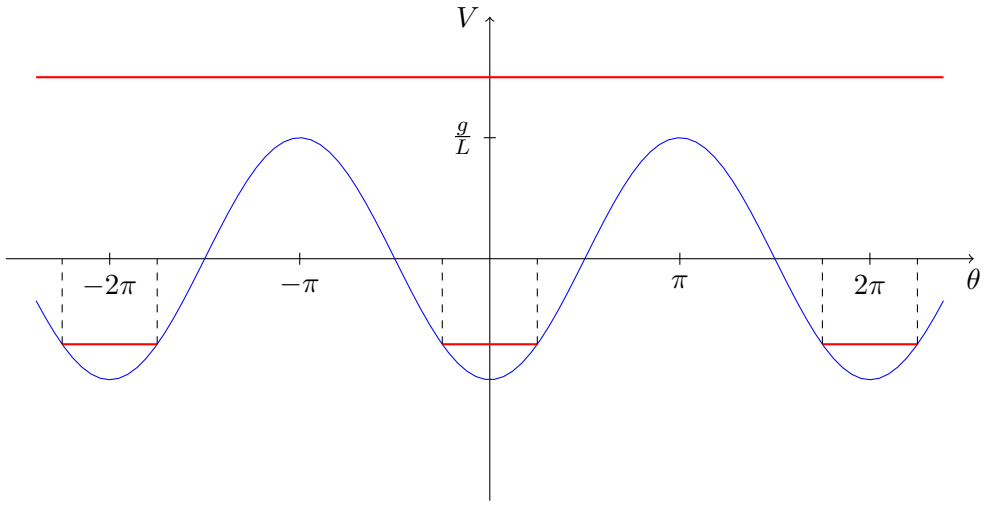


Figure 4.6: The potential $V = -\frac{g}{L}\cos\theta$ for the pendulum (in blue) and the allowed ranges for θ for two different values of the energy E (in red). In the upper case, the pendulum runs through all values of θ, in the lower one θ is confined to one of the intervals indicated by the dashed lines.

The second term is due to the gravitational force, while the first term is commonly referred to as the centrifugal force. In particular, the rod has to support a force greater than the gravitational force on the mass at the endpoint. The second equation in (4.33) can be rewritten as

$$\ddot{\theta} = -\frac{g}{L}\sin\theta \tag{4.35}$$

and is thus of the general form (4.1), but with $f(\theta, \dot{\theta}, t) = -\frac{g}{L}\sin\theta$ only depending on the position θ (and not on time or the velocity $\dot{\theta}$). In particular, with the potential $V(\theta) = -\frac{g}{L}\cos\theta$, we obtain the constant of motion

$$E = \frac{1}{2}\dot{\theta}^2 - \frac{g}{L}\cos\theta, \tag{4.36}$$

and we can distinguish the two regimes $-\frac{g}{L} \leq E < \frac{g}{L}$ and $\frac{g}{L} < E$, c.f. Fig. 4.6 for a sketch of the potential and the allowed values for the angle θ in the two regimes. In the first case, there are turning points at solutions to $\cos\theta = -\frac{LE}{g}$, such that the motion is confined to one of the intervals $[2\pi n - \arccos\frac{LE}{g}, 2\pi n + \arccos\frac{LE}{g}]$ for $n \in \mathbb{Z}$. This is the periodic motion known from a clock pendulum or a swing. In the second case ($\frac{g}{L} < E$), there are no turning points, i.e., $\dot{\theta}$ remains either positive or negative and runs through all angles. This corresponds to a rotation either in the clockwise or the anti-clockwise direction (depending on the sign of $\dot{\theta}$). These types of motion correspond precisely to some of the integral curves shown in Fig. 4.2. The "wavy" lines at the top and the bottom correspond to the rotation in the clockwise and the anti-clockwise direction, while the "circular" lines correspond to the periodic orbits.

Up to now, we considered qualitative features of the solutions to the dynamical equation. Can we also solve it, i.e., determine $\theta(t)$ for given initial conditions? Imposing for simplicity the initial condition $\theta(t = 0) = 0$ (with $\dot{\theta}(t = 0)$ fixed by the energy, up to the sign), we have,

according to the discussion in the previous section, to first determine $t(\theta)$ by

$$t(\theta) = \int_0^\theta \frac{1}{\sqrt{2(E + k\cos\theta')}} \mathrm{d}\theta', \tag{4.37}$$

where we introduced the abbreviation $k = \frac{g}{L}$. When we consider the periodic case, i.e., $|\frac{E}{k}| < 1$, we can, by a change of variable, bring this to the form

$$t(\theta) = \frac{1}{\sqrt{k}} \int_0^{\arcsin \frac{\sin\frac{\theta}{2}}{\sin\frac{\theta_{\mathrm{tp}}}{2}}} \frac{1}{\sqrt{1 - \sin^2\frac{\theta_{\mathrm{tp}}}{2}\sin^2 x}} \mathrm{d}x, \tag{4.38}$$

with θ_{tp} the angle at the turning point, defined by $\cos\theta_{\mathrm{tp}} = -\frac{E}{k}$. The integral cannot be expressed in terms of elementary functions. In fact, it defines the *incomplete elliptic integral of the first kind*. However, one can get useful approximations for small oscillations, i.e., $\theta_{\mathrm{tp}} \ll 1$. To illustrate this, consider the period T of the motion, which, by the symmetry of the motion under $\theta \to -\theta$, is obtained as four times the time it takes to reach the turning point θ_t from the equilibrium position $\theta = 0$:

$$T = \frac{4}{\sqrt{k}} \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1 - \sin^2\frac{\theta_{\mathrm{tp}}}{2}\sin^2 x}} \mathrm{d}x. \tag{4.39}$$

As a "zeroth order" approximation, we can now consider the limit $\theta_{\mathrm{tp}} \to 0$, for which the integrand equals 1, so that

$$\lim_{\theta_{\mathrm{tp}}\to 0} T = \frac{2\pi}{\sqrt{k}}. \tag{4.40}$$

Later, when we introduce the Taylor series, we will compute corrections to this result. We will then also show that the period diverges in the limit $\theta_t \to \pi$, as expected (if one could reach the unstable equilibrium position in finite time such that one has vanishing velocity there, the solution to the dynamical equation would be non-unique).

Exercise 4.3. *Perform the change of variables to get from* (4.37) *to* (4.38). *For this, express* E *in terms of* θ_{tp}, *use the identity* $\cos\theta = 1 - 2\sin^2\frac{\theta}{2}$ *and the variable transformation*

$$\sin x = \frac{\sin\frac{\theta'}{2}}{\sin\frac{\theta_{\mathrm{tp}}}{2}}. \tag{4.41}$$

Using the pendulum, one can also illustrate the importance of choosing a suitable variable to describe the motion. Above, we chose the displacement angle θ and arrived at the rather simple equation (4.35). Let us consider the equation we would have obtained by describing the motion instead by using the distance $x = L\sin\theta$ of the mass to the vertical axis. Of course, with this choice we are already restricting ourselves to displacement angles $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$, but to describe (not so small) oscillations, this is completely sufficient. It turns out that in terms of x, the dynamical equation takes the complicated form

$$\ddot{x} = -\frac{x\dot{x}^2}{L^2(1 - \frac{x^2}{L^2})} - \frac{g}{L}x\sqrt{1 - \frac{x^2}{L^2}}. \tag{4.42}$$

In particular, we have generated a dependence on the velocity $\dot{x}$. This illustrates that a dependence of the "force" f on the velocity does not necessarily imply friction or drag forces, but can also be an artefact of an unsuitable choice of the dynamical variable.

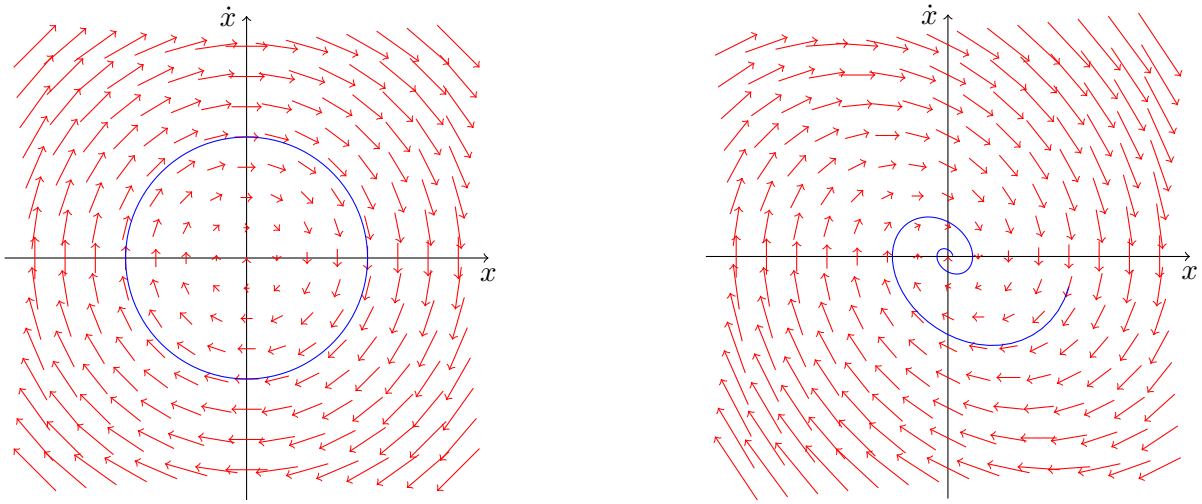


Figure 4.7: The flow field for the oscillator (in red) for $\beta = 1$, $\gamma = 0$ (left side) and $\beta = 1$, $\gamma = \frac{1}{4}$ (right side). Also shown (in blue) is an integral curve, i.e., a solution to the dynamical equation.

Exercise 4.4. *Consider the change of variable $x = L\sin\theta$ from the displacement angle θ to x as discussed above.*

a) Show that

$$\dot{x} = L\dot{\theta}\cos\theta, \qquad \ddot{x} = L\ddot{\theta}\cos\theta - L\dot{\theta}^2\sin\theta. \tag{4.43}$$

b) Using these, derive (4.42) from (4.35) (or vice versa).

4.4 The oscillator

The oscillator equation has the form

$$\ddot{x} = -\beta x - 2\gamma\dot{x}, \tag{4.44}$$

with real coefficients β and γ. Typically, β and γ will be non-negative, but it will also be instructive to consider negative values. A prototypical example is a mass m attached to a spring of spring constant k, in which case $\beta = \frac{k}{m}$ and (if friction is neglected) $\gamma = 0$.

To get a first idea of the possible dynamics, one may plot the corresponding flow field $\mathbf{F}(x, \dot{x}) = (-\dot{x}, -\beta x - 2\gamma\dot{x})$ in the $(x, \dot{x})$-plane, c.f. Fig. 4.7. Also shown are integral curves, i.e., solutions to dynamical equation. For $\gamma = 0$, the motion is periodic, while for $\gamma > 0$, the oscillation is damped, so that the solutions converge to $(x, \dot{x}) = (0, 0)$.

To really prove this behaviour (and to obtain the integral curves shown in the figures, one has to solve (4.44). This is greatly facilitated by the fact that it is a *linear* equation, i.e., when writing it in the form $E(x, \dot{x}, \ddot{x}) = 0$, then the function E is linear,

$$E(\lambda_1 x_1 + \lambda_2 x_2, \lambda_1\dot{x}_1 + \lambda_2\dot{x}_2, \lambda_1\ddot{x}_1 + \lambda_2\ddot{x}_2) = \lambda_1 E(x_1, \dot{x}_1, \ddot{x}_1) + \lambda_2 E(x_2, \dot{x}_2, \ddot{x}_2), \tag{4.45}$$

where $\lambda_1, \lambda_2 \in \mathbb{R}$. For later purposes, it is important to note that E is even complex linear, i.e., if we allow for complex $x(t) \in \mathbb{C}$, then the above also holds for $\lambda_1, \lambda_2 \in \mathbb{C}$. The linearity of

the dynamical equation implies that if $x_1(t)$ and $x_2(t)$ both solve the dynamical equation, i.e., $E(x_1, \dot{x}_1, \ddot{x}_1) = 0$ and $E(x_2, \dot{x}_2, \ddot{x}_2) = 0$, then also their linear combination

$$x(t) = \lambda_1 x_1(t) + \lambda_2 x_2(t) \tag{4.46}$$

(in particular, $x(t) = 0$ is always a solution). This *superposition principle* is very useful for finding the general solution to the dynamical equation. Assume we have found two solutions $x_1(t)$, $x_2(t)$ which are linearly independent, i.e., there is no $\lambda \in \mathbb{R}$ such that $x_1(t) = \lambda x_2(t)$ for all t. Then the general solution to the dynamical equation is their linear combination (4.46). To see this, one can note that, as solutions $x(t)$ are uniquely determined in terms of initial data $(x(t_0), \dot{x}(t_0))$, and the solutions $x_1(t)$ and $x_2(t)$ are linearly independent, so must be their initial data $(x_1(t_0), \dot{x}_1(t_0))$, $(x_2(t_0), \dot{x}_2(t_0))$. But the space of possible initial data is a two-dimensional vector space, so any possible initial data $(x(t_0), \dot{x}(t_0))$ can be expressed uniquely as a linear combination of $(x_1(t_0), \dot{x}_1(t_0))$ and $(x_2(t_0), \dot{x}_2(t_0))$. Hence, in order to obtain the general solution (4.46) it suffices to find two linearly independent solutions.

But coming up with two linearly independent solution to a linear dynamical equation such as (4.44) is very simple. We use the exponential ansatz

$$x(t) = e^{-i\omega t}, \tag{4.47}$$

with ω to be determined (the factor of $-i$ is introduced for notational convenience, as will become clearer below). Note that we allow here for complex solutions. Plugging this ansatz into (4.44), we obtain

$$\left(\omega^2 + 2i\gamma\omega - \beta\right) e^{-i\omega t} = 0, \tag{4.48}$$

leading to the quadratic equation

$$\omega^2 + 2i\gamma\omega - \beta = 0, \tag{4.49}$$

with solutions

$$\omega_\pm = -i\gamma \pm \sqrt{\beta - \gamma^2}. \tag{4.50}$$

Hence, as long as $\beta \neq \gamma^2$ (the case $\beta = \gamma^2$ will be discussed below), the general solution to the dynamical equation (4.44) is

$$x(t) = \lambda_+ e^{-i\omega_+ t} + \lambda_- e^{-i\omega_- t} = e^{-\gamma t}\left(\lambda_+ e^{-i\sqrt{\beta-\gamma^2}t} + \lambda_- e^{i\sqrt{\beta-\gamma^2}t}\right), \tag{4.51}$$

with complex coefficients $\lambda_\pm$.

We still need to arrive at real solutions $x(t)$. One can do this in different (but equivalent) manners. One possibility is to find conditions on $\lambda_\pm$ which guarantee that $x(t)$ is real. For $\beta > \gamma^2$, this is the case for $\lambda_- = \bar{\lambda}_+$. Writing $\lambda_\pm = \frac{1}{2} C e^{\pm i\phi}$ (with real C, ϕ), we arrive at

$$x(t) = C e^{-\gamma t} \cos\left(\sqrt{\beta - \gamma^2} t - \phi\right), \tag{4.52}$$

with two real parameters C, ϕ. For $\beta < \gamma^2$, we have $\sqrt{\beta - \gamma^2} = i\sqrt{\gamma^2 - \beta}$, so that also the exponentials are real, so that $x(t)$ is real if and only if both λ_+ and λ_- are. Then the general solution is

$$x(t) = \lambda_+ e^{-(\gamma - \sqrt{\gamma^2-\beta})t} + \lambda_- e^{-(\gamma + \sqrt{\gamma^2-\beta})t}. \tag{4.53}$$

Another possibility to obtain a real solution is to take the real part of the general complex solution $x(t)$. As the coefficients in the linear function $E(x, \dot{x}, \ddot{x})$ are real, it follows that for a complex solution $x(t)$ to $E(x, \dot{x}, \ddot{x}) = 0$, we have

$$0 = \operatorname{Re} E(x, \dot{x}, \ddot{x}) = E(\operatorname{Re} x, \operatorname{Re} \dot{x}, \operatorname{Re} \ddot{x}), \tag{4.54}$$

so that also $\operatorname{Re} x(t)$ is a solution. Computing the real part of (4.51) for the case $\beta > \gamma^2$, we obtain

$$\operatorname{Re} x(t) = e^{-\gamma t}\left(\operatorname{Re}(\lambda_+ + \lambda_-)\cos\sqrt{\beta-\gamma^2}t + \operatorname{Im}(\lambda_+ - \lambda_-)\sin\sqrt{\beta-\gamma^2}t\right). \tag{4.55}$$

As the complex numbers $\lambda_\pm$ can be chosen arbitrarily, any values for the two real coefficients $\operatorname{Re}(\lambda_+ + \lambda_-)$ and $\operatorname{Im}(\lambda_+ - \lambda_-)$ are possible, so that we can express the general real solution also as

$$x(t) = e^{-\gamma t}\left(A\cos\sqrt{\beta-\gamma^2}t + B\sin\sqrt{\beta-\gamma^2}t\right). \tag{4.56}$$

Using the trigonometric identity

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta, \tag{4.57}$$

one can uniquely relate (C,ϕ) in (4.52) to (A,B) in (4.57) and vice versa.

Having determined the general solution (up to the case $\beta = \gamma^2$), we now turn to the discussion of the physics of these solutions for different ranges of the parameters.

$\beta > 0, \gamma = 0$: This case is called the *undamped oscillator equation*, or sometimes also just the *oscillator equation*. The general solution

$$x(t) = C\cos(\omega t - \phi) \tag{4.58}$$

with the *angular frequency* $\omega = \sqrt{\beta}$ is periodic with period $T = \frac{2\pi}{\omega}$. One calls C the *amplitude* and ϕ the *phase* of the solution. It is interesting to now that the period is the same for all solutions, in particular independent of the amplitude. This fact is practically important for example in the design of mechanical clockworks. In the $(x,\dot{x})$ plane, the solution

$$\begin{pmatrix} x(t) \\ \dot{x}(t) \end{pmatrix} = \begin{pmatrix} C\cos(\omega t - \phi) \\ -\omega C\sin(\omega t - \phi) \end{pmatrix} \tag{4.59}$$

is an ellipse with major/minor axes along the x and $\dot{x}$ axes.

$\beta > \gamma^2, \gamma > 0$: This case is called the *undercritical damped oscillator equation*. In the general solution

$$x(t) = Ce^{-\gamma t}\cos(\omega t - \phi) \tag{4.60}$$

with $\omega = \sqrt{\beta-\gamma^2}$, we multiply the oscillating $\cos(\omega t - \phi)$ with an exponential damping factor. The angular frequency $\omega = \sqrt{\beta-\gamma^2}$ of the oscillation is smaller than that of the corresponding undamped oscillator (same β). The decay rate due to the exponential is γ, the damping coefficient in the oscillator equation.

$0 < \beta < \gamma^2, \gamma > 0$: This case is called the *overcritical damped oscillator equation*. The general solution (4.53) is a linear combination of two decaying exponentials. Their decay rates are given by

$$\gamma_\pm = \gamma \mp \sqrt{\gamma^2 - \beta}. \tag{4.61}$$

For *generic* initial conditions (for which the coefficient λ_+ of the term $e^{-\gamma_+ t}$ does not happen to vanish), the leading behaviour at late times will be $x(t) \sim e^{-\gamma_+ t}$. It is interesting to note that the decay rate γ_+ *decreases* as γ increases. In fact, for $\gamma^2 \gg \beta$, one has[3]

$$\gamma_+ = \gamma\left(1 - \sqrt{1 - \frac{\beta}{\gamma^2}}\right) \sim \frac{\beta}{2\gamma}. \tag{4.62}$$

[3]This can be shown using Taylor approximation, as discussed below.

Hence, increasing the damping coefficient γ beyond the critical value $\gamma^2 = \beta$ slows down the relaxation of the system to $x = 0$.

$\beta = \gamma^2, \gamma > 0$: This is called the *critically damped oscillator equation*. As $\omega_\pm$ as given by (4.50) coincide in this case, we up to now only have a single solution $x(t) = e^{-\gamma t}$. A general method to arrive at a further, linearly independent, solution is *variation of constants*. One multiplies the know solution with a function of t, i.e., considers $x(t) = y(t)e^{-\gamma t}$. The dynamical equation now yields an equation for $y(t)$, namely $\ddot{y} = 0$, the general solution of which is $y(t) = A + Bt$. The solution with $B = 0$ is of course the original solution, so we arrive at the general solution

$$x(t) = (A + Bt)e^{-\gamma t}. \tag{4.63}$$

The critically damped oscillator thus has decay rate γ (the linear growth in the first factor is always beaten by the exponential decay for late times). Hence, if for a fixed value of β one wants to optimize the system for quick return to the equilibrium position (such as in the damping mechanism of a swing door), one will choose γ close to criticality.

$\beta = 0, \gamma \neq 0$: In this case, the general solution is a linear combination of the constant solution $x(t) = 1$, and $x(t) = e^{-\gamma t}$. For $\gamma > 0$, this means that any solution will converge with decay rate γ to a constant solution (with limiting value determined by the initial condition), while for $\gamma < 0$ this means that the generic solution will exponentially grow with growth rate $-\gamma$.

$\beta = 0, \gamma = 0$: In this case of free motion, the general solution is $x(t) = A + Bt$.

$\beta < 0$: The general solution is a linear combination of $e^{-\gamma_\pm t}$ with $\gamma_\pm$ as in (4.61). Irrespective of the sign of γ, we have $\gamma_+ < 0$ and $\gamma_- > 0$. Hence, the generic solution will have exponential growth with rate $-\gamma_+ = \sqrt{\gamma^2 - \beta} - \gamma$. In the special case $\gamma = 0$, the general solution is $x(t) = \lambda_+ e^{\sqrt{-\beta}t} + \lambda_- e^{-\sqrt{-\beta}t}$.

$\beta > 0, \gamma < 0$: In this case, one can again distinguish three cases depending on whether β is greater, equal, or smaller than γ^2. The solutions have the same form as for the undercritical, critical, and overcritical damped oscillator, except that due to the sign of γ there is exponential growth instead of deacy.

We have seen that the generic solution (i.e., without fine-tuning of initial data) has exponential growth whenever either $\beta < 0$ or $\gamma < 0$.

4.5 The driven oscillator

We now consider a variant of the oscillator, the driven oscillator, for which the dynamical equation is

$$\ddot{x} = -\beta x - 2\gamma \dot{x} + f(t), \tag{4.64}$$

i.e., we add an explicitly time-dependent driving force $f(t)$. Importantly, this driving force is assumed to be independent of x and $\dot{x}$. The dynamical equation is thus an *inhomogeneous linear differential equation*, schematically of the form

$$E(x, \dot{x}, \ddot{x}) = F(t), \tag{4.65}$$

with E linear in the sense discussed above. Such an inhomogeneous linear differential equation has the following important property: If $x(t)$ is a solution to the inhomogeneous equation (4.65)

and $x_0(t)$ a solution to the associated homogeneous equation $E(x, \dot{x}, \ddot{x}) = 0$, then $x(t) + x_0(t)$ also solves the inhomogeneous equation (4.65), due to the linearity of E:

$$E(x + x_0, \dot{x} + \dot{x}_0, \ddot{x} + \ddot{x}_0) = E(x, \dot{x}, \ddot{x}) + E(x_0, \dot{x}_0, \ddot{x}_0) = F(t). \tag{4.66}$$

A further useful fact is that if $x_1(t)$ and $x_2(t)$ are solutions to the inhomogeneous equations (4.65) with inhomogeneities $F_1(t)$ and $F_2(t)$ (for the same linear function E), then $x(t) = x_1(t) + x_2(t)$ is a solution to the inhomogeneous equation (4.65) with inhomogeneity $F(t) = F_1(t) + F_2(t)$:

$$E(x_1 + x_2, \dot{x}_1 + \dot{x}_2, \ddot{x}_1 + \ddot{x}_2) = E(x_1, \dot{x}_1, \ddot{x}_1) + E(x_2, \dot{x}_2, \ddot{x}_2) = F_1(t) + F_2(t). \tag{4.67}$$

Similarly, assuming that the linear function E has real coefficients, but $F(t)$ is complex, the real part of $x(t)$ for a solution of (4.65) is a solution to (4.65) with inhomogeneity $\operatorname{Re} F(t)$:

$$\operatorname{Re} F = \operatorname{Re} E(x, \dot{x}, \ddot{x}) = E(\operatorname{Re} x, \operatorname{Re} \dot{x}, \operatorname{Re} \ddot{x}). \tag{4.68}$$

The strategy to find the general solution of an inhomogeneous linear differential equation is thus to i) find one (particular) solution of the inhomogeneous, and ii) find the general solution of the associated homogeneous equation. The general solution of the inhomogeneous equation is then the sum of the two.

For the oscillator, we already found the general solution in the previous section. It thus remains to find one particular solution of the driven oscillator. We do this for the case of a periodic driving force $f(t) = \alpha \cos \omega t$ of real amplitude α and angular frequency ω. It is convenient to write this as the real part of $\alpha e^{-i\omega t}$ and then find a solution of the inhomogeneous equation for this complex driving force. A solution for the original equation is then obtained by taking the real part, as explained above.

A convenient ansatz for a driving force $f(t) = \alpha e^{-i\omega t}$ is a solution of the same periodicity, i.e., $x(t) = Ce^{-i\omega t}$ with C a complex coefficient. This ansatz indeed solves (4.64) for

$$C = \frac{\alpha}{\beta - \omega^2 - 2i\gamma\omega} = -\frac{\alpha}{(\omega - \omega_+)(\omega - \omega_-)}, \tag{4.69}$$

with $\omega_\pm$ as given by (4.50). It is convenient to write this complex number in terms of modulus and phase as

$$C = \frac{\alpha}{\sqrt{(\beta - \omega^2)^2 + 4\gamma^2\omega^2}} e^{i\phi} \tag{4.70}$$

with

$$\phi = \arccos \frac{\beta - \omega^2}{\sqrt{(\beta - \omega^2)^2 + 4\gamma^2\omega^2}}. \tag{4.71}$$

Here we used that (under the assumptions $\gamma, \omega \geq 0$) that the imaginary part of $\beta - \omega^2 + 2i\gamma\omega$ is positive, so that phase runs between 0 and π, a range on which the cosine is invertible. A real particular solution is thus obtained as

$$x(t) = \operatorname{Re}\left[Ce^{-i\omega t}\right] = \frac{\alpha}{\sqrt{(\beta - \omega^2)^2 + 4\gamma^2\omega^2}} \cos(\omega t - \phi). \tag{4.72}$$

The general solution is obtained by adding the general solution to the corresponding homogeneous problem, i.e., the oscillator as studied in the previous section. For $\beta > 0$ and $\gamma > 0$, the general solution of the oscillator exhibits exponential decay, so that at late times, any solution approaches the particular solution (4.72). Let us analyze this solution in detail, in particular

its dependence on the driving frequency ω. For large ω, the amplitude decays as ω^{-2}, and one easily checks that, for $\beta > 2\gamma^2$, it is maximal at[4]

$$\omega = \sqrt{\beta - 2\gamma^2}. \tag{4.73}$$

This is called *resonance*. For small damping, $\gamma^2 \ll \beta$, resonance takes place close to the oscillation frequency $\sqrt{\beta - \gamma^2}$. That resonance can lead to huge amplitudes can easily be seen by considering the limit $\gamma \to 0$ of no damping. Then the amplitude diverges as ω approaches the proper angular frequency $\sqrt{\beta}$ of the oscillator. In practice, resonance can have both desired and undesired consequences. An example for the latter are oscillations of footbridges excited by passengers marching in step.[5] An example for the former is a swing, where by applying a small but periodic force (close to the oscillation frequency) large oscillations can be generated.

Let us now examine the consequences of the phase ϕ and how it depends on the parameters. In the limit $\omega \to 0$ of vanishing frequency of the driving force, the phase ϕ vanishes. Then displacement $x(t)$ is in phase with the driving force $f(t)$, i.e., the driving force is maximal when the displacement x is maximal, and in the same direction as the displacement. If ϕ is increased, the motion lags behind the force, or, put differently, the maximum of the force precedes the maximum of the displacement: The force is maximal at $t = 0$, while the displacement reaches its maximum at $t = \frac{\phi}{\omega}$. Small positive values for ϕ are obtained for small values of the driving frequency ω. As ω is increased, the phase ϕ increases until it reaches $\phi = \frac{\pi}{2}$ at $\omega = \sqrt{\beta}$, the oscillation frequency in the absence of damping. In this case, the force is in phase with the velocity $\dot{x}(t)$, as $-\sin(\alpha - \frac{\pi}{2}) = -\cos\alpha$, i.e., the force is maximal when the velocity is maximal, and acts in the direction of the motion. Further increasing ω, the phase increases further, until reaching π in the limit $\omega \to \infty$. In this limiting case, the driving force is acting in the direction opposite to that of the displacement.

A special case occurs in the absence of damping ($\gamma = 0$), when the driving frequency ω coincides with the oscillation frequency of the oscillator ($\omega = \sqrt{\beta}$). Then C as given by (4.69) is ill-defined. A suitable ansatz is then

$$x(t) = Bte^{-i\omega t}. \tag{4.74}$$

Exercise 4.5. *Show that with an appropriate choice of B, the ansatz (4.74) solves the dynamical equation for the driven oscillator (with $\gamma = 0$ and $\omega = \sqrt{\beta}$). Determine the real part of the solution and determine the phase shift between $x(t)$ and $f(t) = \alpha\cos\omega t$ (ignore the factor of t in the amplitude of $x(t)$ for the determination of the phase).*

4.6 Taylor approximation

Taylor approximation is a technique to approximate, near some point x_0, a function $f(x)$ by a power series in the displacement $x - x_0$ around that point. The basic idea is the following: The function $f(x)$ and the constant function $f_0(x) := f(x_0)$ coincide at x_0. If we define

$$f_1(x) := f(x_0) + (x - x_0)f'(x_0), \tag{4.75}$$

[4] For $\beta < \gamma^2$, the maximum occurs at $\omega = 0$.

[5] After the collapse of a footbridge caused by marching soldiers in 1831, it is (at least in Germany) forbidden to march in step over a bridge. However, as the example of London's millennium bridge showed, the synchronization of passengers can also be caused by oscillations of the bridge. In such a case, additional damping mechanisms are applied in order to move the resonance frequency out of the range relevant for walking.

then not only f and f_1 coincide at $x = x_0$, but also their first derivatives. The reason is that the supplementary term vanishes at $x = x_0$ and so does not spoil the fact that $f(x)$ and $f_1(x)$ coincide at $x = x_0$. But the supplementary term has precisely the correct slope in order for the first derivative of $f_1(x)$ to coincide with $f(x)$ at $x = x_0$ as $f_1'(x) = f'(x_0)$. One calls $f_1(x)$ the *first order Taylor approximation* of $f(x)$ near $x = x_0$ (and likewise $f_0(x)$ the zeroth order Taylor approximation).

A polynomial in $(x - x_0)$ which coincides up to two derivatives with $f(x)$ is

$$f_2(x) := f(x_0) + (x - x_0)f'(x_0) + \frac{1}{2}(x - x_0)^2 f''(x_0). \tag{4.76}$$

The supplementary term does not affect the value of the function and its first derivative at $x = x_0$, but produces precisely the correct second order derivative at $x = x_0$: $f_2''(x) = f''(x_0)$. Hence, the zeroth, first, and second derivates of $f(x)$ and $f_2(x)$ coincide near $x = x_0$. If $f(x)$ is smooth, i.e., arbitrarily often differentiable, then this can be continued to arbitrarily high order: The function

$$f_n(x) := \sum_{k=0}^{n} \frac{1}{k!}(x - x_0)^k f^{(k)}(x_0) \tag{4.77}$$

coincides with $f(x)$ up to its nth derivative at $x = x_0$. Here $f^{(k)}$ denotes the kth derivative of f.

As an example, the Taylor approximation of $\sin x$ is shown in Figure 4.8. We see that around x_0 the function is better and better approximated for higher orders of the Taylor approximation. The mathematically precise statement is that for any smooth functions $f(x)$, the remainder term

$$R_n(x) := f(x) - f_n(x), \tag{4.78}$$

is such that the limit $x \to x_0$ of $(x - x_0)^{-(n+1)} R_n(x)$ exists. One usually writes this as

$$f(x) = f_n(x) + \mathcal{O}((x - x_0)^{n+1}), \tag{4.79}$$

where $\mathcal{O}$ stands for "order". This notation is very useful to control the corrections in a (Taylor) approximation. One can calculate with it as if $\mathcal{O}(x^n)$ would be Cx^n with an undetermined coefficient C. For example, we have

$$(1 + x + \mathcal{O}(x^2))^2 = 1 + 2x + \mathcal{O}(x^2), \tag{4.80}$$

as the first undetermined coefficient the expansion of the square is that of x^2 (through the product of 1 and $\mathcal{O}(x^2)$).

In some cases, the limit $n \to \infty$ of the Taylor approximation is well defined, giving rise to the Taylor series

$$\sum_{k=0}^{\infty} \frac{1}{k!}(x - x_0)^k f^{(k)}(x_0). \tag{4.81}$$

In particular, for the exponential function (and thus also the trigonometric functions), the Taylor series coincides with the usual power series representation around $x_0 = 0$:

$$\exp(x) = \sum_{k=0}^{\infty} \frac{1}{k!} x^k. \tag{4.82}$$

As an application, consider a dynamical law of the form

$$\ddot{x} = f(x) \tag{4.83}$$

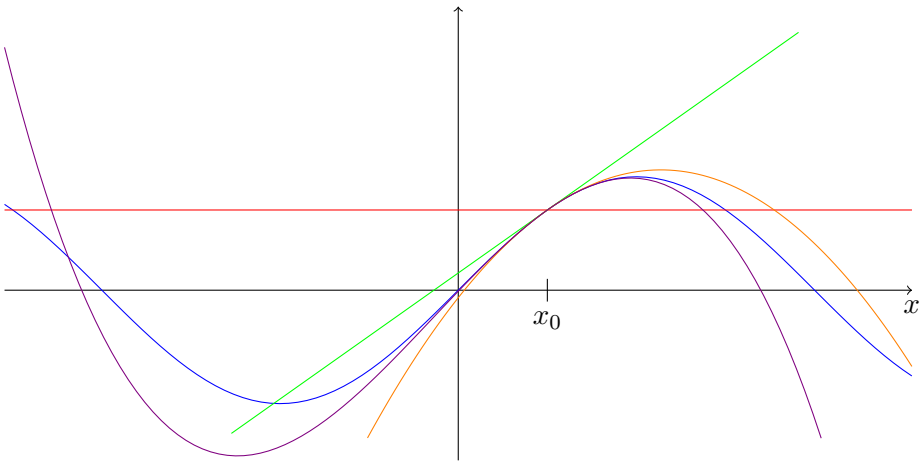


Figure 4.8: Taylor approximation of $f(x) = \sin(x)$ (in blue) around $x_0 = \frac{\pi}{4}$. Shown are the zeroth (red), first (green), second (orange), and third (violet) order Taylor approximations.

and assume that $f(x)$ vanishes at $x = x_0$. In particular $x(t) = x_0$ is a solution of the dynamical equation. We call this an equilibrium position. We want to study small perturbations of this equilibrium. With a first order Taylor approximation of the force, have

$$\ddot{x} = f'(x_0)(x - x_0) + \mathcal{O}((x - x_0)^2), \tag{4.84}$$

or, expressed in terms of $y = x - x_0$, and with $\beta = -f'(x_0)$,

$$\ddot{y} = -\beta y + \mathcal{O}(y^2). \tag{4.85}$$

For small deviations, i.e., disregarding the $\mathcal{O}(y^2)$ corrections, this is the dynamical equation for the oscillator with vanishing damping. For $\beta > 0$, we thus find oscillatory motion around the equilibrium. Such an equilibrium is called *stable* against small perturbations, as the perturbed solution remains close to the equilibrium. On the other hand, for $\beta < 0$, we found, for generic initial data, exponential growth. Hence, such an equilibrium is *unstable*, as small perturbations do in general not remain small. Phrased in terms of the potential $V(x)$ defined in Section 4.2 (by the requirement that $V'(x) = -f(x)$), the stability condition $f'(x_0) < 0$ translates to $V''(x_0) > 0$, i.e., the equilibrium is stable if it is a local minimum of the potential. Conversely, if the equilibrium is a local maximum of the potential, then it is unstable.

In the case of the pendulum discussed in Section 4.3, we dealt with the force $f(x) = -\frac{g}{L}\sin x$, which vanishes at $x = n\pi$ for integer n. Its first derivative is $f'(x) = -\frac{g}{L}\cos x$, which is negative for even n and positive for odd n. Hence, by the above analysis, the equilibria at $x = n\pi$ for even n are stable, while those for odd n are unstable. Furthermore, in the former case, we identify $\beta = \frac{g}{L}$, so that small oscillations around a stable equilibrium have angular frequency $\omega = \sqrt{g/L}$, in agreement with the result (4.40) for the period of oscillations in the limit of small displacements.

As a further application, we can compute corrections to the period of the pendulum, for which we obtained the expression (4.39). Reinserting $k = \frac{g}{L}$ and introducing $y := \sin^2 \frac{\theta_{\text{tp}}}{2}$, we can consider the period as a function

$$T(y) = 4\sqrt{\frac{L}{g}} \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1 - y\sin^2 x}} \mathrm{d}x \tag{4.86}$$

of y (and thus the maximal displacement angle θ_{tp}). We already evaluated it at $\theta_{\text{tp}} = 0$, i.e., $y = 0$, c.f. (4.40). We consider the first order correction to this expression. For this, we need evaluate the first derivative

$$T'(y) = 4\sqrt{\frac{L}{g}} \int_0^{\frac{\pi}{2}} \frac{1}{2} \sin^2 x \frac{1}{(1 - y \sin^2 x)^{\frac{3}{2}}} \mathrm{d}x \tag{4.87}$$

at $y = 0$, obtaining

$$\begin{aligned} T'(0) &= 2\sqrt{\frac{L}{g}} \int_0^{\frac{\pi}{2}} \sin^2 x \mathrm{d}x \\ &= 2\sqrt{\frac{L}{g}} \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 - \cos 2x) \mathrm{d}x \\ &= \sqrt{\frac{L}{g}} \frac{\pi}{2}. \end{aligned} \tag{4.88}$$

As the first order approximation, we thus get

$$T = 2\pi\sqrt{\frac{L}{g}} \left(1 + \frac{1}{4} y\right) + \mathcal{O}(y^2) = 2\pi\sqrt{\frac{L}{g}} \left(1 + \frac{1}{16} \theta_{\text{tp}}^2\right) + \mathcal{O}(\theta_{\text{tp}}^4), \tag{4.89}$$

where in the last step we used that $\sin x = x + \mathcal{O}(x^3)$, so $\sin^2 x = x^2 + \mathcal{O}(x^4)$.

We can also use Taylor expansion to discuss the approach to the unstable equilibrium at $\theta = \pm\pi$, i.e., for energy $E = k$. In that case, the integrand in (4.37) becomes $\frac{1}{\sqrt{2k}} \frac{1}{\sqrt{1+\cos\theta'}}$. We are interested in the regime of θ' approaching π. For the Taylor expansion of $1 + \cos\theta'$ near π, we obtain

$$1 + \cos\theta' = \frac{1}{2}(\theta' - \pi)^2 + \mathcal{O}((\theta' - \pi)^3). \tag{4.90}$$

Hence, the integrand is $k^{-\frac{1}{2}}(\pi - \theta')^{-1}(1 + \mathcal{O}(\pi - \theta'))$, so that as the upper end θ of the integration approaches π, we get

$$t(\theta) = -\frac{1}{\sqrt{k}} \ln(\pi - \theta) + \mathcal{O}(1). \tag{4.91}$$

Hence, by inverting the logarithm, we see that the equilibrium position is approached with an exponential fall-off at a rate $\sqrt{k}$, i.e.,

$$\theta(t) \sim \pi - Ce^{-\sqrt{k}t} \tag{4.92}$$

for large t and some constant C.

5 Coupled oscillations

In the previous chapter, we studied motion in one dimension, and saw that near equilibria the dynamical equation can be approximated by the oscillator equation, by which one can distinguish stable and unstable equilibria. In general, motion takes place in more than one dimension and more than one mass point is involved. Hence, in general a mechanical system is to be described not by a single function $u(t)$, but by several ones, $u_i(t)$, with i ranging from 1 to n.[1] To describe the three-dimensional motion of p point particles, we would need $n = 3p$. But we can (and will) also study the one-dimensional motion of two point particles coupled by a spring, in which case $n = 2$. We collect all the coordinates u_i in a vector $\mathbf{u} \in \mathbb{R}^n$. Then general form of the dynamical equation is

$$\ddot{u}_i(t) = f_i(\mathbf{u}(t), \dot{\mathbf{u}}(t), t). \tag{5.1}$$

Under reasonable conditions on the functions f_i, there is, for initial data $\mathbf{u}(t_0)$, $\dot{\mathbf{u}}(t_0)$ at some time t_0, a unique solution to (5.1). In this chapter, we will be concerned with f_i which do not depend explicitly on time t. Furthermore, we either restrict to f_i which are linear in $\mathbf{u}$, $\dot{\mathbf{u}}$ or to the consideration of perturbations of equilibria $\mathbf{u}_0$ (characterized by $f_i(\mathbf{u}_0, 0) = 0$). Upon linearization (see below) the second case then reduces to the first case.

For our analysis, we need some mathematical tools, in particular analysis of functions of several variables and linear algebra, which we will first introduce.

5.1 Interlude: Analysis of functions of several variables

We frequently consider functions $f(x_1, \ldots, x_n)$ of several variables. For a function $f(x)$ of a single variable, its first and second order derivative give important information on extrema and also enter the Taylor approximation. What are the generalizations of these concepts for a function of several variables? A crucial part of the answer is the concept of a *partial derivative*, defined by

$$\frac{\partial}{\partial x_i} f(x_1, \ldots, x_n) := \lim_{\varepsilon \to 0} \frac{f(x_1, \ldots, x_i + \varepsilon, \ldots, x_n) - f(x_1, \ldots, x_i, \ldots, x_n)}{\varepsilon}. \tag{5.2}$$

Here the ith coordinate is varied while the others are held fixed. In geometric terms, this gives the slope of the function at $(x_1, \ldots, x_n)$ as one moves in the ith coordinate direction.

The rules for differentiation of functions $f(x)$ of a single variable have straightforward generalizations to partial derivatives. As the usual derivative, the partial derivative is linear, i.e.,

$$\frac{\partial}{\partial x_i} (\lambda_1 f_1 + \lambda_2 f_2) = \lambda_1 \frac{\partial}{\partial x_i} f_1 + \lambda_2 \frac{\partial}{\partial x_i} f_2. \tag{5.3}$$

The product (Leibniz) rule generalizes to

$$\frac{\partial}{\partial x_i} (fg) = \left(\frac{\partial}{\partial x_i} f \right) g + f \left(\frac{\partial}{\partial x_i} g \right). \tag{5.4}$$

[1] As in our discussion of one-dimensional motion in Chapter 4, we do not assume that the u_i are particle's coordinates in a Euclidean coordinate system.

To formulate the generalization of the chain rule, consider $x_1, \ldots, x_n$ as functions of another variable, say t. Then

$$\frac{\mathrm{d}}{\mathrm{d}t} f(x_1(t), \ldots, x_n(t)) = \dot{x}_1(t) \frac{\partial}{\partial x_1} f(x_1(t), \ldots, x_n(t)) + \cdots + \dot{x}_n(t) \frac{\partial}{\partial x_n} f(x_1(t), \ldots, x_n(t)), \quad (5.5)$$

where the dot denotes derivative w.r.t. t. In particular when the x_i are coordinates in a Euclidean coordinate system, it is convenient to write the sum on the r.h.s. as the scalar product of $\dot{\mathbf{x}}(t)$ and the *gradient*

$$\boldsymbol{\nabla} f = \begin{pmatrix} \frac{\partial}{\partial x_1} f \\ \vdots \\ \frac{\partial}{\partial x_n} f \end{pmatrix} \quad (5.6)$$

of the function f. The symbol $\boldsymbol{\nabla}$ used there is called "nabla". We can thus write

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{x}(t)) = \dot{\mathbf{x}}(t) \cdot \boldsymbol{\nabla} f(\mathbf{x}(t)). \quad (5.7)$$

This is also called "differentiation along a curve", i.e., given a curve $\mathbf{x}(t)$ on $\mathbb{R}^n$ and a function $f(\mathbf{x})$ on $\mathbb{R}^n$, we can define a function $f(\mathbf{x}(t))$ of a single variable t. Then the above differentiation w.r.t. t gives the slope of $f(\mathbf{x})$ along the curve $\mathbf{x}(t)$.

For a function $f(\mathbf{r})$ on $\mathbb{R}^3$, we can write the gradient in terms of the Euclidean basis as

$$\boldsymbol{\nabla} f = \hat{\mathbf{x}} \partial_x f + \hat{\mathbf{y}} \partial_y f + \hat{\mathbf{z}} \partial_z f, \quad (5.8)$$

where we also use the convenient abbreviated notation $\partial_x = \frac{\partial}{\partial x}$.

Exercise 5.1. *Verify the gradients (*$\mathbf{a}$ *is a constant vector)*

$$\boldsymbol{\nabla}(\mathbf{a} \cdot \mathbf{r}) = \mathbf{a}, \quad (5.9)$$

$$\boldsymbol{\nabla}(\mathbf{r} \cdot \mathbf{r}) = 2\mathbf{r}, \quad (5.10)$$

$$\boldsymbol{\nabla}|\mathbf{r}| = \frac{\mathbf{r}}{|\mathbf{r}|} = \hat{r}. \quad (5.11)$$

By the chain rule

$$\boldsymbol{\nabla} f(g(\mathbf{x})) = \boldsymbol{\nabla} g(\mathbf{x}) f'(g(\mathbf{x})), \quad (5.12)$$

and (5.11), one sees that the gradient of a function depending only on the distance $r = |\mathbf{r}|$ to the origin is given by

$$\boldsymbol{\nabla} f(r) = \hat{\mathbf{r}} f'(r). \quad (5.13)$$

Conventionally, if $f(\mathbf{r}, t)$ is a function of the position $\mathbf{r}$ and time t, then the gradient collects only the partial derivatives w.r.t. the spatial coordinates. For a curve $\mathbf{r}(t)$, we then have

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{r}(t), t) = \dot{\mathbf{r}} \cdot \boldsymbol{\nabla} f + \frac{\partial}{\partial t} f. \quad (5.14)$$

The derivative on the l.h.s. is then called a *total (time) derivative*, in distinction to the partial time derivative on the right hand side. The two terms on the r.h.s. can be understood as stemming from i) the change in f due to the change in the position $\mathbf{r}$ with time and ii) the change in f due to the change in time (with position fixed).

A generalization of the above is a "change of coordinates" situation where x_i are functions of another set of coordinates $y_1, \ldots, y_n$. In that case, we have

$$\frac{\partial}{\partial y_i} f(\mathbf{x}(\mathbf{y})) = \frac{\partial x_1(\mathbf{y})}{\partial y_i} \frac{\partial}{\partial x_1} f(\mathbf{x}(\mathbf{y})) + \cdots + \frac{\partial x_n(\mathbf{y})}{\partial y_i} \frac{\partial}{\partial x_n} f(\mathbf{x}(\mathbf{y})). \tag{5.15}$$

It is then customary to collect all the partial derivatives $\frac{\partial x_j}{\partial y_i}$ into the *Jacobian matrix*

$$J_x = \begin{pmatrix} \frac{\partial x_1}{\partial y_1} & \cdots & \frac{\partial x_1}{\partial y_n} \\ \vdots & & \vdots \\ \frac{\partial x_n}{\partial y_1} & \cdots & \frac{\partial x_n}{\partial y_n} \end{pmatrix}. \tag{5.16}$$

We can then write the above in the form

$$\frac{\partial}{\partial y_i} f = \sum_{j=1}^{n} (J_x)_{ji} \frac{\partial}{\partial x_j} f, \tag{5.17}$$

or, even more compact, in a the form of a matrix multiplication as $\boldsymbol{\nabla}_y f = J_x^T \boldsymbol{\nabla}_x f$, where $\boldsymbol{\nabla}_y$ and $\boldsymbol{\nabla}_x$ denote the gradients w.r.t. the variables x, y and J_x^T denotes the transpose of J_x. For a change of coordinates $x(y)$ in one dimension, we know that it is invertible near y_0 when $x'(y_0) \neq 0$. This derivative also occurs when passing from integration over x to integration over y. Analogously, for functions of n variables, the change of variables is invertible near $\mathbf{y}_0$ when the *Jacobian*, the determinant of the Jacobian matrix, does not vanish. The Jacobian also occurs when passing from integration over $\mathbf{x}$ to integration over $\mathbf{y}$.

We apply the above to coordinate systems which are often very useful:

Polar coordinates (ρ, ϕ) on $\mathbb{R}^2$, with $\rho \geq 0$ and $\phi \in [0, 2\pi)$ are defined via

$$\mathbf{r} = \begin{pmatrix} x \\ y \end{pmatrix} = \rho \begin{pmatrix} \cos\phi \\ \sin\phi \end{pmatrix}, \tag{5.18}$$

i.e., ρ is the distance to the origin and ϕ the angle to the x-axis (in counterclockwise direction). The corresponding Jacobian matrix is

$$J = \begin{pmatrix} \frac{\partial x}{\partial \rho} & \frac{\partial x}{\partial \phi} \\ \frac{\partial y}{\partial \rho} & \frac{\partial y}{\partial \phi} \end{pmatrix} = \begin{pmatrix} \cos\phi & -\rho\sin\phi \\ \sin\phi & \rho\cos\phi \end{pmatrix}, \tag{5.19}$$

so the Jacobian is ρ. This implies that the change of coordinates is invertible for $\rho > 0$. That $\rho = 0$ should be excluded is a consequence of the fact that the angle ϕ is not well-defined at $\rho = 0$. Furthermore, the integration measure on the plane in polar coordinates is $\rho \mathrm{d}\rho \mathrm{d}\phi$.

Parameterizing a curve $\mathbf{r}(t)$ by $(\rho(t), \phi(t))$, we obtain the velocity

$$\dot{\mathbf{r}} = \dot{\rho}\hat{\boldsymbol{\rho}} + \dot{\phi}\rho\hat{\boldsymbol{\phi}}, \tag{5.20}$$

where

$$\hat{\boldsymbol{\rho}} = \begin{pmatrix} \cos\phi \\ \sin\phi \end{pmatrix}, \qquad \hat{\boldsymbol{\phi}} = \begin{pmatrix} -\sin\phi \\ \cos\phi \end{pmatrix}, \tag{5.21}$$

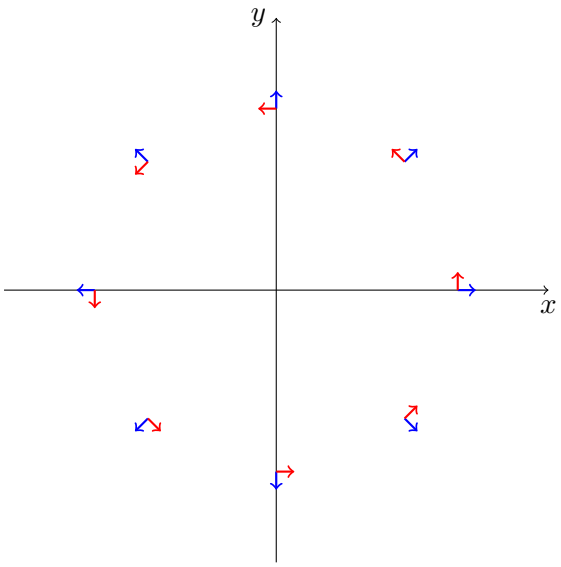


Figure 5.1: Illustration of the basis vectors $\hat{\rho}$ (in blue) and $\hat{\phi}$ (in red) associated to polar coordinates.

see also (3.30) for the corresponding vectors in three dimensions. These vector fields are sketched in Fig. 5.1. We see that $\hat{\boldsymbol{\rho}}$ always points radially outwards, and $\hat{\boldsymbol{\phi}}$ points to counterclockwise direction. For each ϕ, these two vectors form an ONB. In particular, we obtain for the square of the velocity

$$|\dot{\mathbf{r}}|^2 = \dot{\rho}^2 + \rho^2\dot{\phi}^2. \tag{5.22}$$

Comparing the derivative along a curve of the function $f(\rho(t), \phi(t))$, i.e.,

$$\frac{\mathrm{d}}{\mathrm{d}t}f = \dot{\rho}\frac{\partial}{\partial\rho}f + \dot{\phi}\frac{\partial}{\partial\phi}f, \tag{5.23}$$

with $\dot{\mathbf{r}} \cdot \boldsymbol{\nabla} f$, we obtain the gradient in polar coordinates as[2]

$$\boldsymbol{\nabla} f = \hat{\boldsymbol{\rho}}\frac{\partial}{\partial\rho}f + \hat{\boldsymbol{\phi}}\frac{1}{\rho}\frac{\partial}{\partial\phi}f. \tag{5.24}$$

Cylindrical coordinates (ρ, ϕ, z) on $\mathbb{R}^3$, with $\rho \geq 0$ and $\phi \in [0, 2\pi)$ are defined via

$$\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \rho\cos\phi \\ \rho\sin\phi \\ z \end{pmatrix}, \tag{5.25}$$

i.e., ρ is the distance to the z-axis and ϕ the angle to the x-axis of the projection of $\mathbf{r}$ to the $x-y$-plane. The corresponding Jacobian matrix is

$$J = \begin{pmatrix} \cos\phi & -\rho\sin\phi & 0 \\ \sin\phi & \rho\cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{5.26}$$

[2]For this, we may expand $\boldsymbol{\nabla} f$ w.r.t. the basis $\{\hat{\boldsymbol{\rho}}, \hat{\boldsymbol{\phi}}\}$ as $\boldsymbol{\nabla} f = (\boldsymbol{\nabla} f)_\rho \hat{\boldsymbol{\rho}} + (\boldsymbol{\nabla} f)_\phi \hat{\boldsymbol{\phi}}$, compute the scalar product with $\dot{\mathbf{r}}$ as given by (5.20), and then compare the coefficients of $\dot{\rho}$ and $\dot{\phi}$ (as $\frac{\mathrm{d}}{\mathrm{d}t}f(\mathbf{r}(t)) = \dot{\mathbf{r}} \cdot \boldsymbol{\nabla} f$ must hold for all curves, it must in particular hold for curves where either ϕ or ρ are held fixed, i.e., $\dot{\phi} = 0$ or $\dot{\rho} = 0$).

so the Jacobian is ρ. This implies that the change of coordinates is invertible for $\rho > 0$, i.e., away from the z-axis where ϕ is ill-defined. The integration measure in cylindrical coordinates is thus $\rho \mathrm{d}\rho\mathrm{d}\phi\mathrm{d}z$.

Parameterizing a curve $\mathbf{r}(t)$ by $(\rho(t), \phi(t), z(t))$, we obtain the velocity

$$\dot{\mathbf{r}} = \dot{\rho}\hat{\boldsymbol{\rho}} + \dot{\phi}\rho\hat{\boldsymbol{\phi}} + \dot{z}\hat{\mathbf{z}}, \tag{5.27}$$

with the ONB $\{\hat{\boldsymbol{\rho}}, \hat{\boldsymbol{\phi}}, \hat{\mathbf{z}}\}$ as given in (3.30). Now $\hat{\boldsymbol{\rho}}$ always points away from the z-axis, $\hat{\boldsymbol{\phi}}$ points to counterclockwise direction around the z axis and $\hat{\mathbf{z}}$ points in z-direction. The square of the velocity of the curve is

$$|\dot{\mathbf{r}}|^2 = \dot{\rho}^2 + \rho^2\dot{\phi}^2 + \dot{z}^2. \tag{5.28}$$

Comparing the derivative along a curve of the function $f(\rho(t), \phi(t), z(t))$, i.e.,

$$\frac{\mathrm{d}}{\mathrm{d}t}f = \dot{\rho}\frac{\partial}{\partial\rho}f + \dot{\phi}\frac{\partial}{\partial\phi}f + \dot{z}\frac{\partial}{\partial z}f \tag{5.29}$$

with $\dot{\mathbf{r}} \cdot \boldsymbol{\nabla} f$, we obtain the gradient in cylindrical coordinates as

$$\boldsymbol{\nabla} f = \hat{\boldsymbol{\rho}}\frac{\partial}{\partial\rho}f + \hat{\boldsymbol{\phi}}\frac{1}{\rho}\frac{\partial}{\partial\phi}f + \hat{\mathbf{z}}\frac{\partial}{\partial z}f. \tag{5.30}$$

Spherical coordinates (r, θ, ϕ) on $\mathbb{R}^3$, with $r \geq 0$, $\theta \in [0, \pi]$, and $\phi \in [0, 2\pi)$ are defined via

$$\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = r \begin{pmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{pmatrix}, \tag{5.31}$$

i.e., r is the distance to the origin, θ the angle between $\mathbf{r}$ and the positive z-axis, and ϕ the angle to the x-axis of the projection of $\mathbf{r}$ to the $x-y$-plane, c.f. Fig. 5.2. The corresponding Jacobian is $r^2\sin\theta$, which vanishes on the z-axis (where ϕ is ill-defined). The integration measure in spherical coordinates is thus $r^2\sin\theta\mathrm{d}r\mathrm{d}\theta\mathrm{d}\phi$.

Parameterizing a curve $\mathbf{r}(t)$ by $(r(t), \theta(t), \phi(t))$, we obtain the velocity

$$\dot{\mathbf{r}} = \dot{r}\hat{\mathbf{r}} + \dot{\theta}r\hat{\boldsymbol{\theta}} + \dot{\phi}r\sin\theta\hat{\boldsymbol{\phi}}, \tag{5.32}$$

with the ONB $\{\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}\}$ as given in (3.29). Now $\hat{\mathbf{r}}$ points away from the origin, $\hat{\boldsymbol{\theta}}$ points along the sphere from the north to the south pole, and $\hat{\boldsymbol{\phi}}$ points to counterclockwise direction around the z axis. The square of the velocity of the curve is

$$|\dot{\mathbf{r}}|^2 = \dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2. \tag{5.33}$$

In complete analogy to the case of cylindrical coordinates, we obtain

$$\boldsymbol{\nabla} f = \hat{\mathbf{r}}\frac{\partial}{\partial r}f + \hat{\boldsymbol{\theta}}\frac{1}{r}\frac{\partial}{\partial\theta}f + \hat{\boldsymbol{\phi}}\frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}f \tag{5.34}$$

for the gradient in spherical coordinates.

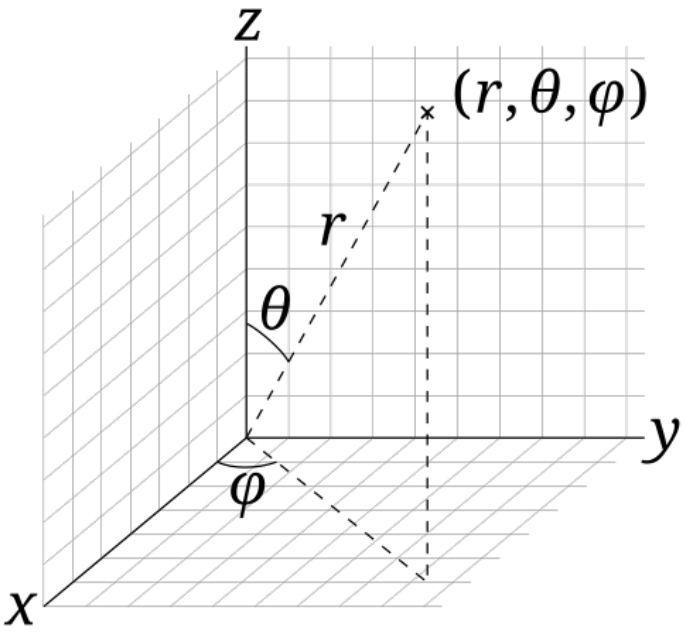


Figure 5.2: Illustration of spherical coordinates. Source: Andeggs via Wikipedia

We now turn back to the general discussion of partial derivatives. Partial derivatives can be composed, i.e., after differentiating w.r.t. x_i, one can further differentiate w.r.t. a possibly different coordinate x_j. A function such that all Nth derivatives exist and are continuous is said to be N times continuously differentiable, or C^N. An important rule regarding composition of derivatives is *Schwarz's theorem*

$$\frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j}f = \frac{\partial}{\partial x_j}\frac{\partial}{\partial x_i}f \tag{5.35}$$

for any C^2 function f, i.e., the order in which the partial derivatives are taken does not matter.

We saw that Taylor approximation is a useful tool for the approximation of a function $f(x)$ near y in terms of a polynomial in $x - y$, using the derivatives of f at y. This can straightforwardly be generalized to functions of several variables. As for us at most approximation to second order will be relevant, we restrict to this order:

$$f(\mathbf{x}) = f(\mathbf{y}) + \sum_{i=1}^{n}(x_i - y_i)\frac{\partial}{\partial x_i}f(\mathbf{y}) + \frac{1}{2}\sum_{i,j=1}^{n}(x_i - y_i)(x_j - y_j)\frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j}f(\mathbf{y}) + \mathcal{O}((\mathbf{x}-\mathbf{y})^3). \tag{5.36}$$

While the second term on the r.h.s. can be expressed in term of the gradient of f, the third term can be expressed in term of the *Hessian matrix*

$$H_f = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n} \end{pmatrix}. \tag{5.37}$$

Namely, we can write the above as

$$f(\mathbf{x}) = f(\mathbf{y}) + (\mathbf{x} - \mathbf{y}) \cdot \boldsymbol{\nabla} f(\mathbf{y}) + \frac{1}{2}(\mathbf{x} - \mathbf{y}) \cdot H_f(\mathbf{y})(\mathbf{x} - \mathbf{y}) + \mathcal{O}((\mathbf{x} - \mathbf{y})^3), \tag{5.38}$$

where in the third term the matrix $H_f(\mathbf{y})$ acts on the vector $\mathbf{x}-\mathbf{y}$ (yielding another vector, which is then "scalar multiplied" with $\mathbf{x}-\mathbf{y}$). For later purposes, it is important to notice that, due Schwarz's theorem (5.35), the Hessian is a real symmetric matrix.

For a function $f(x)$ of a single variable, an extremum at y is indicated by the vanishing of the derivative at y, i.e., $f'(y)=0$, meaning that the slope of f vanishes at y. In terms of the Taylor approximation, it means that the linear term in

$$f(x)=f(y)+(x-y)f'(y)+\frac{1}{2}(x-y)^2f''(y)+\mathcal{O}((x-y)^3) \tag{5.39}$$

vanishes. Analogously, for a local extremum at $\mathbf{y}$ of a function $f(\mathbf{x})$ of several variables, one requires that the first order Taylor coefficient vanishes, i.e., the gradient $\boldsymbol{\nabla} f(\mathbf{y})$ vanishes. Equivalently, for any curve $\mathbf{x}(t)$ through $\mathbf{y}$, the slope at $\mathbf{y}$ vanishes, i.e., if t_0 is such that $\mathbf{x}(t_0)=\mathbf{y}$, then

$$\frac{\mathrm{d}}{\mathrm{d}t}f(\mathbf{x}(t))|_{t=t_0}=0. \tag{5.40}$$

Here the symbol $|_{t=t_0}$ stands for the evaluation of the preceding function at $t=t_0$.[3]

As for the character of a local extremum of a function $f(x)$ of a single variable at y, it is determined by the sign of $f''(y)$. In terms of the second order Taylor approximation (5.39), this determines whether second order term gives a positive or negative contribution, irrespective of the sign of $(x-y)$. For a function of several variable, the second order term of the Taylor expansion is more complicated, c.f. (5.38). Using the abbreviations $\mathbf{z}=\mathbf{x}-\mathbf{y}$ and $H=H_f(\mathbf{y})$, we can write it as $\frac{1}{2}\mathbf{z}\cdot H\mathbf{z}$. Now the Hessian H is a real symmetric matrix, so it can be diagonalized as $H=O^TDO$, with $D=\mathrm{diag}(h_1,\dots,h_n)$ a diagonal matrix with the eigenvalues h_i of H on the diagonal, and O an orthogonal matrix, i.e., $OO^T=\mathbb{1}$. Writing $\mathbf{z}'=O\mathbf{z}$, the quadratic term in the Taylor expansion can thus be written as

$$\frac{1}{2}\mathbf{z}'\cdot D\mathbf{z}'=\frac{1}{2}\left(h_1z_1'^2+\cdots+h_nz_n'^2\right). \tag{5.41}$$

Hence, if all the eigenvalues h_i of H are positive, then this is positive definite, i.e., positive expect for $\mathbf{z}'=0$. As O is invertible, $\mathbf{z}=O^T\mathbf{z}'$, this is the case iff $\mathbf{z}=0$. Hence, the quadratic term of the Taylor approximation is positive for all $\mathbf{z}=\mathbf{x}-\mathbf{y}\neq\mathbf{0}$ iff all eigenvalues of $H=H_f(\mathbf{y})$ are positive. This is thus the condition for a local minimum of f at $\mathbf{y}$. If all eigenvalues of $H_f(\mathbf{y})$ are negative, then we are dealing with a local maximum. When $H_f(\mathbf{y})$ has both negative and positive eigenvalues, then one speaks of a *saddle point* at $\mathbf{y}$, c.f. Fig. 5.3 for sketches of a local minimum and a saddle point.

To conclude this section, we discuss the geometric interpretation of the gradient, when $\mathbf{x}$ are Cartesian coordinates on $\mathbb{R}^n$. Then the gradient $\boldsymbol{\nabla} f(\mathbf{x})$ points in the direction of the highest slope (up the hill), and its length

$$|\boldsymbol{\nabla} f(\mathbf{x})|=\sqrt{\left(\frac{\partial}{\partial x_1}f(\mathbf{x})\right)^2+\cdots+\left(\frac{\partial}{\partial x_n}f(\mathbf{x})\right)^2} \tag{5.42}$$

is the value of the slope. Recalling that $\dot{\mathbf{x}}(t)$ is the tangent vector to the curve at $\mathbf{x}(t)$, we see from (5.7) that the derivative along a curve is given by the scalar product of the tangent vector with the gradient.

[3]The notation $\frac{\mathrm{d}}{\mathrm{d}t}f(\mathbf{x}(t_0))$ is avoided here, because one can read it as differentiating a function w.r.t. t which does not depend on t, thus giving zero for any function f or curve $\mathbf{x}(t)$.

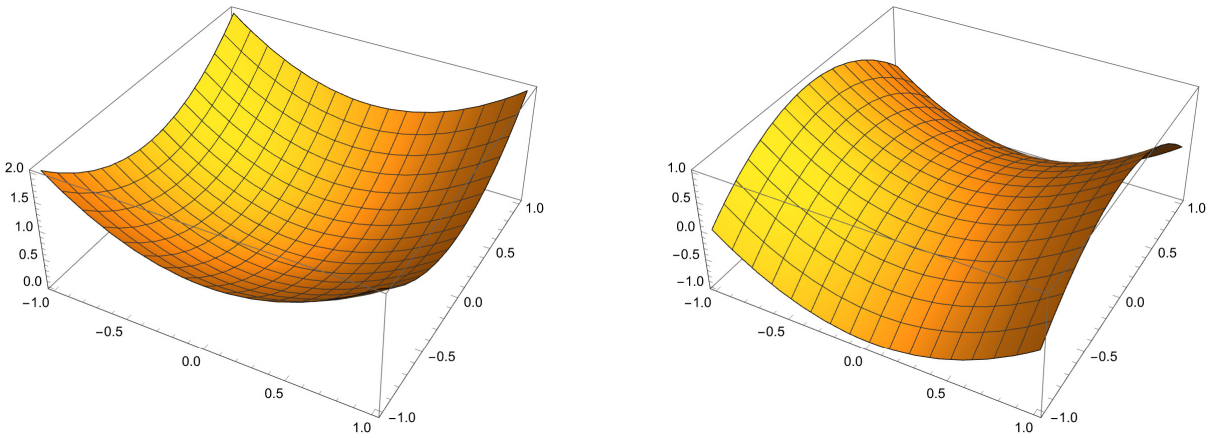


Figure 5.3: Sketch of a local minimum and a saddle point of a function on $\mathbb{R}^2$.

For a continuous function $f(\mathbf{x})$ on $\mathbb{R}^n$, the *level sets* are defined as the sets of points of $\mathbb{R}^n$ where the function takes a specific value:

$$L_{f,y} := \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid f(x_1, \ldots, x_n) = y\}\,. \tag{5.43}$$

For example, one may define the n-dimensional unit sphere as the level set $L_{f,1}$ for the function

$$f(x_1, \ldots, x_{n+1}) = x_1^2 + \cdots + x_{n+1}^2. \tag{5.44}$$

For a function on $\mathbb{R}^2$, the level sets are in general lines (but may also be points or open subsets), which are also called *isolines*. Examples are "height lines" in topographic maps. For a function of $\mathbb{R}^3$, the level sets are in general surfaces, also called *isosurface*.

For a curve $\mathbf{x}(t)$ contained in a level set, i.e., $\mathbf{x}(t) \in L_{f,y}$ for $t \in (t_0, t_1)$ (for example, the curve could be an isoline in $\mathbb{R}^2$), we have

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{x}(t)) = \lim_{\varepsilon \to 0} \frac{f(\mathbf{x}(t+\varepsilon)) - f(\mathbf{x}(t))}{\varepsilon} = \lim_{\varepsilon \to 0} \frac{y - y}{\varepsilon} = 0. \tag{5.45}$$

Recalling (5.7), we see that the gradient is orthogonal to each curve within the level set, i.e., it is orthogonal to the level set.

5.2 Linearization around equilibria

We turn back to the general dynamical equation (5.1), but assume an absence of explicit time dependence, i.e.,

$$\ddot{u}_i(t) = f_i(\mathbf{u}(t), \dot{\mathbf{u}}(t)). \tag{5.46}$$

We further assume that the stationary solution $\mathbf{u}(t) = \mathbf{u}_0$ solves (5.46), i.e., $f_i(\mathbf{u}_0, \mathbf{0}) = 0$ for all i. We now consider small perturbations $\mathbf{w}(t)$ around $\mathbf{u}_0$, i.e.,

$$\mathbf{u}(t) = \mathbf{u}_0 + \mathbf{w}(t). \tag{5.47}$$

Expanding both sides of (5.46) to first order in $\mathbf{w}$ (on the r.h.s. using Taylor expansion), we get

$$\ddot{w}_i(t) = \sum_j \left(w_j \frac{\partial}{\partial u_j} f_i(\mathbf{u}_0, \mathbf{0}) + \dot{w}_j \frac{\partial}{\partial \dot{u}_j} f_i(\mathbf{u}_0, \mathbf{0}) \right) + \mathcal{O}((\mathbf{w})^2). \tag{5.48}$$

When the higher order terms can be neglected (i.e., when the perturbation $\mathbf{w}$ is sufficiently small), then this is a linear system of differential equations, which we can write in the form

$$\ddot{\mathbf{w}} = -B\mathbf{w} - 2C\dot{\mathbf{w}}, \tag{5.49}$$

where the matrices B and C are defined as

$$B_{ij} = -\frac{\partial}{\partial u_j} f_i(\mathbf{u}_0, \mathbf{0}), \qquad C_{ij} = -\frac{1}{2}\frac{\partial}{\partial \dot{u}_j} f_i(\mathbf{u}_0, \mathbf{0}). \tag{5.50}$$

We have written this in a form reminiscent of the notation used for the one-dimensional oscillator, i.e., B reduces to β and C to γ in that case.

Analogously to the exponential ansatz used for the one-dimensional oscillator, a suitable ansatz for a system of linear differential equations such as (5.49) is

$$\mathbf{w}(t) = \mathbf{w}e^{-i\omega t}, \tag{5.51}$$

where one has to solve for the vector $\mathbf{w}$ and the frequency ω simultaneously. Plugging this ansatz into (5.49), one finds

$$\left(B - 2i\omega C - \omega^2 \mathbb{1}\right)\mathbf{w} = 0. \tag{5.52}$$

Hence, we are looking for frequencies ω such that the matrix $B - 2i\omega C - \omega^2\mathbb{1}$ has an eigenvalue zero (the corresponding eigenvector is then $\mathbf{w}$). A matrix has a vanishing eigenvalue iff its determinant vanishes, i.e., we are looking for solutions ω to the equation

$$\det\left(B - 2i\omega C - \omega^2 \mathbb{1}\right) = 0. \tag{5.53}$$

The l.h.s. is a polynomial in ω of order $2n$, so there are $2n$ complex roots ω. We label these as ω_i and the corresponding eigenvectors as $\mathbf{w}_i$. A solution of the form

$$\mathbf{w}(t) = \mathbf{w}_i e^{-i\omega_i t} \tag{5.54}$$

is called a *normal mode* of the system. The general solution to (5.49) is then the linear combination

$$\mathbf{w}(t) = c_1\mathbf{w}_1 e^{-i\omega_1 t} + \cdots + c_n\mathbf{w}_n e^{-i\omega_n t} \tag{5.55}$$

of normal modes. When a root ω of (5.53) is k-fold degenerate, but there are not k linearly independent corresponding eigenvectors $\mathbf{w}$, then the set of solutions is not complete and variation of constants is needed to complement the set of solutions, analogously to the one-dimensional critically damped oscillator.

From the frequencies ω solving (5.53) one can determine the stability properties of the system. If one (or more) of the frequencies has a positive imaginary part, then the equilibrium $\mathbf{u}_0$ is unstable, as a generic small perturbation grows exponentially, and thus does not remain small (leaving the domain of validity of the approximation leading to (5.49)). If such a mode is present, one calls $\operatorname{Im}\omega$ the *Lyapunov exponent* of the instability (in the case of several unstable modes, the maximal imaginary part determines the Lyapunov exponent).

A relevant special case of the above occurs when $C = 0$, i.e., when the acceleration does not depend on the velocity. Then it is convenient to rewrite (5.52) in the form

$$B\mathbf{w} = \omega^2\mathbf{w}, \tag{5.56}$$

so the possible frequencies ω are the square roots (both signs) of the eigenvalues of B and the corresponding $\delta\mathbf{u}$ are the corresponding eigenvectors.

Example 5.1. *We take $n = 2$ and consider the force law*

$$\mathbf{f}(\mathbf{u}, \dot{\mathbf{u}}) = \begin{pmatrix} u_2 \\ -u_1 \end{pmatrix} \tag{5.57}$$

so that the acceleration is independent of the velocity. This vector field is sketched in Fig. 5.4. We can write the equation of motion in the form

$$\ddot{\mathbf{u}} + B\mathbf{u} = \mathbf{0}, \tag{5.58}$$

with

$$B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{5.59}$$

The characteristic polynomial of B is

$$\det(B - \lambda \mathbb{1}) = \lambda^2 + 1, \tag{5.60}$$

so the eigenvalues of B are $\pm i$. The corresponding eigenvectors are easily found to be

$$\mathbf{u}_\pm = \begin{pmatrix} 1 \\ \mp i \end{pmatrix}. \tag{5.61}$$

The frequencies ω (the square roots of these eigenvalues) are

$$\omega_{1/2} = \pm \frac{1}{\sqrt{2}}(1 + i), \qquad \omega_{3/4} = \pm \frac{1}{\sqrt{2}}(1 - i). \tag{5.62}$$

We thus have two frequencies (ω_1 and ω_4) with positive imaginary part, i.e., an instability with Lyapunov exponent $\frac{1}{\sqrt{2}}$. It may be instructive to consider a corresponding solution. The linear combination

$$\mathbf{u}(t) = \mathbf{u}_+ e^{-i\omega_1 t} + \mathbf{u}_- e^{-i\omega_4 t} = \begin{pmatrix} e^{-i\frac{t}{\sqrt{2}}} + e^{i\frac{t}{\sqrt{2}}} \\ -ie^{-i\frac{t}{\sqrt{2}}} + ie^{i\frac{t}{\sqrt{2}}} \end{pmatrix} e^{\frac{t}{\sqrt{2}}} = 2e^{\frac{t}{\sqrt{2}}} \begin{pmatrix} \cos\frac{t}{\sqrt{2}} \\ -\sin\frac{t}{\sqrt{2}} \end{pmatrix} \tag{5.63}$$

is real and also shown in Fig. 5.4. We see that the solution is a rotation in the $u_1 - u_2$ plane with an exponentially growing radius. Note that the vector field plotted in Fig. 5.4 is the force field $\mathbf{f}$, not the flow field $\mathbf{F}$ that we used in the analysis of motion in one dimension in Section 4.2.[4] *In particular, the trajectory need not be tangent to the vector field.*

5.3 The coupled oscillator

We consider the setup sketched in Fig. 5.5, i.e., two masses m_1 and m_2 attached to fixed walls and two each other via springs of spring constants k_1, k_2, k_3. We parameterize the position of the two masses by their deviations u_1, u_2 from the equilibrium position. Equations of motion for the two masses are then

$$m_1 \ddot{u}_1 = -k_1 u_1 + k_3(u_2 - u_1), \tag{5.64}$$

$$m_2 \ddot{u}_2 = -k_2 u_2 + k_3(u_1 - u_2). \tag{5.65}$$

[4] The latter would be a vector field on $\mathbb{R}^4$ (representing two positions u_1, u_2 and two velocities $\dot{u}_1$, $\dot{u}_2$), which makes it challenging to plot.

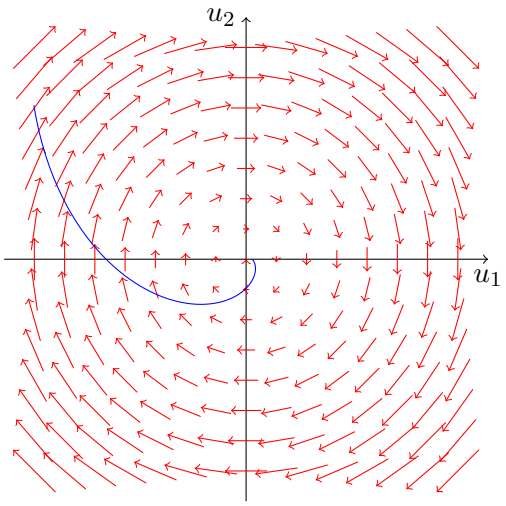


Figure 5.4: The force vector field $\mathbf{f}(\mathbf{u})$ (in red) given by (5.57). The trajectory (5.63) is shown in blue.

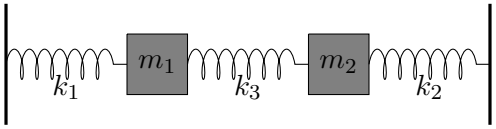


Figure 5.5: The coupled oscillator.

This can be seen as follows: The accelerations vanish for $u_1 = u_2 = 0$, i.e., in the equilibrium position. The first term on the r.h.s. of the first equation is the usual spring force due to the first spring. Likewise for the first term on the r.h.s. of the second equation. As for terms proportional to $u_2 - u_1$: This quantity is the deviation of the length of the middle spring out of its length in equilibrium. If $u_2 - u_1 > 0$, the spring is extended w.r.t. its equilibrium length, giving rise to a force in positive direction on the first and in negative direction on the second mass.

For simplicity, we now assume $m_1 = m_2 = m$ and $k_1 = k_2 = k$. The spring constants of the middle spring will be called $k_3 = k'$. Then the above system can be written as

$$\ddot{\mathbf{u}} = -B\mathbf{u} \tag{5.66}$$

with

$$B = \frac{1}{m}\begin{pmatrix} k+k' & -k' \\ -k' & k+k' \end{pmatrix} = \begin{pmatrix} \beta+\beta' & -\beta' \\ -\beta' & \beta+\beta' \end{pmatrix}, \tag{5.67}$$

where $\beta := \frac{k}{m}$, $\beta' := \frac{k'}{m}$. According to the above general analysis, we need to determine the eigenvalues of B, as their square roots determine the frequencies of the system. The characteristic polynomial of B is

$$\det(B - \lambda\mathbb{1}) = (\beta + \beta' - \lambda)^2 - \beta'^2 = \lambda^2 - 2(\beta+\beta')\lambda + \beta^2 + 2\beta\beta'. \tag{5.68}$$

The roots of this polynomial are

$$\lambda_\pm = \beta + \beta' \mp \sqrt{(\beta+\beta')^2 - \beta^2 - 2\beta\beta'} = \beta + \beta' \mp \beta'. \tag{5.69}$$

And the corresponding eigenvectors are

$$\mathbf{u}_+ = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \mathbf{u}_- = \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{5.70}$$

The general real solution of the coupled oscillator is thus

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = A_+ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cos(\omega_+ t - \phi_+) + A_- \begin{pmatrix} 1 \\ -1 \end{pmatrix} \cos(\omega_- t - \phi_-), \tag{5.71}$$

where

$$\omega_+ = \sqrt{\lambda_+} = \sqrt{\gamma}, \qquad \omega_- = \sqrt{\lambda_-} = \sqrt{\gamma + 2\gamma'}. \tag{5.72}$$

The four parameters $A_\pm$, $\phi_\pm$ specify a particular solution, i.e., they have to be determined by the initial conditions for the four quantities u_1, u_2, $\dot{u}_1$, $\dot{u}_2$. The two vectors $\mathbf{u}_+$, $\mathbf{u}_-$ (together with the corresponding frequencies ω_+, ω_-) are called the *normal modes* of the coupled oscillator.

It is instructive to consider the two possible modes in more detail. In the "+"-mode, the two masses move in parallel, as the displacements u_1 and u_2 coincide at all times. Hence, in this mode, the spring between them is irrelevant, as its extension is fixed. The frequency $\omega_+ = \sqrt{k/m}$ coincides with the expectation for such a mode: As the two masses move in parallel, the mass is doubled to $2m$. The two springs of spring constant k act like one spring of spring constant $2k$, but the supplementary factor 2 in both the mass and the spring constant cancel in the frequency.

In the "−" mode, the masses are moving in opposite phase, i.e., when the first mass is to the left of its equilibrium position, then the second mass is to the right of its equilibrium position, and vice versa. As for the frequency, the result can also be subjected to consistency checks. For $k' = 0$ (corresponding to absence of the middle spring), the two masses oscillate individually on a spring of spring constant k, so the frequency must be $\sqrt{k/m}$. For $k = 0$ (corresponding to absence of the outer spring), we can reduce the situation to a single mass attached to a spring of half the original length. Such a spring has a spring constant of twice of that of the original spring.[5] Hence, the frequency for the case $k = 0$ should be $\sqrt{2k/m}$, consistent with our result.

A periodic driving force can be implemented by considering the dynamical equation

$$\ddot{\mathbf{u}} + B\mathbf{u} = \mathbf{v}e^{-i\omega t}. \tag{5.73}$$

Here ω is the angular frequency of the driving force, and $\mathbf{v}$ specifies how the driving force acts on the two masses. For example, if $\mathbf{v} = (\alpha, 0)$, then the force acts with amplitude α only on the first mass. For $\mathbf{v} = \alpha(1, 1)$, both masses are affected by the same force.[6] With the ansatz $\mathbf{u}(t) = \mathbf{u}e^{-i\omega t}$, the dynamical equation (5.73) is solved for

$$(B - \omega^2 \mathbb{1})\mathbf{u} = \mathbf{v}. \tag{5.74}$$

When $B - \omega^2 \mathbb{1}$ is invertible, i.e., when the driving frequency differs from the oscillation frequencies $\omega_\pm$, this can be solved by

$$\mathbf{u} = (B - \omega^2 \mathbb{1})^{-1}\mathbf{v}. \tag{5.75}$$

In case the driving is at one of the oscillation frequencies, a more general ansatz is required, analogously to driven single oscillator, c.f. Exercise 4.5.

[5] The force a spring exerts stems from the tension due its deformation. In order to deform the original and the halved spring in the same manner (same distance between neighboring coils), the halved spring needs to be compressed by a distance only half as large as for the original spring. As the spring constant is the proportionality between force and distance, the spring constant is doubled upon halving the spring.

[6] By introducing a complex phase between the two components v_1 and v_2, one can also model situations where the forces on the two masses are shifted by a phase.

6 Newtonian mechanics

6.1 Newton's axioms

In order to formulate Newton's axioms, we first need to introduce certain concepts. In this we go beyond the framework actually employed by Newton, in order to prepare for the discussion of Galilean relativity later on (and Special and General Relativity in later courses). The most basic concept that we will use is that of an *event* which means a particular spatial location at a particular time. Note that nothing actually needs to happen at an event, but something possibly could happen (for example, a particle might be at the specific position at the specific time).

The next important concept is that of *coordinate systems*, by which observers label each event unambiguously by four real numbers such as (t, x, y, z). The first coordinate t indicates the time of the event and plays a distinguished role, as will become clear from the following further requirements on coordinate systems:

- All observers agree about which events are simultaneous. This means that if observers $\mathcal{O}$ and $\mathcal{O}'$ attribute coordinates $(t_{A/B}, x_{A/B}, y_{A/B}, z_{A/B})$ and $(t'_{A/B}, x'_{A/B}, y'_{A/B}, z'_{A/B})$ to events A and B, then if $t_A = t_B$ also $t'_A = t'_B$.

- All observers agree about the length of the time interval between two events, i.e., in the notation used above, $t_B - t_A = t'_B - t'_A$ for all events A, B and all admissible coordinate systems.

- Observers can measure the *distance* $d_{A,B}$ between two simultaneous events, for example using rods. All observers agree on the distance of two simultaneous events. Furthermore, there are distinguished *Cartesian coordinate systems* (t, x, y, z) for which

$$d_{A,B} = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2 + (z_A - z_B)^2}. \tag{6.1}$$

Before continuing, some comments on these requirements are in order:

- The absolute notion of simultaneity is fundamental to Newtonian Mechanics. It took Einstein's careful analysis of the operational basis of simultaneity (and his willingness to give up cherished but operationally baseless concepts) to dispose of this absolute notion of simultaneity, leading to Special Relativity.

- Given the absolute notion of simultaneity, the second requirement simply states that observers can agree on a common scale (or unit) for time.

- As we will see, even in Newtonian Mechanics, the distance between two events A, B is observer dependent if we drop the requirement that A and B are simultaneous.

- The assumption that there is a Cartesian coordinate system, i.e., a system in which the distance of any two points is given by (6.1), is very restrictive. It is easy to come up with

counterexamples: For example, there is no Cartesian coordinate system on the sphere (not even locally), i.e., a system of coordinates (x, y) such that the distance of any points A, B (defined as the length of the shortest curve along the sphere connecting the two) is given by $d^2_{A,B} = (x_A - x_B)^2 + (y_A - y_B)^2$. This is due to the sphere being curved. In the language of Differential Geometry, the assumption of the existence of a Cartesian coordinate system is equivalent to the assumption that space is flat.[1] This assumption is given up in General Relativity, but as long as the length scales considered are small w.r.t. the radius of curvature, the assumption of flatness is justified.

A further important concept is that of a *point particle*, which is an object of negligible spatial extent and no (or irrelevant) inner structure, so that it can be completely described by assigning a position $(x(t), y(t), z(t))$ to each time t in some coordinate system (which in particular means that a point particle can not be simultaneously at different locations). The map $t \mapsto (x(t), y(t), z(t))$ is usually called the *trajectory* of the particle. Often, one will treat extended material objects, such as a planet, as a point particle, but whether such an idealization is justified will typically depend on the situation considered. Even if an extended body can be considered as non-deformable (rigid), it may spin, which may have huge impact on its behaviour (think of a table tennis ball hitting the table).

We are now ready to formulate *Newton's first axiom*: There is a Cartesian coordinate system (t, x, y, z), called *inertial frame*, such that, in the absence of external influences, the movement of point particles is *uniform*, in the sense that

$$\mathbf{r}(t) := \begin{pmatrix} x \\ y \\ z \end{pmatrix}(t) = \mathbf{r}_0 + t\mathbf{v}. \tag{6.2}$$

Here we used the standard notation of indicating a vector in $\mathbb{R}^3$ in bold face letters.

In the original formulation of the first axiom, it is stated that free (i.e., in the absence of external influences) motion is uniform. However, this is only true in an inertial frame and the only meaningful way to define such a frame is to refer to the uniformity of free motion. In order break this circularity, we formulated the first axiom as the postulate that (at least one) inertial frame exists.

The observational basis of the first axiom is in fact quite shaky: All material objects are subject to gravity, so it is practically impossible to study particles in the absence of external influences, at least when gravity is considered as an external influence (as is done in Newtonian Mechanics). And indeed when gravity is dealt with appropriately, i.e., within General Relativity, one redefines inertial motion as free fall, i.e., one does not treat gravity as an external influence, but as intrinsic to spacetime. One then finds that inertial frames exist only approximately at time and length scales which are small compared to spacetime curvature. Nevertheless, the assumption that inertial frames (in the sense of Newtonian Mechanics) exist proved to be extremely successful (ultimately due to the small spacetime curvature in the vicinity of Earth).

Given an inertial frame, one calls $\mathbf{r}(t)$ the *position*, $\mathbf{v}(t) := \dot{\mathbf{r}}(t) := \frac{\mathrm{d}}{\mathrm{d}t}\mathbf{r}(t)$ the *velocity* and $\mathbf{a}(t) := \ddot{\mathbf{r}}(t) := \frac{\mathrm{d}^2}{\mathrm{d}t^2}\mathbf{r}(t)$ the *acceleration* of the particle at time t. By the first axiom, the acceleration $\mathbf{a}$ vanishes for free motion in an inertial frame. *Newton's second axiom* now states that the acceleration is proportional to the *force* $\mathbf{F}$ acting on the particle, with proportionality

[1] Apparently Gauß, one of the pioneers of Differential Geometry, was considering to use the triangulations that he was performing for the kingdom of Hanover (for cartographical reasons) to test the flatness of the spatial geometry, but concluded that the measurement errors were too large.

factor the (inverse of the) *inertial mass* m of the particle,

$$\mathbf{a} = \frac{1}{m}\mathbf{F}. \tag{6.3}$$

Some comments on this axiom are in order:

- The inertial mass is a positive number characterizing the "resistance" of the particle to changing its velocity, i.e., its inertia. For the time being, we distinguish it from the *gravitational mass* of the particle, which characterizes how strong it is affected by gravity (and how strong it gravitationally affects other particles). Galileo's finding that the acceleration of bodies in the gravitational field (of Earth) is universal (independent of their properties) means that inertial and gravitational mass are proportional to each other, so that (with an appropriate choice of scale) the two can be identified, so that one speaks only of *mass*.

- When writing the second axiom in the form (6.3), it is assumed that the mass m is constant. As we will see, when this is not the case (for a rocket, for example), it has to be generalized as

  $$\frac{\mathrm{d}}{\mathrm{d}t}(m\mathbf{v}) = \mathbf{F}. \tag{6.4}$$

  The quantity $\mathbf{p} := m\mathbf{v}$ occurring here is called the *momentum*.

- The second axiom might seem somewhat void as long as one does not specify how to determine the inertial mass m or the the force $\mathbf{F}$. One might be tempted to read (6.3) as a mere definition of m and/or $\mathbf{F}$. However, there is more to it. Assume that we know that the forces acting on two particles are equal (or opposite). This may be the case when it is clear that the influence exerted on a body does not depend on the characteristics of the body (such as for a body attached to a spring), or, by the third law to be discussed below, when the mutual force between the particles is the only force acting on them. In that case, the second law predicts that i) the direction of the two accelerations is the same (or opposite), ii) the ratio $\frac{|\mathbf{a}_1|}{|\mathbf{a}_2|}$ between the modulus of the accelerations is fixed, irrespective of the magnitude of the force. In particular, one then has a means to measure the inertial mass m_2 as a multiple of the inertial mass m_1 by $m_2 = \frac{|\mathbf{a}_1|}{|\mathbf{a}_2|}m_1$. One calls this an *operational* definition of the inertial mass (up to the choice of the scale provided by the "reference mass" m_1).

- An assumption that is typically implicitly made about (inertial) mass is that it is *additive*, i.e., when a body is composed of two smaller bodies of masses m_1 and m_2, then the mass of the full body is $m = m_1 + m_2$. This assumption is important when an extended body is partitioned into smaller bodies that are then treated as point particles. The assumption is justified as long as binding energies can be neglected, which ceases to be the case at the nuclear level (the mass of two deuterium nuclei is greater than that of a helium nucleus).

- An implicit assumption that is typically made on the force is that it is additive, i.e., when a body is subject to two separate influences, each exerting a force $\mathbf{F}_1$, $\mathbf{F}_2$, then, when both influences are present, the force on the body is given by $\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2$. Having determined the forces $\mathbf{F}_1$, $\mathbf{F}_2$ individually then yields a prediction for the trajectory under their combined influence. Interatomic (or intermolecular) forces are typically not additive.

- Taking into account that the force may in general depend on the position of the particle (as is the case for spring), the velocity of the particle (as is the case for movement in a

medium), and explicitly on the time t (for example when a force is turned on and off), we may write (6.3) as

$$\ddot{\mathbf{r}}(t) = \frac{1}{m}\mathbf{F}(\mathbf{r}(t), \dot{\mathbf{r}}(t), t). \tag{6.5}$$

This is an (ordinary) differential equation, involving derivatives w.r.t. time. This is typical for equations describing a (continuous) time evolution.

Finally, *Newton's third axiom* states that when body 1 exerts a force $\mathbf{F}_{2,1}$ on body 2, then body 2 exerts a force

$$\mathbf{F}_{1,2} = -\mathbf{F}_{2,1} \tag{6.6}$$

on body 1, which is equal in magnitude, but opposite in direction ("actio = reactio").

An underlying implicit assumption is that all forces arise as (sum of) forces between bodies. This neglects the role of (electromagnetic) fields as intermediary. When a beam of electromagnetic radiation is emitted, there is a force on the charges producing the radiation, but there is no body on which the opposite force acts. When the beam of radiation is absorbed at a target, there is a force on the charges contained in the target. But due to the finite propagation speed of electromagnetic radiation, the forces on emitter and receiver act at different times (and will in general have different magnitude). In order to restore the balance of forces (or rather conservation of momentum) at each time, one has to understand the electromagnetic field as an independent entity to which a momentum (density) can be assigned.

6.2 Galilei transformations

Newton's first axiom asserts that there is at least one inertial frame, i.e., a Cartesian coordinate system $(t, \mathbf{r})$ such that free motion is uniform. However, we will see that it is not unique. The possible transformations from one inertial frame to a different one will be called *Galilei transformations*. Let us nail down the possible coordinate transformations $(t'(t,\mathbf{r}), \mathbf{r}'(t,\mathbf{r}))$ which are compatible with our axioms. Here $(t'(t,\mathbf{r}), \mathbf{r}'(t,\mathbf{r}))$ denote the coordinates in the new inertial frame of the event that was labelled by coordinates $(t, \mathbf{r})$ in the original inertial frame.

In terms of the coordinates of an inertial frame, the set of events "swiped over" by the free motion of a point particle is a straight line in "space-time" $\mathbb{R}^4$, i.e., in the language of linear algebra a one-dimensional affine subspace. By a transformation to a different inertial frame, all such one-dimensional affine subspaces must be mapped to one-dimensional affine subspaces, and the set of such maps is the set of affine linear maps, i.e., of the form

$$\begin{pmatrix} t' \\ \mathbf{r}' \end{pmatrix} = \begin{pmatrix} a & \mathbf{w}^T \\ -\mathbf{v}_0 & R \end{pmatrix} \begin{pmatrix} t \\ \mathbf{r} \end{pmatrix} - \begin{pmatrix} t_0 \\ \mathbf{r}_0 \end{pmatrix} \tag{6.7}$$

$$= \begin{pmatrix} at + \mathbf{w}\cdot\mathbf{r} - t_0 \\ -\mathbf{v}_0 t + R\mathbf{r} - \mathbf{r}_0 \end{pmatrix}. \tag{6.8}$$

Here we decomposed the 4×4 matrix into a number a, two vectors $\mathbf{v}_0$ and $\mathbf{w}$, and a 3×3 matrix R. The signs have been chosen for convenience and the superscript T stands for the transpose. However, not all such transformations are compatible with the other axioms. Consider two events A, B, described in the original inertial frame by coordinates $(t_A, \mathbf{r}_A)$, $(t_B, \mathbf{r}_B)$, so that the duration between the two is $\Delta t = t_A - t_B$. For the above transformation, the duration between the two events in the "primed" inertial frame, we find

$$\Delta t' = at_A + \mathbf{w}\cdot\mathbf{r}_A - t_0 - (at_B + \mathbf{w}\cdot\mathbf{r}_B - t_0) = a\Delta t + \mathbf{w}\cdot(\mathbf{r}_A - \mathbf{r}_B). \tag{6.9}$$

As this should coincide with Δt for all events A, B, we must have $a = 1$ and $\mathbf{w} = 0$.

Now consider the distance of two simultaneous events (described in the original frame by $(t, \mathbf{r}_A)$ and $(t, \mathbf{r}_B)$) as measured in the primed inertial frame. We first compute

$$\mathbf{r}'_A - \mathbf{r}'_B = -\mathbf{v}_0 t + R\mathbf{r}_A - \mathbf{r}_0 - (-\mathbf{v}_0 t + R\mathbf{r}_B - \mathbf{r}_0) = R(\mathbf{r}_A - \mathbf{r}_B). \tag{6.10}$$

It is noteworthy that a further term $\mathbf{v}_0(t_B - t_A)$ would be present if we had not restricted attention to simultaneous events. As we require the distance between the two events, as measured by the modulus of $\mathbf{r}_A - \mathbf{r}_B$ and $\mathbf{r}'_A - \mathbf{r}'_B$, should coincide in the two inertial frames, it follows that the 3×3 matrix R should be such that it preserves the length of any vector $\mathbf{r}$ it acts on. The set of such matrices is the set $O(3)$ of *orthogonal matrices*, characterized by

$$R^T R = \mathbb{1}. \tag{6.11}$$

An (trivial) example for such a matrix is $R = \mathbb{1}$, but we will discuss more interesting examples (and their interpretation) below. But before, we want to prove the above claim

$$|R\mathbf{v}| = |\mathbf{v}| \ \forall \mathbf{v} \in \mathbb{R}^3 \iff R \in O(3) \tag{6.12}$$

that the set of matrices R preserving lengths coincides with the set of orthogonal matrices. The implication from right to left is straightforward:

$$|R\mathbf{v}|^2 = \langle R\mathbf{v}, R\mathbf{v}\rangle = \langle \mathbf{v}, R^T R\mathbf{v}\rangle = |\mathbf{v}|^2. \tag{6.13}$$

To prove the converse direction, we first notice the identity (c.f. (3.23))

$$|\mathbf{v} + \mathbf{w}|^2 - |\mathbf{v} - \mathbf{w}|^2 = 4\langle \mathbf{v}, \mathbf{w}\rangle, \tag{6.14}$$

which allows to express the scalar product (containing information about the angle) in terms of lengths only. To prove the implication from left to right, we now assume that R preserves lengths to obtain

$$4\langle \mathbf{v}, \mathbf{w}\rangle = |\mathbf{v} + \mathbf{w}|^2 - |\mathbf{v} - \mathbf{w}|^2 = |R(\mathbf{v} + \mathbf{w})|^2 - |R(\mathbf{v} - \mathbf{w})|^2 = 4\langle R\mathbf{v}, R\mathbf{w}\rangle = 4\langle \mathbf{v}, R^T R\mathbf{w}\rangle. \tag{6.15}$$

This is possible for all $\mathbf{v}$, $\mathbf{w}$ only if $R^T R = \mathbb{1}$. A side result is that orthogonal matrices not only preserve the lengths of vectors, but also their scalar product, and thus their angle.

By the above analysis, the most general coordinate transformation preserving the desired structure (inertial, Cartesian, absolute simultaneity), is parameterized by $(R, \mathbf{v}_0, t_0, \mathbf{r}_0)$ (with R an orthogonal matrix, $\mathbf{v}_0$ and $\mathbf{r}_0$ vector of $\mathbb{R}^3$, and t_0 a number) and transforms the coordinates $(t, \mathbf{r})$ of an event to coordinates

$$t' = t - t_0, \qquad \mathbf{r}' = R\mathbf{r} - \mathbf{v}_0 t - \mathbf{r}_0. \tag{6.16}$$

The trivial transformation, which leaves the coordinates invariant, is given by $R = \mathbb{1}$ and all other parameters vanishing.

Let us interpret the above parameters of the transformation, starting with t_0. Obviously, it corresponds to a shift of the time coordinate, also called a *time translation*. What may be less obvious is the direction of the shift. To investigate this, consider the origin of time in the two frames. The origin of time in the primed frame, $t' = 0$, corresponds to $t = t_0$ in the original frame. Hence, if $t_0 > 0$, we have shifted the origin of time to the future when passing from the

original to the primed system. Such transformations reflect the possibility to choose an origin of time. In order to simplify the discussion, we will in the following assume $t_0 = 0$, so that $t' = t$.

The parameter $\mathbf{r}_0$ can be interpreted likewise. Setting $R = \mathbb{1}$ and $\mathbf{v}_0 = 0$, we see that the coordinates are only shifted. One calls this a *(spatial) translation*. By an analysis analogous to that for t_0, we see that in going to the primed system, we have shifted the origin of the spatial coordinates by $\mathbf{r}_0$. Such transformations may be useful when a certain point is distinguished, as the case of a central potential. It is then convenient to choose coordinates such that the distinguished point is a the origin.

To interpret R, we now consider $\mathbf{r}_0 = 0$ and $\mathbf{v}_0 = 0$, but $R \neq \mathbb{1}$. We first note that a consequence of (6.11) is $R^{-1} = R^T$, so in particular (6.11) is equivalent to

$$RR^T = \mathbb{1}, \tag{6.17}$$

which implies that also $R^T = R^{-1}$ is an orthogonal matrix. Furthermore, the relations $\det(AB) = \det(A)\det(B)$ and $\det(A) = \det(A^T)$ imply that $\det(R) = \pm 1$. The subset of $O(3)$ with $\det(R) = 1$ is the set $SO(3)$ of *special orthogonal matrices*. An example for an orthogonal matrix R with $\det(R) = \pm 1$ is $R = \pm\mathbb{1}$. While $R = \mathbb{1}$ is a trivial transformation, $R = -\mathbb{1}$ corresponds to a reflection of the coordinate system at the origin, which in particular maps a right-handed to a left handed system. Any orthogonal matrix R with $\det(R) = -1$ can be written as $R = -R'$ with $R' \in SO(3)$.

To understand the effect of the action of R, we are looking for $\{\hat{\mathbf{x}}', \hat{\mathbf{y}}', \hat{\mathbf{z}}'\}$ defined by the requirement

$$x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}} = x'\hat{\mathbf{x}}' + y'\hat{\mathbf{y}}' + z'\hat{\mathbf{z}}', \tag{6.18}$$

implying that (x', y', z') are the coefficients w.r.t. the basis $\{\hat{\mathbf{x}}', \hat{\mathbf{y}}', \hat{\mathbf{z}}'\}$ of the point described by coordinates (x, y, z) in the original frame. In order to find $\hat{\mathbf{x}}'$, we simply choose $x' = 1$ and $y' = z' = 0$ and evaluate the l.h.s. of (6.18) (and analogously for $\hat{\mathbf{y}}'$ and $\hat{\mathbf{z}}'$). The corresponding (x, y, z) can be found by inverting the transformation: $\mathbf{r} = R^{-1}\mathbf{r}' = R^T\mathbf{r}'$. Hence,

$$\hat{\mathbf{x}}' = R^T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \hat{\mathbf{y}}' = R^T \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \hat{\mathbf{z}}' = R^T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \tag{6.19}$$

As we saw above, also R^T is an orthogonal matrix, and as such matrices preserve the scalar product, we see that $\{\hat{\mathbf{x}}', \hat{\mathbf{y}}', \hat{\mathbf{z}}'\}$ is in fact an ONB. It is right/left-handed for $\det R = \pm 1$, as the triple product $\hat{\mathbf{x}}' \cdot (\hat{\mathbf{y}}' \times \hat{\mathbf{z}}')$ coincides with the determinant of R^T. Hence, for $\det R = 1$ (i.e., $R \in SO(3)$), R describes a *rotation* of the coordinate frame, while for $\det R = -1$ it describes a rotation composed with a *reflection*. A prototypical example is

$$R = \begin{pmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{6.20}$$

which performs a rotation of the coordinate system by an angle α around the z-axis.

Finally, let us consider the effect of $\mathbf{v}_0 \neq 0$. We set $R = \mathbb{1}$ and $\mathbf{r}_0 = 0$. Now consider the trajectory of the spatial origin in the primed frame, i.e., $\mathbf{r}'(t') = \mathbf{0}$. In the original system, this corresponds to

$$\mathbf{r}(t) = R^{-1}\left(\mathbf{r}'(t') + \mathbf{v}_0 t + \mathbf{r}_0\right) = \mathbf{v}_0 t. \tag{6.21}$$

Hence, we see that the spatial origin of the primed frame moves with constant velocity $\mathbf{v}_0$ w.r.t. the origin of the original frame. A transformation to a frame which is moving with uniform velocity w.r.t. the original frame is called a *boost* (sometimes this term is reserved for the corresponding transformation in Special Relativity).

We have employed here the *passive* interpretation of Galilei transformations, in which $(t', \mathbf{r}')$ and $(t, \mathbf{r})$ are the coordinates of the same event, but in different inertial frames. One can also employ an *active* interpretation, in which the transformation "moves around" events in spacetime, i.e., $(t', \mathbf{r}')$ are interpreted as coordinates (w.r.t. the same frame) of the image under the transformation of the event described by coordinates $(t, \mathbf{r})$.

6.3 Galilean Relativity

Given a particle trajectory described by $\mathbf{r}(t)$ in the original frame, the same trajectory is described by

$$\mathbf{r}'(t') = R\mathbf{r}(t) - \mathbf{v}_0 t - \mathbf{r}_0 = R\mathbf{r}(t') - \mathbf{v}_0 t' - \mathbf{r}_0. \tag{6.22}$$

The corresponding velocity and acceleration is given by

$$\mathbf{v}'(t') := \frac{\mathrm{d}}{\mathrm{d}t'}\mathbf{r}'(t') = R\mathbf{v}(t') - \mathbf{v}_0, \tag{6.23}$$

$$\mathbf{a}'(t') := \frac{\mathrm{d}^2}{\mathrm{d}t'^2}\mathbf{r}'(t') = R\mathbf{a}(t'). \tag{6.24}$$

Using (6.3), we see that also in the primed system the second axiom holds, but with force, expressed in the new coordinate system, given by

$$\mathbf{F}' = R\mathbf{F}, \tag{6.25}$$

i.e., obtained from $\mathbf{F}$ by acting with the rotation matrix also transforming the coordinates of the position. In particular, if the frames are not rotated, $R = \mathbb{1}$, then the two forces coincide.

Now assume that we are given a force law $\mathbf{F}_{1,2} = \mathbf{F}(\mathbf{r}_1, \dot{\mathbf{r}}_1, \mathbf{r}_2, \dot{\mathbf{r}}_2)$ (such as a gravitational force) determining the force exerted on particle 1 by particle 2 as a function of their positions and velocities. The (Galilean) *relativity principle* requires that such a force law should have the same form in any two inertial frames $(t, \mathbf{r})$, $(t', \mathbf{r}')$, i.e., the forces in the unprimed and the primed system are given by

$$\mathbf{F}_{1,2} = \mathbf{F}(\mathbf{r}_1, \dot{\mathbf{r}}_1, \mathbf{r}_2, \dot{\mathbf{r}}_2), \qquad \mathbf{F}'_{1,2} = \mathbf{F}(\mathbf{r}'_1, \dot{\mathbf{r}}'_1, \mathbf{r}'_2, \dot{\mathbf{r}}'_2). \tag{6.26}$$

One may phrase this as the requirement that the laws of physics must be the same in each inertial frame. As according to (6.25) the two forces are related by a rotation, we must then have

$$\mathbf{F}(\mathbf{r}'_1, \dot{\mathbf{r}}'_1, \mathbf{r}'_2, \dot{\mathbf{r}}'_2) = R\mathbf{F}(\mathbf{r}_1, \dot{\mathbf{r}}_1, \mathbf{r}_2, \dot{\mathbf{r}}_2), \tag{6.27}$$

where on the r.h.s. $\mathbf{r}_i$ and $\dot{\mathbf{r}}_i$ should be seen as being determined by $\mathbf{r}'_i$ and $\dot{\mathbf{r}}'_i$ (by inverting (6.22) and (6.23)).

The meaning of the requirement (6.27) might be best grasped by considering examples. For the gravitational force law

$$\mathbf{F}(\mathbf{r}_1, \mathbf{r}_2) = -Gm_1m_2\frac{\mathbf{r}_1 - \mathbf{r}_2}{|\mathbf{r}_1 - \mathbf{r}_2|^3} \tag{6.28}$$

one easily verifies the requirement:

$$\begin{aligned}\mathbf{F}(\mathbf{r}_1', \mathbf{r}_2') &= \mathbf{F}(R\mathbf{r}_1 - \mathbf{v}_0 t - \mathbf{r}_0, R\mathbf{r}_2 - \mathbf{v}_0 t - \mathbf{r}_0) \\ &= -Gm_1m_2 \frac{R\mathbf{r}_1 - R\mathbf{r}_2}{|R\mathbf{r}_1 - R\mathbf{r}_2|^3} \\ &= -Gm_1m_2 R \frac{\mathbf{r}_1 - \mathbf{r}_2}{|\mathbf{r}_1 - \mathbf{r}_2|^3},\end{aligned} \tag{6.29}$$

where in the last step we used that an orthogonal transformation does not change the length of a vector, $|R\mathbf{r}| = |\mathbf{r}|$.

An example for a force law not fulfilling (6.27) would be the fictitious force law (note that it does conform to Newton's third axiom $\mathbf{F}(\mathbf{r}_1, \mathbf{r}_2) = -\mathbf{F}(\mathbf{r}_2, \mathbf{r}_1)$, due to the antisymmetry of the vector product)

$$\mathbf{F}(\mathbf{r}_1, \mathbf{r}_2) = C\mathbf{r}_1 \times \mathbf{r}_2. \tag{6.30}$$

In order to show that this does not fulfill (6.27), it suffices to consider translations, i.e., Galilei transformations with $R = \mathbb{1}$ and $\mathbf{v}_0 = 0$. Then

$$\begin{aligned}\mathbf{F}(\mathbf{r}_1', \mathbf{r}_2') &= \mathbf{F}(\mathbf{r}_1 - \mathbf{r}_0, \mathbf{r}_2 - \mathbf{r}_0) \\ &= C(\mathbf{r}_1 - \mathbf{r}_0) \times (\mathbf{r}_2 - \mathbf{r}_0) \\ &= C\left(\mathbf{r}_1 \times \mathbf{r}_2 - \mathbf{r}_0 \times \mathbf{r}_2 - \mathbf{r}_1 \times \mathbf{r}_0\right) \\ &= \mathbf{F}(\mathbf{r}_1, \mathbf{r}_2) - C\mathbf{r}_0 \times (\mathbf{r}_2 - \mathbf{r}_1).\end{aligned} \tag{6.31}$$

Due to the supplementary second term on the r.h.s., (6.27) is violated.

Restricting to a force law $\mathbf{F}(\mathbf{r}_1, \mathbf{r}_2)$ depending only on positions (not velocities), we may ask the following: What is the most general form of $\mathbf{F}(\mathbf{r}_1, \mathbf{r}_2)$ compatible with the requirement (6.27) of the Galilean relativity principle? Let us first exploit invariance under translations. It implies that $\mathbf{F}$ can in fact only depend on the difference $\mathbf{r}_1 - \mathbf{r}_2$ of the positions, i.e., $\mathbf{F}(\mathbf{r}_1, \mathbf{r}_2) = \mathbf{F}(\mathbf{r}_1 - \mathbf{r}_2)$. To prove this, we *define* $\mathbf{F}(\mathbf{r}) := \mathbf{F}(\mathbf{r}, \mathbf{0})$. Now for any pair of positions $\mathbf{r}_1$, $\mathbf{r}_2$, we can perform the translation which sends $\mathbf{r}_2$ to $\mathbf{0}$ (by choosing the translation parameter $\mathbf{r}_0 = \mathbf{r}_2$). From (6.27) it then follows that

$$\mathbf{F}(\mathbf{r}_1, \mathbf{r}_2) = \mathbf{F}(\mathbf{r}_1', \mathbf{r}_2') = \mathbf{F}(\mathbf{r}_1 - \mathbf{r}_2, \mathbf{0}) = \mathbf{F}(\mathbf{r}_1 - \mathbf{r}_2), \tag{6.32}$$

as we intended to show.

Now consider rotations. As a rotation does not change the modulus of a vector, it follows from (6.27) (exploiting that $\mathbf{F}$ only depends on $\mathbf{r}_1 - \mathbf{r}_2$) that

$$|\mathbf{F}(\mathbf{r})| = |\mathbf{F}(R\mathbf{r})|. \tag{6.33}$$

As by a rotation any vector $\mathbf{r}$ can be mapped to any other vector of the same length, it follows that $|\mathbf{F}(\mathbf{r})|$ can in fact only depend on the length of $\mathbf{r}$, i.e., we must have $|\mathbf{F}(\mathbf{r})| = F(|\mathbf{r}|)$ for some function F on the positive real line.[2]

Having established that $\mathbf{F}(\mathbf{r}_1, \mathbf{r}_2)$ only depends on $\mathbf{r} := \mathbf{r}_1 - \mathbf{r}_2$ and its modulus only on $|\mathbf{r}|$, we turn to the direction of $\mathbf{F}(\mathbf{r})$. We claim that it must be pointing in direction $\mathbf{r}$, i.e.,

$$\mathbf{F}(\mathbf{r}) = F(r)\hat{\mathbf{r}}, \tag{6.34}$$

[2]Similarly to the above, we may *define* $F(r) := |\mathbf{F}(\bar{\mathbf{r}})|$ by choosing some reference vector $\bar{\mathbf{r}}$ of length r. If now $\mathbf{r}$ is any vector of the same length, then there is a rotation R such that $R\mathbf{r} = \bar{\mathbf{r}}$, so that $|\mathbf{F}(\mathbf{r})| = |\mathbf{F}(R\mathbf{r})| = F(r)$.

with $\hat{\mathbf{r}} = \frac{1}{r}\mathbf{r}$ the unit vector pointing in direction $\hat{r}$.[3] To prove this, we use *reductio ad absurdum*, i.e., we assume that (6.34) does not hold for some $\mathbf{r} \neq 0$ and show that this leads to a contradiction. We choose any nontrivial rotation R around the axis defined by $\mathbf{r}$. Such a rotation leaves $\mathbf{r}$ invariant, $R\mathbf{r} = \mathbf{r}$, but any vector which is not parallel to $\mathbf{r}$ will be changed by the rotation: $R\mathbf{u} \neq \mathbf{u}$ for all nonzero $\mathbf{u} \neq u\hat{\mathbf{r}}$. Then the Galilean relativity principle yields

$$\mathbf{F}(\mathbf{r}) = \mathbf{F}(R\mathbf{r}) = R\mathbf{F}(\mathbf{r}) \neq \mathbf{F}(\mathbf{r}), \tag{6.35}$$

where in the last step we used the assumption that $\mathbf{F}(\mathbf{r})$ is not parallel to $\hat{\mathbf{r}}$ (and thus changes under R). Hence the assumption that there is some $\mathbf{r} \neq 0$ such that (6.34) does not hold leads to a contraction. We have thus established that a force law depending only on the positions and complies to the Galilean relativity principle (6.27) is of the form

$$\mathbf{F}(\mathbf{r}_1, \mathbf{r}_2) = F(|\mathbf{r}_1 - \mathbf{r}_2|)\frac{\mathbf{r}_1 - \mathbf{r}_2}{|\mathbf{r}_1 - \mathbf{r}_2|}. \tag{6.36}$$

Such a force is called a *central force*. Obviously, Newton's gravitational force law (6.28) is of that form.

Regarding force laws also depending on the velocities, it follows, by exploiting invariance under boosts in complete analogy to the above, that these in fact only depend on the difference $\dot{\mathbf{r}}_2 - \dot{\mathbf{r}}_1$ of the velocities. An example for a force law depending on positions and velocities and consistent with Galilean Relativity (if one restricts to transformations not changing the orientation, i.e., for which $\det R = 1$) is the *Lorentz force* between two particles of charge q_1 and q_2:

$$\mathbf{F}(\mathbf{r}_1, \dot{\mathbf{r}}_1, \mathbf{r}_2, \dot{\mathbf{r}}_2) = q_1 q_2 \frac{(\mathbf{r}_1 - \mathbf{r}_2) \times (\dot{\mathbf{r}}_1 - \dot{\mathbf{r}}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|^3}. \tag{6.37}$$

Remark 6.1. *It should be emphasized that the force* law *is required to be the same in any inertial frame by the Galilean relativity principle. The force as such of course depends on the inertial frame, as also seen in* (6.25)*. Above, we studied two-body interactions, but in practice it is often convenient to consider some of the bodies as fixed or not influenced by the motion of the others. For example, when studying the motion of a small (test) charge in an electromagnetic field, one typically assumes the electromagnetic field as given and not influenced by the test charge. This corresponds to fixing the other charges or currents and subsuming their influence on the test charge in an* external force*. In such a situation, there then is a preferred inertial frame, the one in which the charges (and currents) producing the electromagnetic field are at rest.*

Remark 6.2. *Einstein's relativity principle is formulated in complete analogy to the Galilean one. The difference is in the transformations between inertial frames. The absolute notion of simultaneity is dropped and replaced by the requirement that the speed of light is the same in any inertial frame.*[4]

6.4 Conservative forces and work

We noticed in Section 4.2 that in one-dimensional motion governed by a force $f(u)$ depending only on the position, the energy

$$E = \frac{1}{2}\dot{u}^2 + V(u) \tag{6.38}$$

[3] Note that we do not exclude $\mathbf{F}(\mathbf{r})$ pointing in the direction opposite to $\mathbf{r}$, this possibility is covered by $F(r)$ being negative.

[4] As the speed of light can be derived from Maxwell's equations of electromagnetism, this can in fact be seen as a consequence of the relativity requirement.

is conserved, where the potential $V(u)$ is such that $V'(u) = -f(u)$. This turned out to be very useful in solving concrete problems (such as the pendulum). A natural question is thus whether such a conserved quantity also exists in three-dimensional motion in Newtonian Mechanics. We first restrict to the motion of a single particle of mass m. A generalization of the above to three dimensions would be[5]

$$E = \frac{m}{2}|\dot{\mathbf{r}}|^2 + V(\mathbf{r}). \tag{6.39}$$

Differentiating w.r.t. t now yields

$$\frac{\mathrm{d}}{\mathrm{d}t}E = m\ddot{\mathbf{r}} \cdot \dot{\mathbf{r}} + \dot{\mathbf{r}} \cdot \boldsymbol{\nabla} V(\mathbf{r}) = \dot{\mathbf{r}} \cdot (m\ddot{\mathbf{r}} + \boldsymbol{\nabla} V(\mathbf{r}))\,, \tag{6.40}$$

where we used (5.7). Hence, the quantity E is conserved whenever the force $\mathbf{F}(\mathbf{r})$ is a *gradient field*, i.e., it can be written as (-1 times) the gradient of a function:

$$\mathbf{F}(\mathbf{r}) = -\boldsymbol{\nabla} V(\mathbf{r}). \tag{6.41}$$

A force $\mathbf{F}(\mathbf{r})$ for which such a *potential* $V(\mathbf{r})$ exists is called *conservative*. If it exists, the potential is unique up to the addition of a constant (the difference $V_1(\mathbf{r}) - V_2(\mathbf{r})$ must have a vanishing gradient, but a function on $\mathbb{R}^3$ with a vanishing gradient must be a constant).

An important class of forces for which a potential exists are central forces, i.e., of the form $\mathbf{F}(\mathbf{r}) = F(r)\hat{\mathbf{r}}$ (force in the radial direction with magnitude depending only on distance to the origin). Such forces always admit a potential, as follows from (5.13): Choosing $V(\mathbf{r}) = V(r)$ we fulfill (6.41) when

$$V'(r) = -F(r). \tag{6.42}$$

Hence, we only need to integrate $-F(r)$ to obtain $V(r)$.

Example 6.1. *The gravitational force on a mass m in $\mathbf{r}$ due to a mass M in $\mathbf{0}$ is $\mathbf{F}(\mathbf{r}) = -\frac{GmM}{r^2}\hat{\mathbf{r}}$, so $F(r) = -\frac{GmM}{r^2}$. It is convenient and customary to define the potential such that it vanishes at infinity, so that*

$$V(r) = \int_r^\infty F(r')\mathrm{d}r' = -\frac{GmM}{r}. \tag{6.43}$$

A necessary and sufficient condition for a force to be conservative is the vanishing of its curl. The curl of a vector field $\mathbf{F}(\mathbf{r})$ is defined as

$$\boldsymbol{\nabla} \times \mathbf{F} = \begin{pmatrix} \partial_y F_z - \partial_z F_y \\ \partial_z F_x - \partial_x F_z \\ \partial_x F_y - \partial_y F_x \end{pmatrix}. \tag{6.44}$$

One easily checks that the curl of a gradient field vanishes,

$$\boldsymbol{\nabla} \times \boldsymbol{\nabla} f(\mathbf{r}) = 0, \tag{6.45}$$

so in particular the curl of a conservative force vanishes. The proof of the converse statement, i.e., that if the curl of a force field vanishes, then it is conservative (a gradient field), is left for future courses.

[5] Here we include the mass m in the first term, as this turns out to be useful when considering systems of many particles in the next section. We did not do so in Section 4.2 as at that point the inertial mass was not yet introduced (implicitly included in the force f).

Exercise 6.1. *Show that the force (this corresponds to the force considered in Example 5.1)*

$$\mathbf{F}(\mathbf{r}) = \begin{pmatrix} y \\ -x \\ 0 \end{pmatrix} \tag{6.46}$$

is not conservative.

In one-dimensional motion, conservation of energy was sufficient to solve the dynamical equation (to the extent that it could be reduced to evaluating a definite integral and inverting a function). In three-dimensional motion, conservation of energy is not sufficient to achieve this. Further conserved quantities, such as angular momentum, are necessary to achieve such a simplification. Nevertheless, conservation of energy in three-dimensional motion already gives important qualitative information about the motion. For example, given initial position $\mathbf{r}_0$ and velocity $\mathbf{v}_0$, at $t = 0$, one can determine the energy E. As this quantity is conserved and $|\mathbf{v}|^2$ non-negative, we know that the particle motion is constrained to the energetically allowed region $\{\mathbf{r} \in \mathbb{R}^3 \mid V(\mathbf{r}) \leq E\}$.

We call the first term of our expression (6.39) for the energy, depending only on the velocity, the *kinetic energy*

$$T := \frac{m}{2}\mathbf{v} \cdot \mathbf{v}. \tag{6.47}$$

This is an interesting quantity to consider even when the force is not conservative, i.e., the energy as given by (6.39) is not defined (by lack of a potential V). For the rate of change of the kinetic energy, we compute

$$\frac{\mathrm{d}}{\mathrm{d}t}T = m\mathbf{a} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{F}(\mathbf{r}, \mathbf{v}, t), \tag{6.48}$$

where in the second step we used Newton's equation. We can thus compute the change of the kinetic energy between times t_1 and t_2 by integration:

$$W_{1,2} := T_2 - T_1 = \int_{t_1}^{t_2} \dot{\mathbf{r}}(t) \cdot \mathbf{F}(\mathbf{r}(t), \dot{\mathbf{r}}(t), t)\mathrm{d}t. \tag{6.49}$$

This quantity we call the *work* done on the particle on its trajectory from $\mathbf{r}(t_1)$ to $\mathbf{r}(t_2)$. In the case when the force $\mathbf{F}$ only depends on the position (not on the velocity or the time), we obtain

$$W_{1,2} = \int_{t_1}^{t_2} \mathbf{F}(\mathbf{r}(t)) \cdot \dot{\mathbf{r}}(t)\mathrm{d}t. \tag{6.50}$$

The integral on the r.h.s. is known as the *line integral* (or *path integral*) of the vector field $\mathbf{F}(\mathbf{r})$ along the path $\mathbf{r}(t)$ from t_1 to t_2. An illustration of the line integral is shown in Fig. 6.1: At each point along the curve, we take the scalar product of $\mathbf{F}$ and $\dot{\mathbf{r}}$ and integrate this up.

Example 6.2. *We revisit Example 5.1. For the force*

$$\mathbf{f}(\mathbf{u}) = \begin{pmatrix} u_2 \\ -u_1 \end{pmatrix} \tag{6.51}$$

we found the trajectories

$$\mathbf{u}(t) = 2e^{t/\sqrt{2}} \begin{pmatrix} \cos t/\sqrt{2} \\ -\sin t/\sqrt{2} \end{pmatrix} \tag{6.52}$$

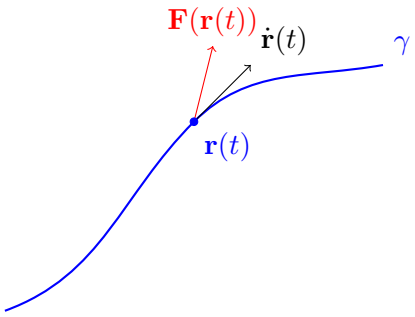


Figure 6.1: Illustration of the line integral.

as possible solutions. The corresponding velocity is

$$\dot{\mathbf{u}}(t) = \sqrt{2}e^{t/\sqrt{2}} \begin{pmatrix} \cos t/\sqrt{2} - \sin t/\sqrt{2} \\ -\sin t/\sqrt{2} - \cos t/\sqrt{2} \end{pmatrix}, \tag{6.53}$$

so that the work performed during times t_0 and t_1 is

$$\begin{aligned} \int \mathbf{f} \cdot \mathrm{d}\mathbf{u} &= \int_{t_0}^{t_1} \mathbf{f} \cdot \dot{\mathbf{u}} \mathrm{d}t \\ &= \int_{t_0}^{t_1} 2e^{t/\sqrt{2}} \begin{pmatrix} -\sin t/\sqrt{2} \\ -\cos t/\sqrt{2} \end{pmatrix} \cdot \sqrt{2}e^{t/\sqrt{2}} \begin{pmatrix} \cos t/\sqrt{2} - \sin t/\sqrt{2} \\ -\sin t/\sqrt{2} - \cos t/\sqrt{2} \end{pmatrix} \mathrm{d}t \\ &= 2\sqrt{2} \int_{t_0}^{t_1} e^{\sqrt{2}t} \mathrm{d}t \\ &= 2\left(e^{\sqrt{2}t_1} - e^{\sqrt{2}t_0}\right). \end{aligned} \tag{6.54}$$

This can easily be checked to coincide with the change in kinetic energy $\frac{1}{2}|\dot{\mathbf{u}}|^2$ between times t_0 and t_1 (recall that in Chapter 5 we included the mass into the force, effectively setting $m = 1$).

The line integral fulfills two important properties. The first one can be seen as a generalization of the fundamental theorem of calculus. Namely, if the vector field $\mathbf{F}$ which is integrated is a gradient field, $\mathbf{F} = \boldsymbol{\nabla} f$, then its line integral is the difference of the values of $f(\mathbf{r})$ evaluated at the endpoints:

$$\int_{t_1}^{t_2} \boldsymbol{\nabla} f(\mathbf{r}(t)) \cdot \dot{\mathbf{r}}(t) \mathrm{d}t = \int_{t_1}^{t_2} \frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{r}(t)) \mathrm{d}t = f(\mathbf{r}(t_2)) - f(\mathbf{r}(t_1)). \tag{6.55}$$

Here we used (5.7) in the first and the fundamental theorem of calculus in the second step. In particular, the work done by a conservative force is equal to the difference of the potential at the two points:

$$W_{1,2} = V(\mathbf{r}(t_1)) - V(\mathbf{r}(t_2)). \tag{6.56}$$

This must of course be the case as the energy E is conserved.

The second important property of the line integral is that it only depends on the path taken, not on the velocity. To see this, consider a strictly increasing function $s(t)$ with inverse function $t(s)$. Then $s \mapsto \mathbf{r}(t(s))$ is a curve along the same points and in the same direction, but the velocities will differ unless $t(s) = s$:

$$\frac{\mathrm{d}}{\mathrm{d}s} \mathbf{r}(t(s)) = \frac{\mathrm{d}t}{\mathrm{d}s} \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}. \tag{6.57}$$

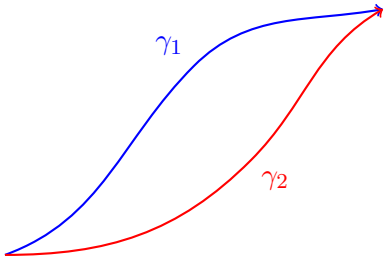


Figure 6.2: Two paths connecting the same points.

Now perform the line integral using the curve $s \mapsto \mathbf{r}(t(s))$, with $s_1 = s(t_1)$ and $s_2 = s(t_2)$:

$$\begin{aligned}\int_{s_1}^{s_2} \mathbf{F}(\mathbf{r}(t(s))) \cdot \frac{\mathrm{d}}{\mathrm{d}s}\mathbf{r}(t(s))\mathrm{d}s &= \int_{s_1}^{s_2} \mathbf{F}(\mathbf{r}(t(s))) \cdot \frac{\mathrm{d}}{\mathrm{d}t}\mathbf{r}(t(s))\frac{\mathrm{d}t}{\mathrm{d}s}\mathrm{d}s \\ &= \int_{t_1}^{t_2} \mathbf{F}(\mathbf{r}(t)) \cdot \dot{\mathbf{r}}(t)\mathrm{d}t. \end{aligned} \tag{6.58}$$

Here we used a transformation of variables in the last step. In the final result, we recovered the line integral corresponding to the original path $\mathbf{r}(t)$. One says that the line integral is *independent of the parametrization*. Above, we considered a strictly increasing function $s(t)$, so that the path is covered in the same direction. Changing the direction, i.e., choosing a decreasing function $s(t)$, the line integral changes it sign. As the line integral only depends on the path taken (including its direction), but not on how it is parameterized, one also used the notation

$$\int_\gamma \mathbf{F} \cdot \mathrm{d}\mathbf{r} \tag{6.59}$$

for the line integral, where γ denotes the path (including the direction). One also calls $\mathrm{d}\mathbf{r} := \dot{\mathbf{r}}\mathrm{d}t$ the *infinitesimal line element*.

We derived the expression (6.50) for the work by considering the change in kinetic energy for a trajectory $\mathbf{r}(t)$ fulfilling the equation of motion. However, the expression (6.50) makes sense for arbitrary paths $\mathbf{r}(t)$ (not necessarily fulfilling the equation of motion). The expression can now be interpreted as follows: Assume the particle is subject to the force $\mathbf{F}$ and a further (unspecified) force such that the trajectory resulting from the sum of these forces is $\mathbf{r}(t)$. Then the work (6.50) is the contribution of the force $\mathbf{F}$ to the change of kinetic energy between the two endpoints. As we just proved, if $\mathbf{F}$ only depends on the position, this quantity does not depend on how this path was covered (with which velocity, for example).

If the force $\mathbf{F}$ is conservative, then, as we already saw, the work done by it along a path only depends on the two endpoints of the path and is given by (6.56). Hence, for a conservative force, the work is path-independent. But also the converse holds: If the work is path-independent, i.e., if for any two paths γ_1, γ_2 with the same endpoints (see Fig. 6.2 for an illustration) we have

$$\int_{\gamma_1} \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \int_{\gamma_2} \mathbf{F} \cdot \mathrm{d}\mathbf{r}, \tag{6.60}$$

then the force $\mathbf{F}$ is conservative. The relation between this criterion for a conservative force and the one provided above (vanishing of the curl of $\mathbf{F}$) is provided by *Stokes' theorem*, which will be covered in TP2.

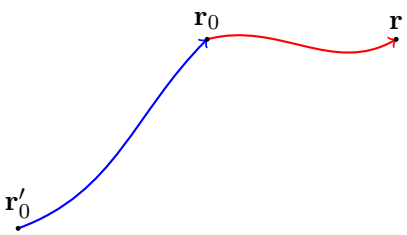


Figure 6.3: The path from $\mathbf{r}'_0$ to $\mathbf{r}$ is split into a path from $\mathbf{r}'_0$ to $\mathbf{r}_0$, and one from $\mathbf{r}_0$ to $\mathbf{r}$.

Now if the work is path-independent, we can directly use the line integral to define the potential V. For that, fix an arbitrary reference point $\mathbf{r}_0$. Then define

$$V(\mathbf{r}) := -\int_{\gamma_{\mathbf{r}_0\to\mathbf{r}}} \mathbf{F}\cdot \mathrm{d}\mathbf{r}, \tag{6.61}$$

where $\gamma_{\mathbf{r}_0\to\mathbf{r}}$ is any path from $\mathbf{r}_0$ to $\mathbf{r}$ (by the assumed path-independence, it is irrelevant which path one chooses). To motivate the definition, we recall that if $\mathbf{F}$ is conservative, i.e., $\mathbf{F} = -\boldsymbol{\nabla} V$, then the r.h.s. gives, by (6.55), $V(\mathbf{r}) - V(\mathbf{r}_0)$. Hence, we recover the potential V, up to a shift by $V(\mathbf{r}_0)$.

By definition, the potential as defined by (6.61) vanishes at the reference point $\mathbf{r}_0$, i.e., $V(\mathbf{r}_0) = 0$ (for $\gamma_{\mathbf{r}_0\to\mathbf{r}_0}$ we can choose the constant path $\mathbf{r}(t) = \mathbf{r}_0$, for which $\dot{\mathbf{r}} = \mathbf{0}$). Hence, the reference point fixes the zero of the potential. If we define the potential $V'(\mathbf{r})$ as in (6.61), but using a different reference point $\mathbf{r}'_0$, then this new potential is related to the old one by a shift: $V'(\mathbf{r}) = V(\mathbf{r}) - V(\mathbf{r}'_0)$. To see this, we can choose the path $\gamma_{\mathbf{r}'_0\to\mathbf{r}}$ as the composition of two paths, one from $\mathbf{r}'_0$ to $\mathbf{r}_0$, and one from $\mathbf{r}_0$ to $\mathbf{r}$, c.f. Fig. 6.3. The line integral then splits into a sum of two line integrals:

$$V'(\mathbf{r}) = -\int_{\gamma_{\mathbf{r}'_0\to\mathbf{r}}} \mathbf{F}\cdot \mathrm{d}\mathbf{r} = -\int_{\gamma_{\mathbf{r}'_0\to\mathbf{r}_0}} \mathbf{F}\cdot \mathrm{d}\mathbf{r} - \int_{\gamma_{\mathbf{r}_0\to\mathbf{r}}} \mathbf{F}\cdot \mathrm{d}\mathbf{r} = -V(\mathbf{r}'_0) + V(\mathbf{r}). \tag{6.62}$$

In the last step, we used that under a change of the direction of the path, the line integral changes its sign.

It remains to show that path-independence of the work (or equivalently vanishing of the curl of $\mathbf{F}$) implies that $\mathbf{F}$ is conservative, i.e., that $V(\mathbf{r})$ as defined above fulfills the desired property $\boldsymbol{\nabla} V(\mathbf{r}) = -\mathbf{F}(\mathbf{r})$. This will be the case iff $\mathbf{a}\cdot\boldsymbol{\nabla} V = -\mathbf{a}\cdot\mathbf{F}$ for all $\mathbf{a}$. By (5.7), the l.h.s. of this equation can be computed as $\frac{\mathrm{d}}{\mathrm{d}s} V(\mathbf{r}+s\mathbf{a})|_{s=0}$. To evaluate this, we need to know V along the curve through $\mathbf{r}$ in the direction $\mathbf{a}$. It is convenient to use $\mathbf{r}$ as the reference point,[6] and use the path $\mathbf{r}(t) = \mathbf{r} + t\mathbf{a}$ from $\mathbf{r}(0) = \mathbf{r}$ to $\mathbf{r}(s) = \mathbf{r} + s\mathbf{a}$. Thus, we compute

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}s} V(\mathbf{r}+s\mathbf{a})|_{s=0} &= -\frac{\mathrm{d}}{\mathrm{d}s}\int_0^s \mathbf{F}(\mathbf{r}+t\mathbf{a})\cdot\mathbf{a}\mathrm{d}t|_{s=0}\\ &= -\mathbf{F}(\mathbf{r})\cdot\mathbf{a},\end{aligned} \tag{6.63}$$

which is the desired equation.

[6] For any other reference point, the potential would be shifted by a constant, which drops out upon differentiating w.r.t. s.

Remark 6.3. *The path-independence of the line integral is equivalent to the vanishing of the line integral for all closed paths, i.e.,*

$$\oint \mathbf{F} \cdot \mathrm{d}\mathbf{r} = 0 \tag{6.64}$$

for all closed path γ (the circle through the integral sign emphasizes that this is a line integral along a closed path). To see this, we note that any two paths γ_1, γ_2 from $\mathbf{r}_1$ to $\mathbf{r}_2$ can be composed as $(-\gamma_2) \circ \gamma_1$ to a closed path from $\mathbf{r}_1$ to $\mathbf{r}_1$, by first going along γ_1 to $\mathbf{r}_2$ and then backwards along γ_2 to $\mathbf{r}_1$. As the inversion of the direction corresponds to the change of sign, we see that vanishing of the line integral along closed paths implies path independence. To prove the converse implication, it suffices to note that line integral along the constant path always vanishes, so path-independence implies that all paths with the same start- and end-point must have a vanishing line integral.

6.5 Conservation laws

In many important cases, conservation laws apply, which greatly facilitate the treatment of the problem (recall how conservation of energy could be used to solve dynamics in one dimension).

We are now considering systems of point particles, with masses m_i at positions $\mathbf{r}_i$ (i labels the different particles). Each point mass m_i may be subject to an *external force* $\mathbf{F}_i(\mathbf{r}_i, \dot{\mathbf{r}}_i, t)$, depending only on the position and velocity of the particle (and possibly time), and *interaction forces* due to the other particles. We assume that only two-body forces are present and denote the forces exerted by the jth particle on the ith particle by $\mathbf{F}_{ij}(\mathbf{r}_i, \mathbf{r}_j, \dot{\mathbf{r}}_i, \dot{\mathbf{r}}_j, t)$. In most cases, the interaction will only depend on $\mathbf{r}_i - \mathbf{r}_j$ (as discussed in Section 6.3), but for the time being we allow a general dependence on the positions and velocities of the two particles, as well as time. As an example for the type of system we have in mind, consider a gas. The molecules of the gas are subject to gravity and forces exerted by the wall of the container of the gas, as well as intermolecular (van-der-Waals) forces. The former are external, while the latter are interaction forces.

Depending further assumptions on the external and interaction forces, there are important conservation laws. *Energy conservation* holds whenever all external and interaction forces are conservative in the sense that

$$\mathbf{F}_i(\mathbf{r}_i) = -\boldsymbol{\nabla} V_i(\mathbf{r}_i), \qquad \mathbf{F}_{ij}(\mathbf{r}_i - \mathbf{r}_j) = -\boldsymbol{\nabla} V_{ij}(|\mathbf{r}_i - \mathbf{r}_j|). \tag{6.65}$$

For simplicity, we have here assumed that the interaction potential only depends on the distance of the two particles, thus giving rise to a central force (recall Section 6.3). We note that Newton's third law implies that $V_{ij}(r) = V_{ji}(r)$, as follows from[7]

$$\mathbf{F}_{ij} = -\hat{\mathbf{r}}_{ij} V'_{ij}(r_{ij}), \qquad \mathbf{F}_{ji} = -\hat{\mathbf{r}}_{ji} V'_{ji}(r_{ij}), \tag{6.66}$$

where we introduced the notation

$$\mathbf{r}_{ij} := \mathbf{r}_i - \mathbf{r}_j, \qquad r_{ij} = |\mathbf{r}_{ij}|, \qquad \hat{\mathbf{r}}_{ij} = \frac{\mathbf{r}_{ij}}{r_{ij}}. \tag{6.67}$$

[7] Strictly speaking, this only shows that $V'_{ij}(r) = V'_{ji}(r)$, but this implies $V_{ij}(r) = V_{ji}(r)$ up to a constant, which can be set to zero.

The conserved energy is then given by

$$E = \sum_i \left(\frac{m_i}{2} |\dot{\mathbf{r}}_i|^2 + V_i(\mathbf{r}_i) \right) + \sum_{i,j \text{ s.t. } i<j} V_{ij}(r_{ij}). \tag{6.68}$$

In order to avoid double counting, we here only include the interaction potentials for $i < j$.[8] To prove this conservation law, we first note that, due to the chain rule,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} V(r_{ij}) &= \frac{\mathrm{d}}{\mathrm{d}t} r_{ij} V'(r_{ij}) \\ &= \left(\dot{\mathbf{r}}_i \cdot \boldsymbol{\nabla}_i r_{ij} + \dot{\mathbf{r}}_j \cdot \boldsymbol{\nabla}_j r_{ij} \right) V'(r_{ij}) \\ &= \left(\dot{\mathbf{r}}_i \cdot \hat{\mathbf{r}}_{ij} + \dot{\mathbf{r}}_j \cdot \hat{\mathbf{r}}_{ji} \right) V'(r_{ij}) \\ &= -\dot{\mathbf{r}}_i \cdot \mathbf{F}_{ij} - \dot{\mathbf{r}}_j \cdot \mathbf{F}_{ji}. \end{aligned} \tag{6.69}$$

Here $\boldsymbol{\nabla}_i$ denotes the gradient w.r.t. $\mathbf{r}_i$. Using this result, we verify

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} E &= \sum_i \left(m_i \ddot{\mathbf{r}}_i \cdot \dot{\mathbf{r}}_i + \dot{\mathbf{r}}_i \cdot \boldsymbol{\nabla} V_i(\mathbf{r}_i) \right) - \sum_{i,j \text{ s.t. } i<j} \left(\dot{\mathbf{r}}_i \cdot \mathbf{F}_{ij} + \dot{\mathbf{r}}_j \cdot \mathbf{F}_{ji} \right) \\ &= \sum_i \left(m_i \ddot{\mathbf{r}}_i \cdot \dot{\mathbf{r}}_i + \dot{\mathbf{r}}_i \cdot \boldsymbol{\nabla} V_i(\mathbf{r}_i) \right) - \sum_{i,j \text{ s.t. } i \neq j} \dot{\mathbf{r}}_i \cdot \mathbf{F}_{ij} \\ &= \sum_i \dot{\mathbf{r}}_i \cdot \left(m_i \ddot{\mathbf{r}}_i - \mathbf{F}_i - \sum_{j \text{ s.t. } j \neq i} \mathbf{F}_{ij} \right) \\ &= 0, \end{aligned} \tag{6.70}$$

where in the last step we used the equation of motion.

Momentum conservation holds whenever the external forces are absent, $\mathbf{F}_i = \mathbf{0}$. Defining the total momentum

$$\mathbf{P} := \sum_i m_i \dot{\mathbf{r}}_i, \tag{6.71}$$

we compute

$$\frac{\mathrm{d}}{\mathrm{d}t} \mathbf{P} = \sum_i m \ddot{\mathbf{r}}_i = \sum_i \sum_{j \text{ s.t. } j \neq i} \mathbf{F}_{ij} = \mathbf{0}. \tag{6.72}$$

In the last step, we used that summing $\mathbf{F}_{ij}$ over all possible combinations i, j with $i \neq j$ is equivalent to summing $\frac{1}{2}(\mathbf{F}_{ij} + \mathbf{F}_{ji})$ over all possible combinations i, j with $i \neq j$, i.e.,

$$\sum_{i,j \text{ s.t. } i \neq j} \mathbf{F}_{ij} = \frac{1}{2} \sum_{i,j \text{ s.t. } i \neq j} (\mathbf{F}_{ij} + \mathbf{F}_{ji}) = \mathbf{0}, \tag{6.73}$$

where in the last step we used Newton's third axiom.

An interesting consequence of momentum conservation is that the *center of mass*, defined by

$$\mathbf{R} := \frac{1}{M} \sum_i m_i \mathbf{r}_i, \qquad\qquad M := \sum_i m_i \tag{6.74}$$

[8] Alternatively, we could replace the last sum by $\frac{1}{2} \sum_{i,j \text{ s.t. } i \neq j} V_{ij}(r_{ij})$ which coincides with the given expression due to $V_{ij}(r) = V_{ji}(r)$.

behaves as a free particle:

$$\ddot{\mathbf{R}} = \frac{1}{M}\dot{\mathbf{P}} = \mathbf{0}. \tag{6.75}$$

Hence, $\mathbf{R}(t) = \mathbf{R}_0 + \frac{1}{M}\mathbf{P}_0 t$.

Finally, *angular momentum conservation* holds, whenever both the external and the interaction forces are central forces, i.e.,

$$\mathbf{F}_i(\mathbf{r}_i) = \hat{\mathbf{r}}_i F_i(r_i), \qquad \mathbf{F}_{ij}(\mathbf{r}_{ij}) = \hat{\mathbf{r}}_{ij} F_{ij}(r_{ij}). \tag{6.76}$$

Then the *total angular momentum*, defined as

$$\mathbf{L} := \sum_i m_i \mathbf{r}_i \times \dot{\mathbf{r}}_i, \tag{6.77}$$

is conserved. To see this, we compute

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{L} &= \sum_i m_i \left(\dot{\mathbf{r}}_i \times \dot{\mathbf{r}}_i + \mathbf{r}_i \times \ddot{\mathbf{r}}_i\right) \\
&= \sum_i \mathbf{r}_i \times \left(\hat{\mathbf{r}}_i F_i(r_i) + \sum_{j \text{ s.t. } j\neq i} \hat{\mathbf{r}}_{ij} F_{ij}(r_{ij})\right) \\
&= -\sum_{i,j \text{ s.t. } i\neq j} \frac{\mathbf{r}_i \times \mathbf{r}_j}{r_{ij}} F_{ij}(r_{ij}).
\end{aligned} \tag{6.78}$$

From the first to the second line, we used that $\mathbf{v} \times \mathbf{v} = \mathbf{0}$ and the equations of motion. In the next step, we used again $\mathbf{v} \times \mathbf{v} = \mathbf{0}$. The desired result can now be established as in (6.73), now using the antisymmetry $\mathbf{v} \times \mathbf{w} = -\mathbf{w} \times \mathbf{v}$ of the vector product:

$$\sum_{i,j \text{ s.t. } i\neq j} \frac{\mathbf{r}_i \times \mathbf{r}_j}{r_{ij}} F_{ij}(r_{ij}) = \frac{1}{2} \sum_{i,j \text{ s.t. } i\neq j} \left(\frac{\mathbf{r}_i \times \mathbf{r}_j}{r_{ij}} F_{ij}(r_{ij}) + \frac{\mathbf{r}_j \times \mathbf{r}_i}{r_{ji}} F_{ji}(r_{ji})\right) = \mathbf{0}. \tag{6.79}$$

Here we used that by the third axiom $F_{ij}(r) = F_{ji}(r)$.

An important consequence of conservation of angular momentum concerns the dynamics of a single particle in a central potential. Then the angular momentum $\mathbf{L} = m\mathbf{r} \times \dot{\mathbf{r}}$ is conserved. But as $\mathbf{L}$ is orthogonal to $\mathbf{r}$ (and $\dot{\mathbf{r}}$), it follows that the position vector $\mathbf{r}$ will always lie in the same plane (perpendicular to $\mathbf{L}$). Hence, we see that motion in a central potential is confined to a plane, which can be conveniently chosen to be the $x-y$-plane (so that $\mathbf{L}$ points in the z-direction). Furthermore, choosing polar coordinates (r, ϕ) in that plane,[9]

$$\mathbf{r}(t) = r(t)\hat{\boldsymbol{\rho}}(\phi(t)), \tag{6.80}$$

with the ONB

$$\hat{\boldsymbol{\rho}}(\phi) = \begin{pmatrix} \cos\phi \\ \sin\phi \\ 0 \end{pmatrix}, \qquad \hat{\boldsymbol{\phi}}(\phi) = \begin{pmatrix} -\sin\phi \\ \cos\phi \\ 0 \end{pmatrix}, \tag{6.81}$$

we compute

$$\dot{\mathbf{r}} = \dot{r}\hat{\boldsymbol{\rho}} + r\dot{\phi}\hat{\boldsymbol{\phi}}. \tag{6.82}$$

[9] It would be more in line with our previous conventions to use ρ instead of r, but for motion in the central potential r is more customary.

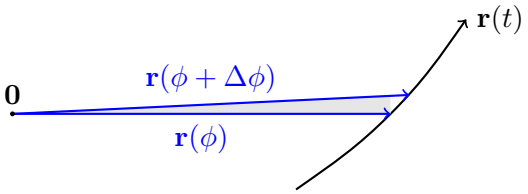


Figure 6.4: The area covered by the blue ray between ϕ and $\phi + \Delta\phi$ is approximated by the gray area, which is $\frac{1}{2}r^2\Delta\phi$. The side lengths of the missing "triangle" are both proportional to $\Delta\phi$, so that the missing piece has area proportional to $(\Delta\phi)^2$, which becomes irrelevant in an integral (think of how the Riemann integral converges to the correct area under the curve in the limit of $\Delta x \to 0$).

Using that $\hat{\boldsymbol{\rho}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}$, we compute

$$L := L_z = mr^2\dot{\phi}. \tag{6.83}$$

This has two interesting consequences: i) When writing this as $\dot{\phi} = \frac{L}{mr^2}$, and using that L is a constant of motion, we see that if we determined $r(t)$, then $\phi(t)$ can be simply obtained by integrating the above. In this manner, we have reduced the problem to the determination of $r(t)$. But this is a one-dimensional problem, and we already learned how to deal with this. We will see this explicitly worked out for the Kepler problem. ii) We can write the above in infinitesimal notation as

$$\frac{1}{2}r^2\mathrm{d}\phi = \frac{L}{2m}\mathrm{d}t. \tag{6.84}$$

The l.h.s. is the infinitesimal area element covered under the angular displacement $\mathrm{d}\phi$ by the line connecting the origin $\mathbf{0}$ and the particle position $\mathbf{r}$, c.f. Fig 6.4. Hence, integrating both sides over a finite interval, we see that the area A covered during time T is $A = \frac{L}{2m}T$, i.e., equal areas are covered in equal times. This is Kepler's second law of planetary motion, which (as our analysis shows) holds for any central force, not just the gravitational.

7 The Kepler problem

We now set out to study planetary motion in the solar system. The following laws were found by Kepler empirically, i.e., by analysing astronomical data:

1. Planetary orbits are ellipses with the sun in one of the focal points.
2. The areas swiped out by a ray from the sun to the planet are equal for time intervals of equal duration.
3. For all planets in the solar system the ratio $\frac{a^3}{T^2}$ is the same, with a the semi-major axis of the orbit and T its period.

The goal of this chapter will be to derive these laws from Newtonian mechanics. As already indicated, the second law is a consequence of conservation of angular momentum, and thus holds for any central force, not necessarily the gravitational one. To derive the other laws, the $\frac{1}{r^2}$ gravitational force law will be essential.

We idealize the situation by considering the system of the sun and a single planet only. This is still a two-body problem, which we can reduce to a one-body problem using conservation of momentum. In the next step, we will analyse the one-body problem, which, using conservation of angular momentum, can be reduced to a one-dimensional problem.

7.1 The two-body problem

We study the dynamics of two point masses m_1, m_2 at $\mathbf{r}_1$, $\mathbf{r}_2$, subject to a central interaction force, i.e., deriving from an interaction potential $V(r_{12})$ (in the terminology used in Section 6.5). We saw that in such a situation, the center of mass $\mathbf{R}(t)$ follows a linear motion. We can thus choose an initial frame such that $\mathbf{R}(t) = 0$. It then follows that $m_1\mathbf{r}_1 = -m_2\mathbf{r}_2$, so in particular also $m_1\ddot{\mathbf{r}}_1 = -m_2\ddot{\mathbf{r}}_2$. It is now convenient to describe the system in terms of $\mathbf{r} := \mathbf{r}_2 - \mathbf{r}_1$, c.f. Fig 7.1. Now $\mathbf{r}$ is subject to a simple dynamical equation:

$$\begin{aligned}\ddot{\mathbf{r}} &= \ddot{\mathbf{r}}_2 - \ddot{\mathbf{r}}_1 \\ &= \left(1 + \frac{m_2}{m_1}\right)\ddot{\mathbf{r}}_2 \\ &= \left(1 + \frac{m_2}{m_1}\right)\frac{1}{m_2}\mathbf{F}_{21}(\mathbf{r}) \\ &= \frac{m_1 + m_2}{m_1 m_2}\mathbf{F}_{21}(\mathbf{r}).\end{aligned} \tag{7.1}$$

Here we used the assumption that the force $\mathbf{F}_{21}$ exterted by m_1 on m_2 depends only on their separation $\mathbf{r}$. Hence, if we introduce the *reduced mass*

$$\mu := \frac{m_1 m_2}{m_1 + m_2}, \tag{7.2}$$

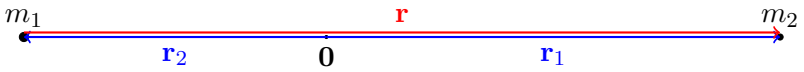


Figure 7.1: The choice of intertial frame in the two-body problem.

we can write the above (dropping the label 21 on the force) as

$$\mu\ddot{\mathbf{r}} = \mathbf{F}(\mathbf{r}). \tag{7.3}$$

We have this reduced the system to an *effective* one-body problem with the same force.

For our later discussions, it is worthwhile to note two special cases for the reduced mass. If one mass (say m_1) is much greater than the other one, then

$$\mu = \frac{m_2}{1+\frac{m_2}{m_1}} = m_2\left(1 - \frac{m_2}{m_1} + \mathcal{O}((m_2/m_1)^2)\right). \tag{7.4}$$

Hence, the reduced mass is close to, but smaller than the smaller mass. On the other hand, for equal masses $m_1 = m_2 = m$, we have $\mu = \frac{m}{2}$, i.e., the reduced mass is half that of the individual masses.

Once we have determined a solution (or the general solution) $\mathbf{r}(t)$ to (7.3), we can obtain $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$ as

$$\mathbf{r}_2(t) = \frac{1}{1+\frac{m_2}{m_1}}\mathbf{r}(t), \qquad \mathbf{r}_1(t) = -\frac{1}{1+\frac{m_1}{m_2}}\mathbf{r}(t). \tag{7.5}$$

In particular, if $m_2 \ll m_1$, then

$$\mathbf{r}_2(t) = \mathbf{r}(t) + \mathcal{O}(m_2/m_1), \qquad \mathbf{r}_1(t) = \mathbf{0} + \mathcal{O}(m_2/m_1). \tag{7.6}$$

In other words, if we neglect terms of $\mathcal{O}(\frac{m_2}{m_1})$, then we can consider the more massive mass to be at rest at the origin, and study only the dynamics of the other mass. In a frame in which the center of mass is not at rest (and fixed to the origin), one still adds $\mathbf{V}_0 t + \mathbf{R}_0$ to both $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$ (with $\mathbf{V}_0$ and $\mathbf{R}_0$ velocity and position of the center of mass at $t = 0$).

7.2 The one-body problem

We have reduced the two-body problem to the effective one-body problem (7.3). This is the equation of motion for a single particle in a central force. We learned in Section 6.5 that then both the energy $E = \frac{\mu}{2}|\dot{\mathbf{r}}|^2 + V(r)$ (with $\mathbf{F}(\mathbf{r}) = -\boldsymbol{\nabla}V(r)$) and the angular momentum $\mathbf{L} = \mu\mathbf{r}\times\dot{\mathbf{r}}$ are conserved. Furthermore, by choosing a coordinate system such that $\mathbf{L}$ is aligned with $\hat{\mathbf{z}}$, we can restrict the motion to the $x-y$-plane and parameterize is by $(r(t), \phi(t))$ as $\mathbf{r}(t) = r(t)\hat{\mathbf{r}}(\phi(t))$. We already computed $\dot{\mathbf{r}} = \dot{r}\hat{\mathbf{r}} + r\dot{\phi}\hat{\boldsymbol{\phi}}$, so that we obtain the energy as

$$E = \frac{\mu}{2}\left(\dot{r}^2 + r^2\dot{\phi}^2\right) + V(r) = \frac{\mu}{2}\dot{r}^2 + \frac{L^2}{2\mu r^2} + V(r), \tag{7.7}$$

where in the second step we used that the conserved angular momentum is $L = \mu r^2\dot{\phi}$. Hence, we have further reduced the problem to an effective one-dimensional dynamical problem involving a particle of mass μ, with position described by r, in the *effective potential*

$$V_{\text{eff}}(r) := \frac{L^2}{2\mu r^2} + V(r). \tag{7.8}$$

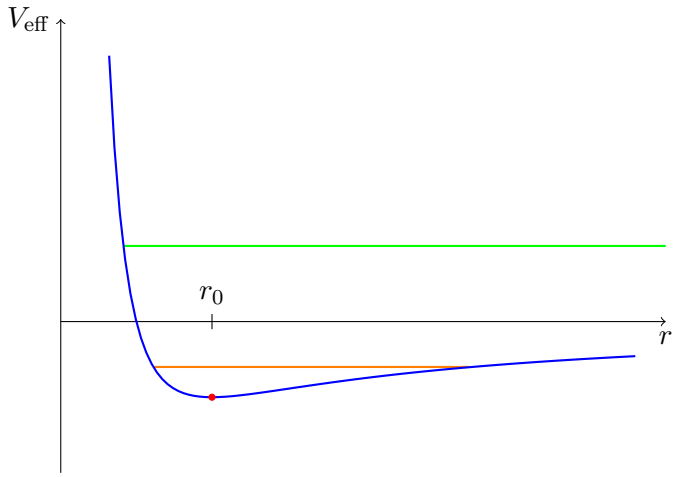


Figure 7.2: The effective potential for the Kepler problem. There is a stable equilibrium at r_0 (red dot), bound orbits for $E < 0$ (orange line), and unbounded orbits (green line).

The first term in the effective potential is called the *centrifugal barrier*, which ensures that (if $L \neq 0$ and the force is not too attractive) the particle cannot reach $r = 0$.

Up to now, we worked with a general central force, but it is both instructive and relevant to restrict to an attractive potential of the form $V(r) = -\frac{\alpha}{r}$, as it occurs in the Kepler problem (recall the gravitational potential (6.43)), or in the Coulomb problem (the hydrogen atom, for example). The resulting effective potential is shown in Figure 7.2. We see that the combination of an attractive long range (gravitational) and a short range repulsive (centrifugal barrier) potential leads to three different types of motion: The potential has a minimum at $r_0 = \frac{L^2}{\mu\alpha}$, i.e., a stable equilibrium with $E = E_{\min} := -\frac{\mu\alpha^2}{2L^2}$. Such a motion at a fixed radius of course corresponds to a circular orbit. Raising the energy above the minimum, for $-E_{\min} < E < 0$, we find bound periodic orbits, with radius r confined to a finite interval. We will see that these correspond to elliptic orbits (as we see them for the planets in the solar system). For $E > 0$, we have unbounded motion: The distance r decreases until it reaches a minimal value, and then again increases indefinitely. We will see that this corresponds to hyperbolic orbits, as they are observed for interstellar objects travelling through the solar system (such as Oumuamua in 2017 or the Voyager satellites).[1]

Let us now study the bound orbits ($E_{\min} < E < 0$) in detail. The two turning points $r_{\min}$ and $r_{\max}$ are given by the roots of

$$E - V_{\text{eff}}(r) = E + \frac{\alpha}{r} - \frac{L^2}{2\mu r^2}, \tag{7.9}$$

which is a quadratic polynomial in $y = r^{-1}$. Hence

$$r^{-1}_{\min/\max} = y_{\max/\min} = \frac{\alpha\mu}{L^2} \pm \sqrt{\frac{\alpha^2\mu^2}{L^4} + 2\frac{E\mu}{L^2}}. \tag{7.10}$$

The precise values are not particularly relevant, but it turns out to be useful to express $E - V_{\text{eff}}(r)$ as

$$E - V_{\text{eff}}(r) = \frac{L^2}{2\mu}(y - y_{\min})(y_{\max} - y). \tag{7.11}$$

[1] The boundary case $E = 0$ corresponds to a parabolic orbit.

We can then solve (7.7) for

$$\dot{r}^2 = \frac{2}{\mu}(E - V_{\text{eff}}(r)) = \frac{L^2}{\mu^2}(y - y_{\min})(y_{\max} - y). \tag{7.12}$$

It is now useful to consider r as a function of ϕ. Then

$$\dot{r} = \frac{\mathrm{d}}{\mathrm{d}t} r(\phi(t)) = \dot{\phi}\frac{\mathrm{d}}{\mathrm{d}\phi} r = \frac{L}{\mu r^2}\frac{\mathrm{d}}{\mathrm{d}\phi} r = -\frac{L}{\mu}\frac{\mathrm{d}}{\mathrm{d}\phi} y. \tag{7.13}$$

Inserting this in the above, we obtain

$$\left(\frac{\mathrm{d}}{\mathrm{d}\phi} y\right)^2 = (y - y_{\min})(y_{\max} - y). \tag{7.14}$$

Using $y_{\max/\min}$ as determined above, this can be rewritten, using

$$z = y - \frac{\alpha\mu}{L^2}, \qquad B = \sqrt{\frac{\alpha^2\mu^2}{L^4} + 2\frac{E\mu}{L^2}}, \tag{7.15}$$

as

$$\left(\frac{\mathrm{d}}{\mathrm{d}\phi} z\right)^2 = (z + B)(B - z) = B^2 - z^2. \tag{7.16}$$

We do not really have to integrate this, as the general solution can be guessed to be

$$z(\phi) = B\cos(\phi - \phi_0). \tag{7.17}$$

Choosing the coordinate system for simplicity such that $\phi_0 = 0$, we thus obtain, in terms of the more readily interpretable $r(\phi)$,

$$\frac{1}{r} = \frac{\alpha\mu}{L^2} + B\cos\phi = \frac{\alpha\mu}{L^2}(1 + \varepsilon\cos\phi), \tag{7.18}$$

with

$$\varepsilon = \left(\frac{\alpha\mu}{L^2}\right)^{-1} B = \sqrt{1 + 2\frac{EL^2}{\mu\alpha^2}}. \tag{7.19}$$

Even though we started our discussion by considering the turning points of closed orbits, the subsequent calculation is in fact valid for all kinds of orbits, i.e., (7.18) holds for all energies $E \geq E_{\min}$. It is only that the interpretation of $y_{\max/\min}$ changes for positive energy $E > 0$. While $y_{\max}$ is still positive (so that its inverse $r_{\min}$ is still a turning point), $y_{\min}$ is negative for $E > 0$. But negative values of y cannot be reached, as $r \to \infty$ for $y \to 0$. Hence, for $E \geq 0$, the values of y are confined to $(0, y_{\max}]$. This can also be seen from (7.18): For $E \geq 0$, we have $\varepsilon \geq 1$, so that the r.h.s. vanishes (corresponding to $r \to \infty$) as

$$\phi \to \pm\arccos(-\varepsilon^{-1}). \tag{7.20}$$

Let us now analyze separately the bound orbits, with $E < 0$ (implying $\varepsilon < 1$), and the unbound orbits, with $E \geq 0$ (implying $\varepsilon \geq 1$):

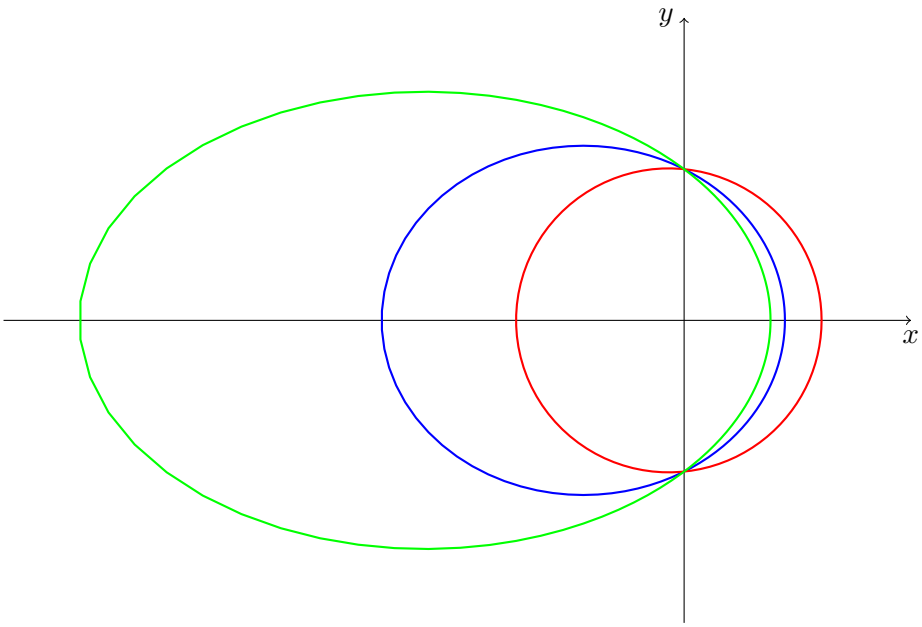


Figure 7.3: Elliptic orbits in the Kepler problem for $\varepsilon = \frac{1}{10}$ (red), $\varepsilon = \frac{1}{2}$ (blue), and $\varepsilon = \frac{3}{4}$ (green). In these plots, $\frac{\alpha\mu}{L^2}$ (i.e., angular momentum L) is kept fixed, while ε (i.e., energy E) is varied.

Bound orbits: To get a first impression, let us plot the bound orbits, i.e., consider the curve

$$\mathbf{r}(\phi) = \frac{L^2}{\alpha\mu}\frac{1}{1+\varepsilon\cos\phi}\begin{pmatrix}\cos\phi\\ \sin\phi\end{pmatrix} \tag{7.21}$$

in the $x-y$-plane. This is shown in Fig. 7.3 for different values of ε. We see that for small value of ε, the orbit is almost circular, while as ε approaches 1 the orbits become more and more eccentric. The minimal/maximal distance to the origin is attained at $r_{\min/\max} = k\frac{1}{1\pm\varepsilon}$ where $k = \frac{L^2}{\alpha\mu}$. One may easily check (homework) that the orbits are indeed ellipses with the origin one of the focal points.[2] The semi-major axis of the ellipse, i.e., the maximal distance from a point on the ellipse to the center, can thus be calculated in terms of $r_{\min/\max}$ as

$$a = \frac{1}{2}\left(r_{\min} + r_{\max}\right) = \frac{k}{1-\varepsilon^2}. \tag{7.22}$$

The parameter ε is the eccentricity of the ellipse.[3]

Once it is established that the orbit is an ellipse, it is straightforward to compute the period T of the motion. We already saw in Section 6.5 that it is related to the area A covered by the motion via $T = \frac{2\mu}{L}A$. The area of an ellipse is $A = \pi ab$, with b the semi-minor axis, which can be determined in terms of ε and a as $b = a\sqrt{1-\varepsilon^2}$. Hence,

$$T = \frac{2\pi\mu}{L}k^2(1-\varepsilon^2)^{-\frac{3}{2}} = \frac{2\pi\mu k^{\frac{1}{2}}}{L}\frac{k^{\frac{3}{2}}}{(1-\varepsilon^2)^{\frac{3}{2}}} = 2\pi\sqrt{\frac{\mu}{\alpha}}a^{\frac{3}{2}}. \tag{7.23}$$

[2] An ellipse can be defined by the property that for each point the sum of the distances to the two focal points is the same.

[3] The eccentricity is defined as the ratio between the distance from the center to a focal point and the semi-major axis.

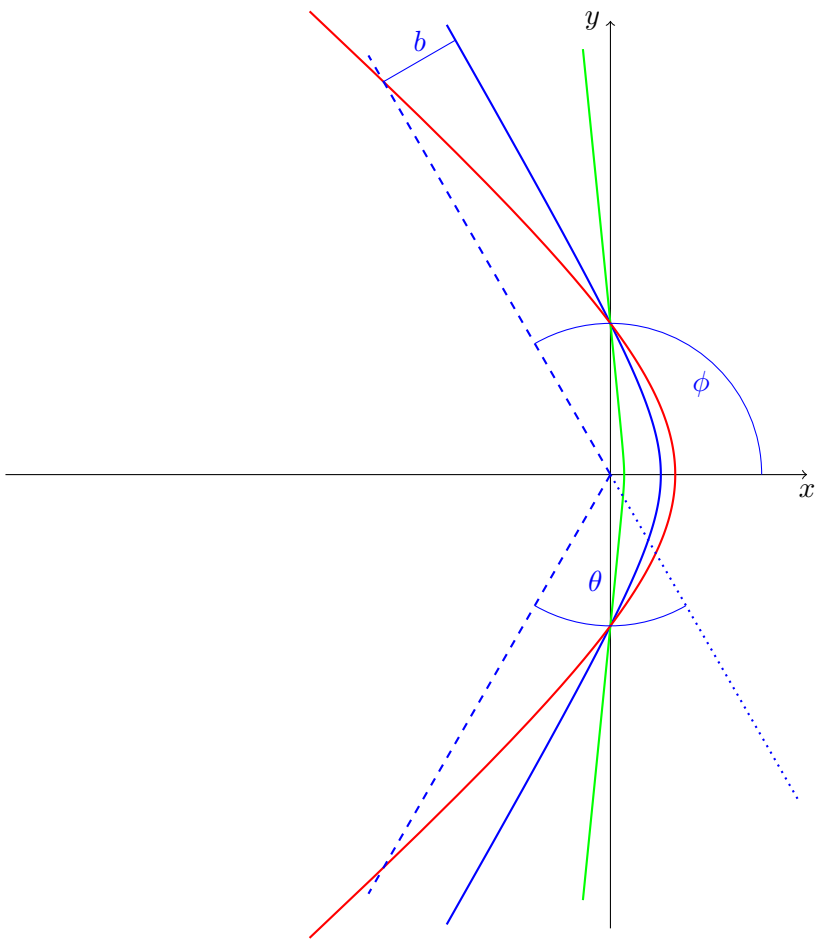


Figure 7.4: Hyperbolic orbits in the Kepler problem for $\varepsilon = \frac{4}{3}$ (red), $\varepsilon = 2$ (blue), and $\varepsilon = 10$ (green). In these plots, $\frac{\alpha\mu}{L^2}$ (i.e., angular momentum L) is kept fixed, while ε (i.e., energy E) is varied. For the blue orbit, also the asymptotic angles $\pm\phi$ are shown, as well as the construction of the scattering angle θ and the impact parameter b.

Unbounded orbits: Examples of unbounded orbits ($\varepsilon \geq 1$) are shown in Fig. 7.4 for different values of ε. We see that as ε increases, the curves become straighter, i.e., the body is less deflected by the gravitational potential. For $\varepsilon > 1$, the orbits are hyperbola with the origin as one of the focal points.[4] For $\varepsilon = 1$, the orbit is a parabola, again with the origin as focal point. In both cases, and as for the bound orbits, the closest distance to the origin is reached at $r_{\min} = k\frac{1}{1+\varepsilon}$.

An interesting quantity to consider for an unbound orbit is the *scattering angle* θ given by

$$\theta = 2\arccos(-\varepsilon^{-1}) - \pi, \tag{7.24}$$

c.f. Fig. 7.4 for the construction. A further relevant quantity is the *impact parameter*

$$b = \frac{L}{\mu v_\infty} = \frac{L}{\sqrt{2\mu E}}, \tag{7.25}$$

[4]The focal points of a hyperbola are such that for any point on the hyperbola, the difference of the distances to the two focal points is the same.

whose significance is also shown in Fig. 7.4. In this expression, v_∞ stands for the velocity in the limit $t \to \pm\infty$, which can be expressed as $v_\infty = \sqrt{2E/\mu}$, as the potential vanishes in that limit.

7.3 Discussion

Let us discuss consequences of the above findings and in particular compare with Kepler's laws of planetary motion. First of all, we see that if terms of $\mathcal{O}(\frac{m_2}{m_1})$ (i.e., the masses of the planets) are neglected, then Kepler's empirical laws can be derived from Newton's laws: We already saw that then $\mathbf{r}_2(t) = \mathbf{r}(t)$, which performs an ellipse with the origin as focal point. But as $\mathbf{r}_1(t) = \mathbf{0}$, this coincides with the sun's center. From this, and the discussion at the end of Section 6.5, then follows the second law. The third law then follows from

$$\frac{\mu}{\alpha} = \frac{m_1 m_2}{(m_1+m_2)Gm_1m_2} = \frac{1}{Gm_1}\left(1+\mathcal{O}(m_2/m_1)\right), \tag{7.26}$$

so that upon neglecting the planet's masses w.r.t. that of the sun, the proportionality factor $\sqrt{\mu/\alpha}$ between T and $a^{\frac{3}{2}}$ is indeed the same for all planets.

Now we want to investigate the modifications to Kepler's law brought about by not neglecting terms of $\mathcal{O}(\frac{m_2}{m_1})$. We consider the system consisting of the sun and a single planet. We have seen that $\mathbf{r}(t)$ traces out an ellipse with the coordinate origin as a focal point. As both $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$ are related to $\mathbf{r}(t)$ by a constant scale factor (recall (7.5)), the same is true for these, too. Hence, both the planet and the sun are on an elliptic orbit with the center of mass in a focal point. For the Jupiter-Sun system, this center of mass lies outside of the sun (at roughly a sun's radius distance from the surface). Is Kepler's statement that Jupiters orbit is an ellipse with the Sun in a focal point thus wrong? Yes and no. The answer depends on the coordinate frame used. In an inertial frame, the statement is wrong (even though, compared to the distance Sun-Jupiter, the deviation is only small). However, Kepler was using a heliocentric coordinate system, in which the sun is fixed to the center. As in an inertial frame, the sun is moving non-uniformly, this is a non-inertial frame (we will learn how to deal with non-inertial frames in a later chapter). But in this non-inertial frame, the distance vector $\mathbf{r}(t) = \mathbf{r}_2(t) - \mathbf{r}_1(t)$ actually describes an ellipse with the origin as focal point. Hence, in the coordinate system that Kepler used, the first law actually holds (for the system of a sun and a single planet).

A similar discussion also applies for the second law. The vector $\mathbf{r}_2(t)$ describing the position of the planet w.r.t. the center of mass is just a rescaling of $\mathbf{r}(t)$, for which we established the area law. Hence, the area law also holds in an inertial frame for the area covered by the vector $\mathbf{r}_2(t)$ (and analogously also for the vector $\mathbf{r}_1(t)$ describing the position of the sun). But the area law also holds in the heliocentric coordinate system, as it holds for $\mathbf{r}(t)$.

For the third law, however, it makes a substantial difference whether m_2 is negligible w.r.t. m_1 or not. In the relation (7.23), a refers to the semi-major axis of the ellipse traced out by $\mathbf{r}(t)$, i.e., it is the semi-major axis of the ellipse traced out by the planet in the heliocentric system. For a planet of mass half that of the sun, $m_2 = \frac{m_1}{2}$, the proportionality $\frac{\mu}{\alpha}$ between T^2 and a^3 is

$$\frac{\mu}{\alpha} = \frac{1}{m_1+m_2}\frac{1}{G} = \frac{2}{3}\frac{1}{Gm_1}, \tag{7.27}$$

i.e., it is reduced by a third w.r.t. the value for negligible planet mass m_2. This also shows that the Galilean equivalence principle has to be taken with a grain of salt: While the orbits of a satellite or an apple circling the Earth are practically indistinguishable (given identical initial conditions), this would not apply, for example, to the Moon: Its mass is about 1% that of Earth,

so that the proportionality factor $\frac{\mu}{\alpha}$ likewise differs at the % level from that of the apple or a satellite. Put differently: While the acceleration due to Earth's gravity of apple, satellite, and Moon are the same, the acceleration of Earth due to apple, satellite, and Moon differs. This affects the dynamics of the two-body system, perceptible for example in the period of orbital motion. Universality of free fall thus only holds for *test masses*, i.e., masses which are so small that their "backreaction" can be neglected.

Remark 7.1. *Using the expressions* (7.24) *and* (7.25)*, one can derive a prediction within Newtonian Gravity for the deflection of light by the sun, by considering the limit* $m_2 \to 0$ *of vanishing mass and identifying the velocity* v_∞ *at infinity with* c*. One find*

$$\theta \simeq 2\frac{GM_\odot}{bc^2}. \tag{7.28}$$

This was first derived by the bavarian astronomer, mathematician, and geometer von Soldner in 1804 (and rediscovered by Einstein in 1911). Within General Relativity, one finds a light deflection of twice this value (Einstein 1916). This first prediction of General Relativity was confirmed a few years later by Eddington during a solar eclipse.

Remark 7.2. *The fact that all bound orbits close, i.e., return to the same value of* r *after a turn* ($\phi \to \phi + 2\pi$)*, in the Kepler problem is very special. This only happens for the* $\frac{1}{r}$ *potential and the spherically symmetric harmonic oscillator potential* $V(r) = kr^2$*. For all other spherically symmetric potentials, the orbits do in general not close. One example for this is the Kepler problem in General Relativity, which can be brought to the same form radial form, but with a modified effective potential. The fact that the orbits then do not close manifests in the perihelion shift, i.e., the slow rotation of the point of closest distance to the sun. The explanation of the perihelion shift of Mercury was the first success of General Relativity.*

8 Rigid bodies

Up to now, we considered point particles, i.e., bodies with negligible extension. Now we want to study rigid bodies, which can be thought of as a collection of point particles whose arrangement is fixed in the following sense: For any pair of point particles, the relative position vectors $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ have constant (in time) length. The general motion of a rigid body is a combination of translation and rotation.

8.1 Rotations

We want to understand how to describe the rotation of a rigid body in a succinct manner. To motivate the following construction, we recall that the rotation by an angle α (in counter-clockwise direction) around the z-axis is described by the application of the rotation matrix[1]

$$R(\alpha) = \begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{8.1}$$

on the position vector $\mathbf{r}$ of a point particle. For the velocity of a particle at $\mathbf{r}$ due rotation at angular velocity ω, we now obtain

$$\mathbf{v} = \frac{\mathrm{d}}{\mathrm{d}t} R(\omega t)\mathbf{r}|_{t=0} = \frac{\mathrm{d}}{\mathrm{d}t} \begin{pmatrix} x\cos\omega t - y\sin\omega t \\ x\sin\omega t + y\cos\omega t \\ z \end{pmatrix} |_{t=0} = \omega \begin{pmatrix} -y \\ x \\ 0 \end{pmatrix}. \tag{8.2}$$

This can be neatly expressed using the vector product as

$$\mathbf{v} = (\omega\hat{\mathbf{z}}) \times \mathbf{r}. \tag{8.3}$$

This suggest that the velocity of a particle at $\mathbf{r}$ due to a rotation at angular velocity ω around some axis through the origin is given by

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}, \tag{8.4}$$

where the *angular velocity vector* $\boldsymbol{\omega}$ points along the axis of rotation and has length ω. One easily checks that this has the desired properties, i.e., it lies in the plane orthogonal to $\boldsymbol{\omega}$, is orthogonal to any vector connecting $\mathbf{r}$ and the rotation axis, and has magnitude $\rho\omega$, with ρ the distance to the rotation axis.

Consider now a rigid body consisting of point masses $m_1, m_2, \ldots$ with positions (at a given time) $\mathbf{r}_1, \mathbf{r}_2, \ldots$. The body is subject to a rotation described by the angular velocity vector $\boldsymbol{\omega}$. Then the total angular momentum is

$$\mathbf{L} = \sum_p m_p \mathbf{r}_p \times \mathbf{v}_p = \sum_p m_p \mathbf{r}_p \times (\boldsymbol{\omega} \times \mathbf{r}_p). \tag{8.5}$$

[1] Note that we are now considering an *active* rotation of the body, not a (passive) rotation of the coordinate system. This leads to the sign change w.r.t. (6.20).

The double vector product can be reexpressed using the baccab identity (3.55), so that

$$\mathbf{L} = \sum_p m_p \left(\boldsymbol{\omega} r_p^2 - \mathbf{r}_p (\boldsymbol{\omega} \cdot \mathbf{r}_p) \right). \tag{8.6}$$

It is convenient to rewrite this in components as

$$\begin{aligned} L_i &= \sum_p m_p \left(\omega_i r_p^2 - r_{p,i} \sum_{j=1}^3 \omega_j r_{p,j} \right) \\ &= \sum_p m_p \sum_{j=1}^3 \left(r_p^2 \delta_{ij} \omega_j - r_{p,i} r_{p,j} \omega_j \right) \\ &= \sum_{j=1}^3 I_{ij} \omega_j \end{aligned} \tag{8.7}$$

with the *inertia tensor*

$$I_{ij} = \sum_p m_p \left(r_p^2 \delta_{ij} - r_{p,i} r_{p,j} \right). \tag{8.8}$$

Seeing this as a matrix, which we can write as

$$I = \sum_p m_p \begin{pmatrix} r_p^2 - x_p^2 & -x_p y_p & -x_p z_p \\ -x_p y_p & r_p^2 - y_p^2 & -y_p z_p \\ -x_p z_p & -y_p z_p & r_p^2 - z_p^2 \end{pmatrix}, \tag{8.9}$$

we can express the angular as the matrix I acting on $\boldsymbol{\omega}$:

$$\mathbf{L} = I\boldsymbol{\omega}. \tag{8.10}$$

It is noteworthy that the angular momentum will in general not have the same direction as the angular velocity vector.

Example 8.1. *We compute the inertia tensor for the body consisting of two equal masses m at $\mathbf{r}_{1/2} = \pm L/2\hat{\mathbf{z}}$ (describing a massless rod of length L with two masses at the ends). For both positions, we have $r_{1/2}^2 = \frac{L^2}{4}$, so that the first term in (8.8) is $\frac{mL^2}{2}\mathbb{1}$. For the second term, we note that for both masses only the third component of $\mathbf{r}_{1/2}$ is nonzero, so that it contributes only to the 33 component. In total, we have*

$$I = \frac{mL^2}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \frac{mL^2}{2} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \frac{mL^2}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{8.11}$$

In particular, a rotation around the z-axis does not give rise to an angular momentum.

The inertia tensor also appears in the expression for the kinetic energy stored in the rotating body. We have

$$\begin{aligned} T &= \sum_p \frac{m_p}{2} |\mathbf{v}_p|^2 \\ &= \sum_k \frac{m_p}{2} (\boldsymbol{\omega} \times \mathbf{r}_p) \cdot (\boldsymbol{\omega} \times \mathbf{r}_p) \\ &= \sum_k \frac{m_p}{2} \left(\omega^2 r_p^2 - (\boldsymbol{\omega} \cdot \mathbf{r}_p)^2 \right), \end{aligned} \tag{8.12}$$

where we used the identity (3.56). We can rewrite this, using first (8.6) and then (8.10), as

$$T = \frac{1}{2}\sum_{i=1}^{3}\omega_i \sum_p m_p \left(r_p^2\omega_i - r_{p,i}(\boldsymbol{\omega}\cdot\mathbf{r}_p)\right) = \frac{1}{2}\boldsymbol{\omega}\cdot\mathbf{L} = \frac{1}{2}\boldsymbol{\omega}\cdot I\boldsymbol{\omega}. \tag{8.13}$$

We see that the kinetic energy for the rotation has a similar form as the one for translation, with the angular velocity vector $\boldsymbol{\omega}$ playing the role of the velocity and the inertia tensor I playing the role of the mass.

Up to now, we considered rotations around the origin and purely rotational motion. To describe general motion, it is convenient use the center of mass (cm) $\mathbf{R}$, its velocity $\mathbf{V}$, the relative positions $\mathbf{r}'_p = \mathbf{r}_p - \mathbf{R}$ of the particles w.r.t. the cm, and the angular velocity vector $\boldsymbol{\omega}$ describing the rotation around an axis through the cm. Then the velocity of each particle is

$$\mathbf{v}_p = \mathbf{V} + \boldsymbol{\omega}\times\mathbf{r}'_p, \tag{8.14}$$

i.e., the sum of the cm velocity and the velocity due to the rotation around the cm. The total angular momentum is then

$$\begin{aligned}\mathbf{L} &= \sum_p m_p\mathbf{r}_p\times\mathbf{v}_p\\ &= \sum_p m_p(\mathbf{R}+\mathbf{r}'_p)\times(\mathbf{V}+\boldsymbol{\omega}\times\mathbf{r}'_p)\\ &= M\mathbf{R}\times\mathbf{V} + \sum_p m_p\mathbf{r}'_p\times(\boldsymbol{\omega}\times\mathbf{r}'_p) + \sum_p m_p\mathbf{r}'_p\times\mathbf{V} + \mathbf{R}\times\left(\boldsymbol{\omega}\times\sum_p m_p\mathbf{r}'_p\right).\end{aligned} \tag{8.15}$$

The last two terms vanish, as the cm is at the origin in the cm system. Hence,

$$\mathbf{L} = M\mathbf{R}\times\mathbf{V} + I_{cm}\boldsymbol{\omega}, \tag{8.16}$$

with I_{cm} the inertia tensor as calculated in the cm system.

Exercise 8.1. *Show that one similarly obtains for the kinetic energy*

$$T = \frac{1}{2}MV^2 + \frac{1}{2}\boldsymbol{\omega}\cdot I_{cm}\boldsymbol{\omega}. \tag{8.17}$$

Often, it is convenient to describe a rigid body not as a collection of point particles, but by a density $\rho(\mathbf{r})$, i.e., the integral of $\rho(\mathbf{r})$ over a volume V gives the total mass contained inside the volume. The expression for the inertia tensor then becomes

$$I_{ij} = \int \rho(\mathbf{r})\left(r^2\delta_{ij} - r_i r_j\right)\mathrm{d}^3 r. \tag{8.18}$$

Some general properties of the inertia tensor (which are also useful in concrete calculations) are the following: By construction, the inertia tensor is symmetric, i.e., $I_{ij} = I_{ji}$. Hence, only six of its nine components are independent and need to be computed. This number is further decreased for bodies with some symmetry. Two types of symmetries will be particularly relevant to us:

- Invariance under the reflection of a single coordinate r_k. In terms of a density $\rho(\mathbf{r})$, this means that $\rho(R_k\mathbf{r}) = \rho(\mathbf{r})$, with R_k the matrix performing the reflection. Let $\mathbf{r}' = R_k\mathbf{r}$. Then

$$\begin{aligned} I_{ij} &= \int \rho(R_k\mathbf{r}) \left(r^2\delta_{ij} - r_i r_j\right) \mathrm{d}^3 r \\ &= \int \rho(\mathbf{r}') \left(r'^2\delta_{ij} - (R_k r')_i (R_k r')_j\right) \mathrm{d}^3 r' \\ &= \begin{cases} I_{ij} & i = j \text{ or } i \neq k, j \neq k \\ -I_{ij} & i = k, j \neq k \text{ or } j = k, i \neq k. \end{cases} \end{aligned} \tag{8.19}$$

  Here we used $\rho(\mathbf{r}) = \rho(R_k\mathbf{r})$, $R_k\mathbf{r}' = \mathbf{r}$, and the fact that the reflection does not change the length of a vector. We see that $I_{ij} = 0$ for $i = k \neq j$ and for $j = k \neq i$. For example, for a body symmetric under reflection along the z-axis, we know that $I_{13} = I_{23} = 0$.

- Invariance under rotations in the $k - l$-plane by $\frac{\pi}{2}$. Under such a rotation, $r'_k = \pm r_l$, $r'_l = \mp r_k$. By an argument analogous to the above, it follows that the inertia tensor is diagonal, $I_{ij} = 0$ for $i \neq j$, and that $I_{kk} = I_{ll}$. Hence, for a body invariant under $\frac{\pi}{2}$ rotations around the z-axis, the inertia tensor is of the form

$$I = \begin{pmatrix} I_{11} & 0 & 0 \\ 0 & I_{11} & 0 \\ 0 & 0 & I_{33} \end{pmatrix}, \tag{8.20}$$

  i.e., only two components of the inertia tensor actually need to be computed.

Example 8.2. *We compute the inertia tensor for a homogeneous cylinder of mass M, radius R, and height h, centered at the origin, and symmetry axis aligned to the z-axis. The density is thus (we use the notation ϱ to avoid confusion with the cylinder coordinate ρ)*

$$\varrho(\mathbf{r}) = \begin{cases} \frac{M}{\pi R^2 h} & \rho < R, |z| < \frac{h}{2}, \\ 0 & \textit{otherwise}, \end{cases} \tag{8.21}$$

where we use cylinder coordinates ρ, ϕ, z. By rotational symmetry, the inertia tensor is of the form (8.20). We begin by calculating

$$I_{33} = \int \varrho(\mathbf{r})(r^2 - z^2)\mathrm{d}^3 r \tag{8.22}$$

$$= \frac{M}{\pi R^2 h} \int_0^R \mathrm{d}\rho \rho^3 \int_0^{2\pi} \mathrm{d}\phi \int_{-h/2}^{h/2} \mathrm{d}z \tag{8.23}$$

$$= \frac{MR^2}{2}, \tag{8.24}$$

where we used $r^2 - z^2 = \rho^2$ and the volume element $\rho \mathrm{d}\rho \mathrm{d}\phi \mathrm{d}z$ in cylinder coordinates. For the other component, it is convenient to write $I_{11} = \frac{1}{2}(I_{11} + I_{22})$, so that

$$I_{11} = \frac{1}{2} \int \varrho(\mathbf{r})(2r^2 - x^2 - y^2)\mathrm{d}^3 r \tag{8.25}$$

$$= \frac{M}{2\pi R^2 h} \left[\int_0^R \mathrm{d}\rho \rho^3 \int_0^{2\pi} \mathrm{d}\phi \int_{-h/2}^{h/2} \mathrm{d}z + 2 \int_0^R \mathrm{d}\rho \rho \int_0^{2\pi} \mathrm{d}\phi \int_{-h/2}^{h/2} \mathrm{d}z z^2 \right] \tag{8.26}$$

$$= \frac{MR^2}{4} + \frac{Mh^2}{12}. \tag{8.27}$$

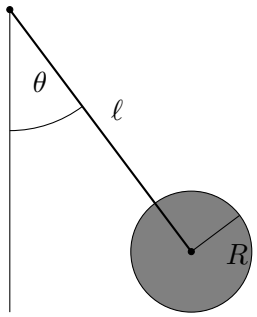


Figure 8.1: A clock pendulum. The disk is fixed to the rod, i.e., it rotates along with it.

We saw above that angular momentum and kinetic energy under rotation and translation is conveniently described in terms of the total mass M, the center of mass $\mathbf{R}$, and the inertia tensor I_{cm} computed in the center of mass frame defined by $\mathbf{r}' = \mathbf{r} - \mathbf{R}$. However, it is sometimes also convenient to work with an inertia tensor I computed w.r.t. the coordinates $\mathbf{r}$, i.e., not in the center of mass frame (one such example, the heavy top, is discussed below). We recall that

$$I = \sum_p m_p \begin{pmatrix} y_p^2 + z_p^2 & -x_p y_p & -x_p z_p \\ -x_p y_p & x_p^2 + z_p^2 & -y_p z_p \\ -x_p z_p & -y_p z_p & x_p^2 + y_p^2 \end{pmatrix}. \tag{8.28}$$

Replacing $x_p = x'_p + X$ (with X, Y, Z denoting the components of $\mathbf{R}$) and analogously for the other coordinates and multiplying out the products, we get three different terms: The terms quadratic in x'_p, y'_p, z'_p, the terms linear in x'_p, y'_p, z'_p, and the terms independent of x'_p, y'_p, z'_p. By the assumption that the center of mass is at the origin of the primed coordinates, we have $\sum_p m_p \mathbf{r}'_p = \mathbf{0}$, so the terms linear in x'_p, y'_p, z'_p cancel out. We are then left with

$$I = I_{cm} + I_{\mathbf{R}} \tag{8.29}$$

with

$$I_{\mathbf{R}} = M \begin{pmatrix} Y^2 + Z^2 & -XY & -XZ \\ -XY & X^2 + Z^2 & -YZ \\ -XZ & -YZ & X^2 + Y^2 \end{pmatrix} \tag{8.30}$$

having the form of the inertia tensor of a particle of mass M at $\mathbf{R}$. The relation (8.29) is also known as the (tensorial version of the) *parallel axis theorem*.

Exercise 8.2. *Compute the angular frequency of small oscillation of the "clock pendulum" sketched in Fig. 8.1. The distance ℓ is measured from the suspension point to the center of the disk. The rod is assumed to be massless and the disk fixed to it (so that it rotates with the rod). Hint: Use (8.29) and the results of Example 8.2 to define the total energy (taking the gravitational potential into account). A time derivative yields equations of motion and by linearization for small θ the oscillation frequency can be determined.*

8.2 Principal axes and Euler angles

As directly follows from its definition, the inertia tensor is symmetric. It follows that one can find an ONB $\{\hat{\boldsymbol{\omega}}_1, \hat{\boldsymbol{\omega}}_2, \hat{\boldsymbol{\omega}}_3\}$ of eigenvectors, i.e., vectors such that

$$I\hat{\boldsymbol{\omega}}_i = I_i \hat{\boldsymbol{\omega}}_i, \qquad \hat{\boldsymbol{\omega}}_i \cdot \hat{\boldsymbol{\omega}}_j = \delta_{ij}. \tag{8.31}$$

The $\hat{\boldsymbol{\omega}}_i$ are called the *principal axes* and the corresponding eigenvalues I_i the *principal moments*. It follows from the above results that if the body is invariant under rotations by $\frac{\pi}{2}$ around any of the coordinate axes, then the coordinate axes are already principal axes.

It is often convenient to express the vectors describing the rotation in terms of the ONB provided by the principal axes, for example expressing the angular velocity vector $\boldsymbol{\omega}$ as

$$\boldsymbol{\omega} = \omega_1 \hat{\boldsymbol{\omega}}_1 + \omega_2 \hat{\boldsymbol{\omega}}_2 + \omega_3 \hat{\boldsymbol{\omega}}_3. \tag{8.32}$$

The corresponding angular momentum is then (expressed in the same basis)

$$\mathbf{L} = I\boldsymbol{\omega} = \omega_1 I_1 \hat{\boldsymbol{\omega}}_1 + \omega_2 I_2 \hat{\boldsymbol{\omega}}_2 + \omega_3 I_3 \hat{\boldsymbol{\omega}}_3. \tag{8.33}$$

In general, this will not point in the same direction as $\boldsymbol{\omega}$. However, if the angular momentum $\boldsymbol{\omega}$ is aligned with one of the principal axes, i.e., $\boldsymbol{\omega} = \omega \hat{\boldsymbol{\omega}}_i$, then

$$\mathbf{L} = \omega I_i \hat{\boldsymbol{\omega}}_i \tag{8.34}$$

has the same direction as the angular velocity vector. In terms of the coefficient ω_1, ω_2, ω_3 also the rotational kinetic energy can be neatly expressed as

$$\frac{1}{2}\boldsymbol{\omega} \cdot I\boldsymbol{\omega} = \frac{1}{2}\left(I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2\right). \tag{8.35}$$

Example 8.3. *When a ball or a cylinder of radius R is rolling, its angular velocity is proportional to the cm velocity, $\omega = V/R$. If I_r is the principal moment corresponding to the rotation axis, then the total kinetic energy is*

$$T = \frac{1}{2}MV^2 + \frac{1}{2}I_r\omega^2 = \frac{1}{2}\left(M + \frac{I_r}{R^2}\right)V^2. \tag{8.36}$$

Hence, when a ball or a cylinder is rolling down a slope, the velocity it acquires by rolling down a height h is

$$V = \sqrt{\frac{2Mgh}{M + I_r/R^2}}, \tag{8.37}$$

which is smaller than the value $v = \sqrt{2gh}$ obtained for a point particle. The reason is that the available potential energy is shared between the translational and rotational kinetic energy.

The ONB consisting of $\hat{\boldsymbol{\omega}}_1, \hat{\boldsymbol{\omega}}_2, \hat{\boldsymbol{\omega}}_3$ can be seen as a body-fixed frame, as it rotates with the body. Each of the three vector has three components, but due to the fact that they are normalized and orthogonal, there is a huge redundancy in the description of the frame in terms of these vectors. In fact, the frame consisting of principal axes (any right orthonormal frame, in fact), can be described by the three *Euler angles* θ, ϕ, ψ. One uses $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$ to parametrize the unit vector $\hat{\boldsymbol{\omega}}_3$ as

$$\hat{\boldsymbol{\omega}}_3 = \begin{pmatrix} \sin\phi \sin\theta \\ -\cos\phi \sin\theta \\ \cos\theta \end{pmatrix} \tag{8.38}$$

in terms of some fixed basis $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$. So θ is the angle between $\hat{\boldsymbol{\omega}}_3$ and $\hat{\mathbf{z}}$ and ϕ is the angle between the $\hat{\mathbf{x}}$ axis and the axis

$$\hat{\mathbf{N}} = \begin{pmatrix} \cos\phi \\ \sin\phi \\ 0 \end{pmatrix}, \tag{8.39}$$

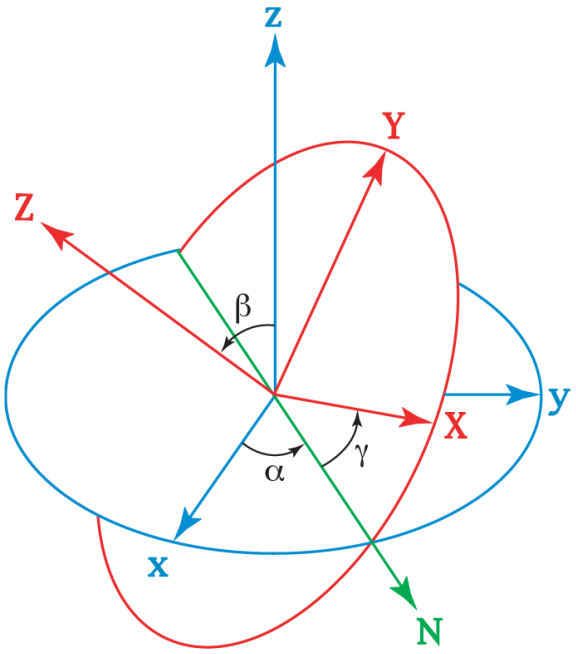


Figure 8.2: Illustration of the Euler angles. To adapt to our notation, replace $\alpha \to \phi$, $\beta \to \theta$, $\gamma \to \psi$, $\mathbf{Z} \to \hat{\boldsymbol{\omega}}_3$, $\mathbf{X} \to \hat{\boldsymbol{\omega}}_1$, and $\mathbf{Y} \to \hat{\boldsymbol{\omega}}_2$. Source: User:PAR via Wikipedia

which is the axis in the $x-y$ plane such that rotation around this axis by θ maps $\hat{\mathbf{z}}$ to $\hat{\boldsymbol{\omega}}_3$, c.f. Fig. 8.2. The angle ψ is now the angle by which one has to rotate around $\hat{\boldsymbol{\omega}}_3$ in order to map $\hat{\mathbf{N}}$ to $\hat{\boldsymbol{\omega}}_1$. With this prescription, one can also express $\hat{\boldsymbol{\omega}}_1$ and $\hat{\boldsymbol{\omega}}_2$ in terms of θ, ϕ, ψ, but this explicit form is not very illuminating and not needed for our purposes. What we will use, however, is the explicit representation of the angular velocity vector $\boldsymbol{\omega}$ in terms of derivatives of the angles,

$$\boldsymbol{\omega} = \dot{\phi}\hat{\mathbf{z}} + \dot{\theta}\hat{\mathbf{N}} + \dot{\psi}\hat{\boldsymbol{\omega}}_3, \tag{8.40}$$

which follows from the fact that the frame rotates in the same manner as the body and the explicit description of the construction of the frame in terms of Euler angles given above.

8.3 Euler's equations

We now want to study how the angular velocity vector $\boldsymbol{\omega}$ and the angular momentum $\mathbf{L}$ change in time. We first compute

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{L} = \sum_p m_p \left(\dot{\mathbf{r}}_p \times \dot{\mathbf{r}}_p + \mathbf{r}_p \times \ddot{\mathbf{r}}_p\right) = \sum_p \mathbf{r}_p \times \mathbf{F}_p =: \boldsymbol{\tau}. \tag{8.41}$$

with $\mathbf{F}_p$ the force acting on the mass m_p. We call $\boldsymbol{\tau}$ the *torque*.

It is now convenient to express $\mathbf{L}$ in terms of the principal axes, i.e., by (8.33). However, one has to keep in mind that the principal axes are rotating with the body, i.e.,

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\boldsymbol{\omega}}_i = \boldsymbol{\omega} \times \hat{\boldsymbol{\omega}}_i = \sum_{j=1}^{3} \omega_j \hat{\boldsymbol{\omega}}_j \times \hat{\boldsymbol{\omega}}_i. \tag{8.42}$$

The ONB $\{\hat{\boldsymbol{\omega}}_1, \hat{\boldsymbol{\omega}}_2, \hat{\boldsymbol{\omega}}_3\}$ can be assumed to be right-handed (by flipping signs, if necessary), so that

$$\hat{\boldsymbol{\omega}}_1 \times \hat{\boldsymbol{\omega}}_2 = \hat{\boldsymbol{\omega}}_3, \qquad \hat{\boldsymbol{\omega}}_2 \times \hat{\boldsymbol{\omega}}_3 = \hat{\boldsymbol{\omega}}_1, \qquad \hat{\boldsymbol{\omega}}_3 \times \hat{\boldsymbol{\omega}}_1 = \hat{\boldsymbol{\omega}}_2. \tag{8.43}$$

Hence,

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\boldsymbol{\omega}}_1 = \omega_3\hat{\boldsymbol{\omega}}_2 - \omega_2\hat{\boldsymbol{\omega}}_3, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\hat{\boldsymbol{\omega}}_2 = \omega_1\hat{\boldsymbol{\omega}}_3 - \omega_3\hat{\boldsymbol{\omega}}_1, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\hat{\boldsymbol{\omega}}_3 = \omega_2\hat{\boldsymbol{\omega}}_1 - \omega_1\hat{\boldsymbol{\omega}}_2. \tag{8.44}$$

We thus obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{L} = \sum_{i=1}^{3}\left(I_i\dot{\omega}_i\hat{\boldsymbol{\omega}}_i + I_i\omega_i\frac{\mathrm{d}}{\mathrm{d}t}\hat{\boldsymbol{\omega}}_i\right) \tag{8.45}$$

$$= \left(I_1\dot{\omega}_1 - I_2\omega_2\omega_3 + I_3\omega_3\omega_2\right)\hat{\boldsymbol{\omega}}_1 + \left(I_2\dot{\omega}_2 - I_3\omega_3\omega_1 + I_1\omega_1\omega_3\right)\hat{\boldsymbol{\omega}}_2 + \left(I_3\dot{\omega}_3 - I_1\omega_1\omega_2 + I_2\omega_2\omega_1\right)\hat{\boldsymbol{\omega}}_3. \tag{8.46}$$

Likewise decomposing $\boldsymbol{\tau}$ as

$$\boldsymbol{\tau} = \tau_1\hat{\boldsymbol{\omega}}_1 + \tau_2\hat{\boldsymbol{\omega}}_2 + \tau_3\hat{\boldsymbol{\omega}}_3, \tag{8.47}$$

we thus obtain *Euler's equations*

$$I_1\dot{\omega}_1 + (I_3 - I_2)\omega_2\omega_3 = \tau_1, \quad I_2\dot{\omega}_2 + (I_1 - I_3)\omega_3\omega_1 = \tau_2, \quad I_3\dot{\omega}_3 + (I_2 - I_1)\omega_1\omega_2 = \tau_3. \tag{8.48}$$

Euler's equations are the rotational analog of Newton's equation $m\dot{\mathbf{v}} = \mathbf{F}$ for translational motion. There is an important difference, though: Unless $\boldsymbol{\omega} \times \mathbf{L} = \mathbf{0}$ (which happens iff $\mathbf{L}$ and $\boldsymbol{\omega}$ are aligned), there are additional terms on the l.h.s. of these equations, quadratic in the angular velocity. One can understand them as a consequence of the fact, if $\boldsymbol{\omega}$ and $\mathbf{L}$ are not aligned, the components of $\mathbf{L}$ w.r.t. the ONB $\{\hat{\boldsymbol{\omega}}_1, \hat{\boldsymbol{\omega}}_2, \hat{\boldsymbol{\omega}}_3\}$ change on account of the rotation of these basis vectors (even in the absence of a torque).

Remark 8.1. *It may seem odd that we have a first order differential equation, instead of a second order differential equation typically encountered in Mechanics. However, also Newtonian Mechanics can be written in first order form (as we discussed in Section 4.2). The equation $m\dot{\mathbf{v}} = \mathbf{F}$ is of first order. The point is that $\mathbf{F}$ will typically depend on the position $\mathbf{r}$, so we also need an evolution equation for this, which is provided by $\dot{\mathbf{r}} = \mathbf{v}$. One thus arrives at a system of first order equations. Similarly, in the rotational case, the torque will in general depend on the orientation of the body, which is tracked by the principal axes $\{\hat{\boldsymbol{\omega}}_1, \hat{\boldsymbol{\omega}}_2, \hat{\boldsymbol{\omega}}_3\}$. Hence, to solve Euler's equations (8.48) in such a case, they have to be supplemented with equations describing the time evolution of these, namely (8.44). This is most conveniently done in terms of Euler angles, using (8.40).*

Example 8.4. *Obviously, in the absence of torque, the rotation around one of the principal axes, so that, for example $\omega_2 = \omega_3 = 0$, $\omega_1 = \Omega = \mathrm{const}$, is a solution to Euler's equations (8.48). If the principal moments are non-degenerate, i.e., pairwise unequal, then these are the only stationary (time-independent) solutions. We want to study the stability of such a rotation, i.e., we consider*

$$\begin{pmatrix}\omega_1(t)\\ \omega_2(t)\\ \omega_3(t)\end{pmatrix} = \begin{pmatrix}\Omega\\ 0\\ 0\end{pmatrix} + \begin{pmatrix}\delta\omega_1(t)\\ \delta\omega_2(t)\\ \delta\omega_3(t)\end{pmatrix}. \tag{8.49}$$

We consider $\delta\omega_i$ as small (compared to Ω), so we expand Euler's equations only up to linear order in $\delta\omega$ (neglecting terms of $\mathcal{O}((\delta\omega)^2)$). The first Euler equation is then

$$I_1\delta\dot{\omega}_1 = 0 + \mathcal{O}((\delta\omega)^2), \tag{8.50}$$

which we can solve, neglecting the terms of $\mathcal{O}((\delta\omega)^2)$, by $\delta\omega_1 = 0$. The equations for $\delta\omega_2$ and $\delta\omega_3$ are coupled:

$$\delta\dot{\omega}_2 + \frac{I_1 - I_3}{I_2}\Omega\delta\omega_3 = \mathcal{O}((\delta\omega)^2), \qquad \delta\dot{\omega}_3 + \frac{I_2 - I_1}{I_3}\Omega\delta\omega_2 = \mathcal{O}((\delta\omega)^2). \tag{8.51}$$

Neglecting the terms of $\mathcal{O}((\delta\omega)^2)$, we can write this in matrix form as

$$\dot{\boldsymbol{\delta\omega}} + B\boldsymbol{\delta\omega} = \mathbf{0}, \tag{8.52}$$

where

$$\boldsymbol{\delta\omega} = \begin{pmatrix}\delta\omega_2\\ \delta\omega_3\end{pmatrix}, \qquad B = \Omega\begin{pmatrix}0 & \frac{I_1-I_3}{I_2}\\ \frac{I_2-I_1}{I_3} & 0\end{pmatrix}. \tag{8.53}$$

This is a linear equation, so we use the "normal mode" ansatz

$$\boldsymbol{\delta\omega}(t) = \boldsymbol{\delta\omega}e^{-i\nu t}, \tag{8.54}$$

leading to

$$(-i\nu\mathbb{1} + B)\,\boldsymbol{\delta\omega} = \mathbf{0}. \tag{8.55}$$

This can also be solved non-trivially if the matrix multiplying $\boldsymbol{\delta\omega}$ has vanishing determinant, i.e.,

$$0 = \det(-i\nu\mathbb{1} + B) = -\nu^2 - \Omega^2\frac{(I_1 - I_3)(I_2 - I_1)}{I_2 I_3}. \tag{8.56}$$

It follows that if $(I_1 - I_3)(I_1 - I_2) \geq 0$, then ν is real and the rotation about the principal axis $\hat{\boldsymbol{\omega}}_1$ is stable. This is the case when I_1 is either the greatest or the smallest of the principal moments. On the other hand, if I_1 is smaller than the largest and greater than the smallest principal moments, i.e., $(I_1 - I_3)(I_1 - I_2) < 0$, then

$$\nu = \pm i\Omega\sqrt{\frac{(I_1 - I_3)(I_2 - I_1)}{I_2 I_3}}. \tag{8.57}$$

As one of the "normal modes" has positive imaginary part, the rotation around such a principal axis is unstable.

8.4 The symmetric top

Major simplifications in Euler's equation occur for a "symmetric top", i.e., a rigid body invariant under rotations around an axis $\hat{\boldsymbol{\omega}}_3$. As we saw above, the other two principal moments then degenerate, $I_1 = I_2 = I$. This means that we can choose any two orthogonal unit vectors $\hat{\boldsymbol{\omega}}_1$, $\hat{\boldsymbol{\omega}}_2$ orthogonal to $\hat{\boldsymbol{\omega}}_3$ as principal axes.

We begin by considering the free top, i.e., in the absence of external forces, and in particular torque. The third Euler equation reduces to $\dot{\omega}_3 = 0$, so the angular velocity of the rotation around the symmetry axis is constant. The other two angular velocity components then fulfill

$$\dot{\omega}_1 = -\Omega\omega_2, \tag{8.58}$$
$$\dot{\omega}_2 = \Omega\omega_1, \tag{8.59}$$

where

$$\Omega = \omega_3 \left(\frac{I_3}{I} - 1 \right). \tag{8.60}$$

We could write this in matrix form and solve it with a normal mode ansatz, but we can also directly see that this is solved by

$$\omega_1 = A\cos(\Omega t - \phi), \qquad \omega_2 = A\sin(\Omega t - \phi) \tag{8.61}$$

Hence, in a frame fixed to the body, the angular velocity vector rotates around the symmetry axis with an angular velocity Ω. Such a motion is called a *precession*. The velocity Ω can be positive ($I_3 > I$, called an oblate top) or negative ($I_3 < I$, called a prolate top). In the first case, the motion in the $\omega_1 - \omega_2$ plane is counter-clockwise, in the second case it is clockwise.

Remark 8.2. *Based on the ellipsoidal form of Earth, Euler predicted a period for the precession of the angular velocity vector of 305 days. Modern measurements of Earth's principal moments suggest a period of 320 days. The actual period, found by the American astronomer Chandler (and thus called* Chandler wobble*) is 433 days. The deviation is due to Earth not being rigid.*[2]

Above, we have worked in a body-fixed frame. As this frame rotates, the components of the angular momentum vector expressed in this frame will do likewise:

$$\mathbf{L} = I\left(\omega_1\hat{\boldsymbol{\omega}}_1 + \omega_2\hat{\boldsymbol{\omega}}_2\right) + I_3\omega_3\hat{\boldsymbol{\omega}}_3. \tag{8.62}$$

On the other hand, in an inertial frame, $\mathbf{L}$ is constant (as no torque is applied). We can use this to study the evolution of ω in an inertial frame. Namely, we write the above as

$$\mathbf{L} = I\boldsymbol{\omega} + (I_3 - I)\omega_3\hat{\boldsymbol{\omega}}_3, \tag{8.63}$$

which can be solved for

$$\boldsymbol{\omega} = \frac{1}{I}\mathbf{L} - \Omega\hat{\boldsymbol{\omega}}_3. \tag{8.64}$$

For the motion of $\hat{\boldsymbol{\omega}}_3$, we thus obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\boldsymbol{\omega}}_3 = \boldsymbol{\omega} \times \hat{\boldsymbol{\omega}}_3 = \frac{1}{I}\mathbf{L} \times \hat{\boldsymbol{\omega}}_3. \tag{8.65}$$

The angular momentum being fixed, this means that the symmetry axis $\hat{\boldsymbol{\omega}}_3$ precesses around the axis defined by $\mathbf{L}$ with angular velocity $\frac{L}{I}$.

We now consider the "heavy symmetric top", i.e., a body as discussed above which is fixed at a point P on the symmetry axis (by a pivot to the table, for example), and subject to gravity. This becomes much simpler if we assume that the angle θ between the symmetry axis $\hat{\boldsymbol{\omega}}_3$ and

[2] This is analogous to the period of a damped oscillator being longer than that of an undamped oscillator. The amplitude of the oscillation is rather small (corresponding to a shift of the poles of about 10 meters), but does (despite damping) not die out as it is constantly excited by atmospheric, oceanic, and geological processes.

the vertical axis $\hat{\mathbf{z}}$ (along which gravity acts) is constant. We thus first concentrate on this case. To get a first idea, we assume that the angular momentum is aligned with the symmetry axis, i.e., $\mathbf{L} = L\hat{\boldsymbol{\omega}}_3 = I_3\omega_3\hat{\boldsymbol{\omega}}_3$. We will see that this is not self-consistent, but it gives us some intuition about the dynamics, which we then use to correct the analysis. As we saw, the torque on a rigid due to gravity can be expressed as $\boldsymbol{\tau} = M\mathbf{R} \times \mathbf{g}$, with $\mathbf{R}$ the position of the center of mass and $\mathbf{g}$ the gravitational acceleration vector. Denoting by d the distance of the center of mass to the pivot point so that $\mathbf{R} = d\hat{\boldsymbol{\omega}}_3$, we can write

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{L} = \boldsymbol{\tau} = \frac{Mgd}{I_3\omega_3}\hat{\mathbf{z}} \times \mathbf{L}. \tag{8.66}$$

We see that $\mathbf{L}$ precesses with angular velocity $\Omega = \frac{Mgd}{I_3\omega_3}$ around the z axis.

The problem with the above derivation is that due to the precession, the angular velocity will not be aligned with the symmetry axis of the top, so that the assumption we based our analysis upon is not fulfilled. However, we will recover the above result as a certain limit of the correct result, which we derive now.

To properly treat the heavy symmetric top (with constant θ), we now consider the angular velocity vector to be a linear combination of the symmetry axis $\hat{\boldsymbol{\omega}}_3$ and the precession axis $\hat{\mathbf{z}}$:

$$\boldsymbol{\omega} = \tilde{\omega}_3\hat{\boldsymbol{\omega}}_3 + \Omega\hat{\mathbf{z}}. \tag{8.67}$$

The second term implements that (with $\frac{\mathrm{d}}{\mathrm{d}t}\hat{\boldsymbol{\omega}}_3 = \boldsymbol{\omega} \times \hat{\boldsymbol{\omega}}_3$) the symmetry axis precesses with angular velocity Ω. The coefficient of the first term was denoted $\tilde{\omega}_3$ to distinguish it from the coefficient ω_3 of $\boldsymbol{\omega}$ w.r.t. a body fixed frame. We now exploit the fact that due to the symmetry of the top, we can choose the principal axis $\hat{\boldsymbol{\omega}}_2$ freely in the plane orthogonal to $\hat{\boldsymbol{\omega}}_3$. We choose it to lie in the plane spanned by $\hat{\boldsymbol{\omega}}_3$ and $\hat{\mathbf{z}}$, so that $\hat{\mathbf{z}} = \cos\theta\hat{\boldsymbol{\omega}}_3 - \sin\theta\hat{\boldsymbol{\omega}}_2$. Then

$$\boldsymbol{\omega} = (\tilde{\omega}_3 + \Omega\cos\theta)\,\hat{\boldsymbol{\omega}}_3 - \Omega\sin\theta\hat{\boldsymbol{\omega}}_2 =: \omega_3\hat{\boldsymbol{\omega}}_3 - \Omega\sin\theta\hat{\boldsymbol{\omega}}_2, \tag{8.68}$$

where we defined ω_3. The corresponding angular momentum is

$$\mathbf{L} = I_3\omega_3\hat{\boldsymbol{\omega}}_3 - I\Omega\sin\theta\hat{\boldsymbol{\omega}}_2 = (I_3\omega_3 - I\Omega\cos\theta)\,\hat{\boldsymbol{\omega}}_3 + I\Omega\hat{\mathbf{z}}, \tag{8.69}$$

so we can express $\hat{\boldsymbol{\omega}}_3$ in terms of $\mathbf{L}$ and $\hat{\mathbf{z}}$ as

$$\hat{\boldsymbol{\omega}}_3 = \frac{1}{I_3\omega_3 - I\Omega\cos\theta}\left(\mathbf{L} - I\Omega\hat{\mathbf{z}}\right). \tag{8.70}$$

This can be used to express the torque as

$$\boldsymbol{\tau} = Mgd\hat{\mathbf{z}} \times \hat{\boldsymbol{\omega}}_3 = \frac{Mgd}{I_3\omega_3 - I\Omega\cos\theta}\hat{\mathbf{z}} \times \mathbf{L}. \tag{8.71}$$

It follows that $\mathbf{L}$ precesses around the z axis with angular velocity $\frac{Mgd}{I_3\omega_3 - I\Omega\cos\theta}$. Due to the linear relation between $\mathbf{L}$, $\hat{\mathbf{z}}$ and $\hat{\boldsymbol{\omega}}_3$, this must coincide with the angular velocity Ω of the precession of $\hat{\boldsymbol{\omega}}_3$. This leads to a quadratic equation for Ω, with solutions

$$\Omega_\pm = \frac{1}{2I\cos\theta}\left(I_3\omega_3 \pm \sqrt{I_3^2\omega_3^2 - 4MgdI\cos\theta}\right). \tag{8.72}$$

Somewhat surprisingly, one finds two solutions. The *slow-precession* frequency Ω_- can be shown to converge, in the limit of large ω_3, to the precession frequency $\Omega = \frac{Mgd}{I_3\omega_3}$ found above.

In particular, $\mathbf{L}$ is then essentially pointing along the symmetry axis, justifying the above treatment. On the other hand, in the same limit of large ω_3, the *fast-precession* frequency Ω_+ tends to $\frac{I_3\omega_3}{I\cos\theta}$, which is independent of g. In this limit, $\mathbf{L}$ tends to $I\Omega_+\hat{\mathbf{z}}$, i.e., the angular momentum is essentially fixed. In such a case, we expect to recover the free top, and indeed the precession frequency can now be expressed as $\frac{L}{I}$, consistent with the result for the free top in an inertial frame.

We now remove the condition that θ is constant. A fully analytic treatment is then not possible, but we can get quite far with the methods developed up to now. The important realization is that we have three conserved quantities: As the torque is both orthogonal to $\hat{\boldsymbol{\omega}}_3$ and $\hat{\mathbf{z}}$, it follows that both ω_3 and L_z are conserved (the first by Euler's equation, the second by (8.41)). The third constant of motion is the energy

$$E = T + Mgd\cos\theta = \frac{I}{2}\left(\omega_1^2+\omega_2^2\right) + \frac{I_3}{2}\omega_3^2 + Mgd\cos\theta. \tag{8.73}$$

as follows from the fact that the gravitational force is conservative. Note that here $I_1 = I_2 = I$ denote the moments of inertia with P as the origin.[3]

We now want to express the ω_i in terms of Euler angles (and their derivatives). We recall that

$$\boldsymbol{\omega} = \dot{\phi}\hat{\mathbf{z}} + \dot{\theta}\hat{\mathbf{N}} + \dot{\psi}\hat{\boldsymbol{\omega}}_3, \tag{8.74}$$

where $\hat{\mathbf{N}}$ is orthogonal both to $\hat{\boldsymbol{\omega}}_3$ and $\hat{\mathbf{z}}$. We again use the freedom to choose, at any fixed time, the basis vector $\hat{\boldsymbol{\omega}}_1$ freely in the place orthogonal to $\hat{\boldsymbol{\omega}}_3$. Namely, we choose $\hat{\boldsymbol{\omega}}_1 = \hat{\mathbf{N}}$. Then $\hat{\boldsymbol{\omega}}_2$, being orthogonal to $\hat{\boldsymbol{\omega}}_1 = \hat{\mathbf{N}}$, lies in the plane spanned by $\hat{\mathbf{z}}$ and $\hat{\boldsymbol{\omega}}_3$, so that

$$\hat{\mathbf{z}} = \cos\theta\hat{\boldsymbol{\omega}}_3 - \sin\theta\hat{\boldsymbol{\omega}}_2. \tag{8.75}$$

It follows that

$$\omega_1^2+\omega_2^2 = \dot{\theta}^2 + \dot{\phi}^2\sin^2\theta, \qquad\qquad \omega_3 = \dot{\psi} + \dot{\phi}\cos\theta. \tag{8.76}$$

The first of these can be inserted in the above expression for the energy, and the second one is a conserved quantity. It remains to express the third conserved quantity, L_z, in term of (derivatives of) Euler angles. With (8.75), we obtain

$$\mathbf{L} = I\left(-\dot{\phi}\sin\theta\hat{\boldsymbol{\omega}}_2 + \dot{\theta}\hat{\boldsymbol{\omega}}_1\right) + I_3\left(\dot{\psi}+\dot{\phi}\cos\theta\right)\hat{\boldsymbol{\omega}}_3 \tag{8.77}$$

and

$$L_z = \hat{\mathbf{z}}\cdot\mathbf{L} = I\dot{\phi}\sin^2\theta + I_3\omega_3\cos\theta. \tag{8.78}$$

We can now use the constants of motion ω_3, L_z to express the energy only in terms of θ and $\dot{\theta}$ as

$$E = \frac{I}{2}\dot{\theta}^2 + \frac{(L_z - I_3\omega_3\cos\theta)^2}{2I\sin^2\theta} + \frac{I_3}{2}\omega_3^2 + Mgd\cos\theta. \tag{8.79}$$

Hence, we are effectively dealing with one-dimensional motion of θ in the effective potential

$$V_{\text{eff}} = \frac{(L_z - I_3\omega_3\cos\theta)^2}{2I\sin^2\theta} + \frac{I_3}{2}\omega_3^2 + Mgd\cos\theta. \tag{8.80}$$

This is plotted in Fig. 8.3. We see a divergence of the effective potential as $\theta\to 0$ and $\theta\to\pi$. In between, there is a single minimum, corresponding to a stable equilibrium at which θ is constant. Raising the energy leads to periodic bounded motion of θ. This is called *nutation*.

[3] The base point P is fixed, so that there is no translational contribution to the kinetic energy. If we would choose the center of mass as the reference point, then the moments of inertia I_1, I_2 would be smaller, but there is a supplementary contribution to the energy from the motion of the center of mass.

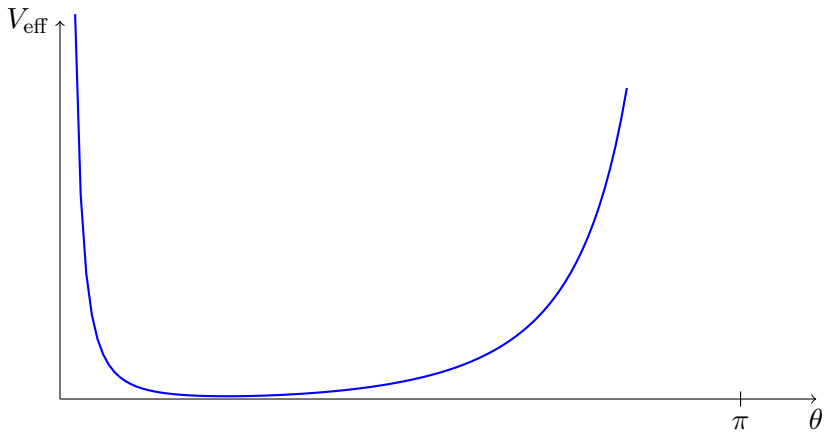


Figure 8.3: The effective potential governing the dynamics of the angle θ between the symmetry and the vertical axis.

9 Accelerated coordinate frames

In the last chapter, we already started considering rotating, i.e., non-inertial, frames. We now want to discuss this in more generality. In Section 6.2, we already derived Galilei transformations as the most general coordinate transformations which i) preserve the duration between two events, $t_A - t_B = t'_A - t'_B$, ii) preserve the Euclidean distance between any two simultaneous events, $|\mathbf{r}_A - \mathbf{r}_B| = |\mathbf{r}'_A - \mathbf{r}'_B|$, and iii) preserve uniform motion. We now drop the last requirement, but keep the first two. Analogously to our discussion in Section 6.2, the most general such transformation can be expressed as

$$t' = t - t_0, \qquad \mathbf{r}' = R(t)\left(\mathbf{r} - \mathbf{r}_0(t)\right), \tag{9.1}$$

with $R(t)$ a (in general time-dependent) orthogonal matrix. We see that the new origin of coordinates $\mathbf{r}' = \mathbf{0}$ is located at $\mathbf{r} = \mathbf{r}_0(t)$, so $\mathbf{r}_0(t)$ is the trajectory of the origin of the non-inertial coordinate system as described in the inertial one. If we assume that the new coordinate frame is also right-handed, we restrict to $R(t) \in SO(3)$. The above reduces to Galilei transformations for constant $R(t) = R$ and affine-linear $\mathbf{r}_0(t) = R^{-1}(\mathbf{v}_0 t + \mathbf{r}_0)$. We will typically refrain from shifting the time, i.e., restrict to $t_0 = 0$.

Let now $\mathbf{r}(t)$ describe a particle trajectory in the inertial frame and let $\mathbf{r}'(t) = R(t)(\mathbf{r}(t) - \mathbf{r}_0(t))$. We begin by determining the velocity $\mathbf{v}'(t)$ as seen in the non-inertial frame. Using that, for any differentiable curve $R(t)$ in the set of orthogonal matrices we have (exercise)

$$\frac{\mathrm{d}}{\mathrm{d}t} R(t)\mathbf{w} = -\boldsymbol{\omega}(t) \times R(t)\mathbf{w} \tag{9.2}$$

with an instantaneous angular velocity vector $\boldsymbol{\omega}(t)$,[1] we obtain

$$\mathbf{v}'(t) := \frac{\mathrm{d}}{\mathrm{d}t}\mathbf{r}'(t) = -\boldsymbol{\omega}(t) \times \mathbf{r}'(t) + R(t)\left(\mathbf{v}(t) - \mathbf{v}_0(t)\right), \tag{9.3}$$

with $\mathbf{v}(t) = \frac{\mathrm{d}}{\mathrm{d}t}\mathbf{r}(t)$ and analogously for $\mathbf{v}_0(t)$. We perform a further derivative to obtain the acceleration in the non-inertial frame:

$$\mathbf{a}'(t) := \frac{\mathrm{d}}{\mathrm{d}t}\mathbf{v}'(t) = -\dot{\boldsymbol{\omega}}(t) \times \mathbf{r}'(t) - \boldsymbol{\omega}(t) \times \mathbf{v}'(t) - \boldsymbol{\omega}(t) \times R(t)\left(\mathbf{v}(t) - \mathbf{v}_0(t)\right) + R(t)\left(\mathbf{a}(t) - \mathbf{a}_0(t)\right). \tag{9.4}$$

The third term on the r.h.s. can now be expressed again in terms of $\mathbf{v}'(t)$, yielding

$$\mathbf{a}'(t) = -\dot{\boldsymbol{\omega}}(t) \times \mathbf{r}'(t) - 2\boldsymbol{\omega}(t) \times \mathbf{v}'(t) - \boldsymbol{\omega}(t) \times \big(\boldsymbol{\omega}(t) \times \mathbf{r}'(t)\big) + R(t)\left(\mathbf{a}(t) - \mathbf{a}_0(t)\right). \tag{9.5}$$

Multiplying by m, we thus get

$$m\mathbf{a}' = R(\mathbf{F} + \mathbf{F}_{\text{trans}}) + \mathbf{F}_{\text{centr}} + \mathbf{F}_{\text{Cor}} + \mathbf{F}_{\text{az}} \tag{9.6}$$

[1] The change of the sign w.r.t. the previous chapter is due to the fact that there we considered active rotations there, whereas her we consider passive ones. The sign is chosen here such that $\boldsymbol{\omega}$ is indeed the angular velocity vector for the rotation of the non-inertial frame w.r.t. the inertial one.

with $\mathbf{F}$ the force in the inertial frame, and the *translational*, *centrifugal*, *Coriolis*, and *azimuthal forces*

$$\mathbf{F}_{\text{trans}} = -m\mathbf{a}_0, \qquad \mathbf{F}_{\text{centr}} = -m\boldsymbol{\omega} \times \left(\boldsymbol{\omega} \times \mathbf{r}'\right), \tag{9.7}$$

$$\mathbf{F}_{\text{Cor}} = -2m\boldsymbol{\omega} \times \mathbf{v}', \qquad \mathbf{F}_{\text{az}} = -m\dot{\boldsymbol{\omega}} \times \mathbf{r}'. \tag{9.8}$$

These are called *fictitious forces*.

For the case of a coordinate transformation between two inertial frames, we found, in Section 6.3, that $m\mathbf{a}' = R\mathbf{F}$, so the first term on the r.h.s. of (9.6) is well-known. The remaining terms only occur for transformations to a non-inertial frame, and indeed we see that they vanish if R is constant (implying $\boldsymbol{\omega} = \mathbf{0}$) and $\mathbf{r}_0(t)$ uniform motion (implying $\mathbf{a}_0 = \mathbf{0}$). Let us discuss the fictitious forces one after another.

The translational force occurs whenever the origin of the non-inertial frame is in non-uniform (accelerated) motion w.r.t. the inertial frame. Its effect is well-known for example from accelerating or braking cars. Another example would be a frame freely falling in a homogeneous gravitational field with gravitational acceleration $\mathbf{g}$ (think of an elevator in free fall). Then the translational force $\mathbf{F}_{\text{trans}} = -m\mathbf{g}$ cancels the gravitational force $\mathbf{F}_g = m\mathbf{g}$. Hence, motion in a freely falling reference frame is free (in the absence of non-gravitational forces). Such Gedanken experiments played a crucial role in the development of General Relativity.

The centrifugal force occurs whenever the non-inertial frame is rotating w.r.t. the inertial frame. It is non-zero, except when $\mathbf{r}'$ and $\boldsymbol{\omega}$ are aligned, i.e., the particle is located on the rotation axis. By the properties of the vector product, it is always orthogonal to the rotation axis $\boldsymbol{\omega}$ and points outwards, with magnitude $F_{\text{centr}} = mr\omega^2 \sin\theta$ (θ being the angle between $\boldsymbol{\omega}$ and $\mathbf{r}'$). Choosing cylindrical coordinates with the cylinder axis aligned with $\boldsymbol{\omega}$, this means

$$\mathbf{F}_{\text{centr}} = m\rho\omega^2\hat{\boldsymbol{\rho}}. \tag{9.9}$$

We experience the centrifugal force when standing on a rotating platform. The centrifugal force can be thought of as causing the oblate ellipsoidal form of Earth and being responsible for the fact that space rocket launches from the equator are most economical. As it is proportional to m, one can write the combined effect of gravitational and centrifugal force as

$$\mathbf{F}_g + \mathbf{F}_{\text{centr}} = m\underbrace{\left(\mathbf{g} - \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}')\right)}_{\mathbf{g}_{\text{eff}}} \tag{9.10}$$

with an effective gravitational acceleration $\mathbf{g}_{\text{eff}}$. The latter defines the direction relevant, for example, in the construction of skyscrapers.

The Coriolis force occurs whenever the particle is in motion w.r.t. a rotating coordinate frame, in a direction not aligned with the rotation axis. Let us get some intuition about this force. Consider a coordinate frame rotating w.r.t. an inertial frame around a fixed axis. Choose this axis as the z-axis and work in polar (or cylindrical coordinates). In the inertial frame, a particle moving radially through the rotation axis with a fixed velocity is in uniform motion, i.e., not subject to a force. In the rotating frame, however, the motion does not appear to be radial, precisely due to the rotation of the frame. From the point of view of the rotating frame, this deviation from radial motion is due to the Coriolis force (the centrifugal force is directed radially, so it cannot explain the deviation from radial motion). With the Coriolis force one can for example explain the different directions of the rotation of hurricanes in the northern and southern hemisphere and the rotation frequency of Foucault's pendulum.

Finally, the azimuthal force occurs whenever the rotation of the accelerated frame w.r.t. the inertial frame changes in time, either in magnitude or direction. It is akin to the translational

force. As an example, consider a person standing on a disk that is set into rotation. In order to stay at the same position on the disk, i.e., stationary in the rotating frame, a frictional force tangential to the direction of rotation is necessary (cancelling the azimuthal force).

10 Lagrangian Mechanics

We conclude the lecture with an outlook to a generalization of Newtonian Mechanics that i) simplifies the solution of many problems considerably (in particular for systems subject to constraints), ii) allows for important structural insights, for example on the relation between symmetries and conservation laws, iii) is applicable also in relativistic and even field theories, and iv) is the starting point of a further reformulation, Hamiltonian Mechanics, which in turn allows for the introduction of concepts of probability theory (leading to Statistical Physics) and is also conceptually close to Quantum Mechanics.

Lagrangian Mechanics was developed in the 18th century by Lagrange (in correspondence with Euler). It can be motivated by the desire to generalize Fermat's principle to the realm of mechanics. We recall that Fermat's principle is the statement that a light ray from A to B takes the path which minimizes (more generally extremizes) the *optical length*, which is nothing but the time it takes to get from A to B, given the speed of light locally given by $v(\mathbf{r}) = \frac{c}{n(\mathbf{r})}$ with $n(\mathbf{r})$ the refractive index of the medium. Mathematically, one can formulate this as the problem to find the curve which minimizes the line integral[1]

$$T = \int \frac{1}{v(\mathbf{r})} \mathrm{d}r = \frac{1}{c} \int n(\mathbf{r}) \mathrm{d}r = \frac{1}{c} \int n(\mathbf{r}(s)) |\dot{\mathbf{r}}(s)| \mathrm{d}s \tag{10.1}$$

among curves connecting A to B. Mathematically, T is a map from the set of curves $\mathbf{r}(s)$ into the real numbers. Such a map is called a *functional*.

Exercise 10.1. *Use Fermat's principle to prove Snell's law $n_1 \sin\theta_1 = n_2 \sin\theta_2$ for the refraction of a light ray at the interface of homogeneous media with refractive indices n_1, n_2. Hint: The shortest path from the start or end point to a point on the interface is a straight line, so one has to minimize the optical length by varying the point on the interface through which the ray passes.*

10.1 Hamilton's principle and the Euler-Lagrange equations

Lagrangian Mechanics provides an analogous prescription for mechanics: The trajectories $\mathbf{r}(t)$ solving Newton's equations arise as extrema (in many cases minima) of the *action functional*[2]

$$S[\mathbf{r}(t)] = \int L(\mathbf{r}(t), \dot{\mathbf{r}}(t), t) \mathrm{d}t, \tag{10.2}$$

with the function $L(\mathbf{r}, \dot{\mathbf{r}}, t)$ the *Lagrangian* (in many relevant situations, it will not depend explicitly on t). We will later see that we can recover the Newtonian equations of motion by

[1] The last equality in this equation can be seen as a definition (of the l.h.s. by the r.h.s). This is a variation of the line integral of a vector field defined in Section 6.4. Again, it is independent of the parametrization (even independent of the orientation) of the curve.

[2] We have written here $S[\mathbf{r}(t)]$ (not $S(\mathbf{r}(t))$) to indicate that S depends on the whole trajectory, and not the position $\mathbf{r}(t)$ at some fixed time.

choosing L to be the difference of kinetic and potential energy, i.e.,[3]

$$L = T - V, \tag{10.3}$$

but for the time being we consider a general $L(\mathbf{r}, \dot{\mathbf{r}}, t)$ and formulate what it precisely means that a trajectory $\mathbf{r}(t)$ extremizes the action. By the *Hamiltonian action principle*, we we are looking for extrema of S among all trajectories $\mathbf{r}(t)$ with the same start and end-points, i.e., $\mathbf{r}(t_0) = \mathbf{r}_0$ and $\mathbf{r}(t_1) = \mathbf{r}_1$, with t_0 and t_1 initial and final time. This looks like a setup quite different from Newtonian Mechanics: Instead of asking

> Given that at time $t = t_0$ the particle is at $\mathbf{r}_0$ with velocity $\mathbf{v}_0$, what is its trajectory $\mathbf{r}(t)$?

we are now asking

> Given that the particle is at $\mathbf{r}_0$ at time t_0 and at $\mathbf{r}_1$ at time t_1, what is its trajectory $\mathbf{r}(t)$ in between?

Nevertheless, we will see that one recovers the Newtonian formulation from the Lagrangian one.

Now what does it mean to extremize a functional? For a function $f(\mathbf{x})$ of several variable, $\mathbf{x}_e$ is an extremum if $\boldsymbol{\nabla} f(\mathbf{x}_e) = \mathbf{0}$, which is equivalent to

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} f(\mathbf{x}_e + \varepsilon \mathbf{y})|_{\varepsilon=0} = 0 \qquad \forall \mathbf{y}. \tag{10.4}$$

We want to use the same principle for defining extrema of a functional, such as the action functional. Let us for simplicity assume a one-dimensional problem, so that the Lagrangian is a function $L(x, \dot{x}, t)$. We say that the trajectory $x_e(t)$ extremizes the action functional if the action for a nearby trajectory, $x(t) = x_e(t) + \varepsilon y(t)$ gives the same result, up to corrections of $\mathcal{O}(\varepsilon^2)$. In other words, we require

$$0 = \frac{\mathrm{d}}{\mathrm{d}\varepsilon} S[x_e(t) + \varepsilon y(t)]|_{\varepsilon=0} = \frac{\mathrm{d}}{\mathrm{d}\varepsilon} \int_{t_0}^{t_1} L(x_e(t) + \varepsilon y(t), \dot{x}_e(t) + \varepsilon \dot{y}(t), t) \mathrm{d}t \Big|_{\varepsilon=0} \tag{10.5}$$

for all functions $y(t)$ with $y(t_0) = y(t_1) = 0$. The latter condition implies that the varied pathes $x(t) = x_e(t) + \varepsilon y(t)$ all have the same starting and end points, as required by the Hamiltonian principle. The above requirement then gives

$$0 = \int_{t_0}^{t_1} \left(\frac{\partial L}{\partial x}(x_e(t), \dot{x}_e(t), t) y(t) + \frac{\partial L}{\partial \dot{x}}(x_e(t), \dot{x}_e(t), t) \dot{y}(t) \right) \mathrm{d}t \tag{10.6}$$

$$= \int_{t_0}^{t_1} \left(\frac{\partial L}{\partial x}(x_e(t), \dot{x}_e(t), t) - \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{x}}(x_e(t), \dot{x}_e(t), t) \right) y(t) \mathrm{d}t, \tag{10.7}$$

where in the second step we used integration by parts and the condition that $y(t_0) = y(t_1) = 0$. Now this equation must hold for arbitrary functions $y(t)$ (subject to the boundary condition). This is only possible if the expression in brackets vanishes. This yields the *Euler-Lagrange equation*

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x} \tag{10.8}$$

[3] This makes an important drawback of the Lagrangian framework obvious: Only conservative forces (and the Lorentz force) can be described. In particular, (kinetic) friction can not be implemented.

which has to be satisfied for trajectories $x(t)$ extremizing the action functional.

It is now high time to see the Hamiltonian principle at work, i.e., to recover Newton's equation by the Euler-Lagrange equation. For this, we consider motion in one dimension, so according to (10.3),

$$L(x, \dot{x}) = \frac{m}{2}\dot{x}^2 - V(x). \tag{10.9}$$

Then

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}, \qquad \frac{\partial L}{\partial x} = -V'(x). \tag{10.10}$$

The Euler-Lagrange equation is thus

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(m\dot{x}\right) = -V'(x), \tag{10.11}$$

which is precisely Newton's equation for motion in one dimension.

What is now the advantage of the Lagrangian formulation? Already in the case of one-dimensional motion, we can use it to simplify the treatment of constrained systems. Consider the case of the pendulum discussed in Section 4.3 in terms of forces. It took a while to derive the equation of motion, as we had to first decompose the forces and determine the tension responsible for keeping the length of the pendulum fixed. In the Lagrangian framework, we can directly implement the constraint that the mass m moves in a plane with a fixed distance ℓ to some point by setting

$$\mathbf{r} = \ell \begin{pmatrix} \sin\theta \\ 0 \\ -\cos\theta \end{pmatrix}, \tag{10.12}$$

implying

$$\dot{\mathbf{r}} = \ell\dot{\theta} \begin{pmatrix} \cos\theta \\ 0 \\ \sin\theta \end{pmatrix}, \tag{10.13}$$

which in turn implies

$$T = \frac{m}{2}|\dot{\mathbf{r}}|^2 = \frac{m\ell^2}{2}\dot{\theta}^2, \qquad V = mgz = -mg\ell\cos\theta. \tag{10.14}$$

It follows that

$$L(\theta, \dot{\theta}) = \frac{m\ell^2}{2}\dot{\theta}^2 + mg\ell\cos\theta. \tag{10.15}$$

The Euler-Lagrange equation is thus

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(m\ell^2\dot{\theta}\right) = -mg\ell\sin\theta, \tag{10.16}$$

which is the same as (4.35), which we found for the pendulum using Newtonian methods.

We now generalize the setup to motion with more than one degree of freedom. We describe the state of the mechanical system by N *generalized coordinates* q_i and the corresponding *generalized velocities* $\dot{q}_i$, where $1 \leq i \leq N$. For the three-dimensional motion of a single particle (q_1, q_2, q_3) can be, for example, the Cartesian coordinates (x, y, z) in an inertial frame, but also the corresponding cylindrical (ρ, ϕ, z) or spherical (r, θ, ϕ) coordinates. For notational convenience, we collect the generalized coordinates and velocities into vectors $\mathbf{q}$ and $\dot{\mathbf{q}}$. The

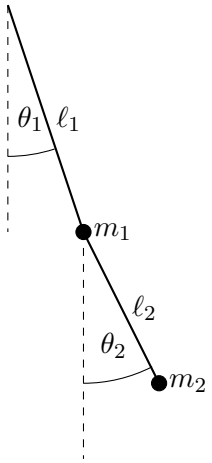


Figure 10.1: The double pendulum.

Lagrangian is now a function of the generalized coordinates and velocities, and possibly of time, i.e., $L = L(\mathbf{q}, \dot{\mathbf{q}}, t)$.

As for the Hamiltonian principle, we again call a trajectory $\mathbf{q}_e(t)$ extremal, if $\frac{\mathrm{d}}{\mathrm{d}\varepsilon} S[\mathbf{q}_e(t) + \varepsilon\boldsymbol{\eta}(t)]|_{\varepsilon=0} = 0$ for all variations $\boldsymbol{\eta}(t)$ which vanish at initial and final time, i.e., $\boldsymbol{\eta}(t_0) = \boldsymbol{\eta}(t_1) = \mathbf{0}$. This leads to

$$0 = \int_{t_0}^{t_1} \sum_{i=1}^{N} \left(\frac{\partial L}{\partial q_i}(\mathbf{q}_e(t), \dot{\mathbf{q}}_e(t), t)\eta_i(t) + \frac{\partial L}{\partial \dot{q}_i}(\mathbf{q}_e(t), \dot{\mathbf{q}}_e(t), t)\dot{\eta}_i(t) \right) \mathrm{d}t \tag{10.17}$$

$$= \int_{t_0}^{t_1} \sum_{i=1}^{N} \left(\frac{\partial L}{\partial q_i}(\mathbf{q}_e(t), \dot{\mathbf{q}}_e(t), t) - \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{q}_i}(\mathbf{q}_e(t), \dot{\mathbf{q}}_e(t), t) \right) \eta_i(t) \mathrm{d}t, \tag{10.18}$$

again using integration by parts and the condition that q_i vanishes at initial and final time. As this must hold for all variations $\boldsymbol{\eta}(t)$, we obtain the Euler-Lagrange equation as

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{q}_i} = \frac{\partial L}{\partial q_i}. \tag{10.19}$$

Example 10.1. *We use the Lagrangian framework to study the* double pendulum*: A mass m_1 is attached to a massless rod of length ℓ_1, to which another massless rod of length ℓ_2 is attached, at whose end there is a mass m_2, c.f. Fig. 10.1. The rods and rotate freely in a common plane, so we can describe the state of the system by two angles ϕ_1, ϕ_2 as indicated in the figure. These are thus our generalized coordinates. The positions of the two masses in Cartesian coordinates in the $x - y$ plane are thus*

$$\mathbf{r}_1 = \ell_1 \begin{pmatrix} -\cos\theta_1 \\ \sin\theta_1 \end{pmatrix}, \qquad \mathbf{r}_2 = \mathbf{r}_1 + \ell_2 \begin{pmatrix} -\cos\theta_2 \\ \sin\theta_2 \end{pmatrix} = \ell_1 \begin{pmatrix} -\cos\theta_1 \\ \sin\theta_1 \end{pmatrix} + \ell_2 \begin{pmatrix} -\cos\theta_2 \\ \sin\theta_2 \end{pmatrix}. \tag{10.20}$$

The corresponding velocities are

$$\dot{\mathbf{r}}_1 = \ell_1\dot{\theta}_1 \begin{pmatrix} \sin\theta_1 \\ \cos\theta_1 \end{pmatrix}, \qquad \dot{\mathbf{r}}_2 = \ell_1\dot{\theta}_1 \begin{pmatrix} \sin\theta_1 \\ \cos\theta_1 \end{pmatrix} + \ell_2\dot{\theta}_2 \begin{pmatrix} \sin\theta_2 \\ \cos\theta_2 \end{pmatrix}, \tag{10.21}$$

so that the kinetic energy is

$$T = \frac{m_1}{2}|\dot{\mathbf{r}}_1|^2 + \frac{m_2}{2}|\dot{\mathbf{r}}_2|^2 \tag{10.22}$$

$$= \frac{m_1}{2}\ell_1^2\dot{\theta}_1^2 + \frac{m_2}{2}\left(\ell_1^2\dot{\theta}_1^2 + \ell_2^2\dot{\theta}_2^2 + 2\ell_1\ell_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1-\theta_2)\right), \tag{10.23}$$

where we used (3.23)*. With the gravitational potential energy, we thus get the Lagrangian*

$$L = \frac{m_1}{2}\ell_1^2\dot{\theta}_1^2 + \frac{m_2}{2}\left(\ell_1^2\dot{\theta}_1^2 + \ell_2^2\dot{\theta}_2^2 + 2\ell_1\ell_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1-\theta_2)\right) + m_1 g\ell_1\cos\theta_1 + m_2 g\left(\ell_1\cos\theta_1 + \ell_2\cos\theta_2\right). \tag{10.24}$$

We compute the partial derivatives

$$\frac{\partial L}{\partial\theta_1} = -m_2\ell_1\ell_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) - g(m_1\ell_1+m_2\ell_2)\sin\theta_1, \tag{10.25}$$

$$\frac{\partial L}{\partial\dot{\theta}_1} = \left(m_1\ell_1^2 + m_2\ell_2^2\right)\dot{\theta}_1 + m_2\ell_1\ell_2\dot{\theta}_2\cos(\theta_1-\theta_2), \tag{10.26}$$

$$\frac{\partial L}{\partial\theta_2} = m_2\ell_1\ell_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) - gm_2\ell_2\sin\theta_2, \tag{10.27}$$

$$\frac{\partial L}{\partial\dot{\theta}_2} = m_2\ell_2^2\dot{\theta}_2 + m_2\ell_1\ell_2\dot{\theta}_1\cos(\theta_1-\theta_2). \tag{10.28}$$

The Euler-Lagrange equations are thus

$$\left(m_1\ell_1^2 + m_2\ell_2^2\right)\ddot{\theta}_1 + m_2\ell_1\ell_2\ddot{\theta}_2\cos(\theta_1-\theta_2) - m_2\ell_1\ell_2\dot{\theta}_2\left(\dot{\theta}_1-\dot{\theta}_2\right)\sin(\theta_1-\theta_2) = -m_2\ell_1\ell_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) - g(m_1\ell_1+m_2\ell_2)\sin\theta_1 \tag{10.29}$$

and

$$m_2\ell_2^2\ddot{\theta}_2 + m_2\ell_1\ell_2\ddot{\theta}_2\cos(\theta_1-\theta_2) - m_2\ell_1\ell_2\dot{\theta}_1\left(\dot{\theta}_1-\dot{\theta}_2\right)\sin(\theta_1-\theta_2) = m_2\ell_1\ell_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1-\theta_2) - gm_2\ell_2\sin\theta_2, \tag{10.30}$$

which simplifies to

$$\left(m_1\ell_1^2 + m_2\ell_2^2\right)\ddot{\theta}_1 + m_2\ell_1\ell_2\ddot{\theta}_2\cos(\theta_1-\theta_2) + m_2\ell_1\ell_2\dot{\theta}_2^2\sin(\theta_1-\theta_2) = -g(m_1\ell_1+m_2\ell_2)\sin\theta_1, \tag{10.31}$$

$$m_2\ell_2^2\ddot{\theta}_2 + m_2\ell_1\ell_2\ddot{\theta}_2\cos(\theta_1-\theta_2) - m_2\ell_1\ell_2\dot{\theta}_1^2\sin(\theta_1-\theta_2) = -gm_2\ell_2\sin\theta_2. \tag{10.32}$$

This is a coupled system of differential equations. Even though the derivation may seem tedious, it is nevertheless considerably simpler (and less error-prone), than a derivation based on decomposition of forces (the student is invited to try...).

The double pendulum exhibits chaotic behaviour, i.e., small perturbations of initial conditions in general grow exponentially in time. This behaviour is not apparent for small perturbations of the equilibrium position: Restricting to oscillations of small amplitude, i.e., ignoring terms of second order in θ_i, $\dot{\theta}_i$, the above linearizes to the system

$$\left(m_1\ell_1^2 + m_2\ell_2^2\right)\ddot{\theta}_1 + m_2\ell_1\ell_2\ddot{\theta}_2 = -g(m_1\ell_1+m_2\ell_2)\theta_1, \tag{10.33}$$

$$m_2\ell_2^2\ddot{\theta}_2 + m_2\ell_1\ell_2\ddot{\theta}_2 = -gm_2\ell_2\theta_2, \tag{10.34}$$

which can be written in matrix form as

$$A\ddot{\boldsymbol{\theta}} + B\boldsymbol{\theta} = 0 \tag{10.35}$$

with

$$A = \begin{pmatrix} m_1\ell_1^2 + m_2\ell_2^2 & m_2\ell_1\ell_2 \\ m_2\ell_1\ell_2 & m_2\ell_2^2 \end{pmatrix}, \qquad B = \begin{pmatrix} gm_1\ell_1 + gm_2\ell_2 & 0 \\ 0 & gm_2\ell_2 \end{pmatrix}. \tag{10.36}$$

A normal mode ansatz $\boldsymbol{\theta}(t) = \boldsymbol{\theta}e^{-i\omega t}$ *leads to*

$$\left(-\omega^2 A + B\right)\boldsymbol{\omega} = \mathbf{0}, \tag{10.37}$$

which can be solved iff

$$0 = \det\left(-\omega^2 A + B\right) \tag{10.38}$$

$$= \left(gm_1\ell_1 + gm_2\ell_2 - \omega^2(m_1\ell_1^2 + m_2\ell_2^2)\right)\left(gm_2\ell_2 - \omega^2 m_2\ell_2^2\right) - \omega^4 m_2^2\ell_1^2\ell_2^2. \tag{10.39}$$

Assuming, for simplicity, $\ell_1 = \ell_2 = \ell$, $m_1 = m_2 = m$, *this simplifies to*

$$0 = 2m^2\ell^2(g - \omega^2\ell)^2 - \omega^4 m^2\ell^4, \tag{10.40}$$

which is solved by

$$\omega_\pm^2 = \frac{g}{\ell}\left(2 \pm \sqrt{2}\right). \tag{10.41}$$

One easily checks that the corresponding "normal vectors" are given by

$$\boldsymbol{\theta}_\pm = \begin{pmatrix} 1 \\ \mp\sqrt{2} \end{pmatrix}. \tag{10.42}$$

Hence, for the smaller oscillation frequency ω_-, *the two pendulums are precisely in phase (with the amplitude of the second one being* $\sqrt{2}$ *times greater than that of the first one), while for the greater oscillation frequency* ω_+ *they are in opposite phase.*

10.2 Conserved quantities

A comprehensive analysis of conserved quantities (and their relation to symmetries) will be provided by Noether's theorem, in a later lecture course. Here, we focus on two simple cases in which conserved quantities occur. First, we introduce a further terminology. Namely, we call

$$p_i := \frac{\partial L}{\partial \dot{q}_i} \tag{10.43}$$

the *generalized momentum* associated to the generalized coordinate q_i. In case of a single body moving in a potential $V(\mathbf{r})$ with configuration described by Cartesian coordinates r_1, r_2, r_3, we have $L = \frac{m}{2}(\dot{r}_1^2 + \dot{r}_2^2 + \dot{r}_3^2) - V(\mathbf{r})$, and so

$$p_i = \frac{\partial L}{\partial \dot{r}_i} = m\dot{r}_i, \tag{10.44}$$

corresponding to the usual notion of momentum. However, for the (single) pendulum, the generalized momentum is

$$p = m\ell^2\dot{\theta}, \tag{10.45}$$

while for the double pendulum, we get

$$p_1 = \left(m_1\ell_1^2 + m_2\ell_2^2\right)\dot{\theta}_1 + m_2\ell_1\ell_2\dot{\theta}_2\cos(\theta_1 - \theta_2), \tag{10.46}$$

$$p_2 = m_2\ell_2^2\dot{\theta}_2 + m_2\ell_1\ell_2\dot{\theta}_1\cos(\theta_1 - \theta_2). \tag{10.47}$$

We note that both of these not only depend on both velocities $\dot{q}_1$, $\dot{q}_2$, but also on the positions θ_1, θ_2. The time-evolution of the generalized momentum is, by the Euler-Lagrange equations, given by

$$\frac{\mathrm{d}}{\mathrm{d}t}p_i = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} = \frac{\partial L}{\partial q_i} \tag{10.48}$$

(so that one can think of $\frac{\partial L}{\partial q_i}$ as generalized forces).

If the Lagrangian L does not depend explicitly on time, $L(\mathbf{q}, \dot{\mathbf{q}}, t) = L(\mathbf{q}, \dot{\mathbf{q}})$, then a conserved quantity is given by the *Hamiltonian*, defined by

$$H = \sum_{i=1}^{N} \dot{q}_i p_i - L. \tag{10.49}$$

Conservation follows from

$$\frac{\mathrm{d}}{\mathrm{d}t}H = \sum_{i=1}^{N}\left(\ddot{q}_i\frac{\partial L}{\partial \dot{q}_i} + \dot{q}_i\dot{p}_i - \dot{q}_i\frac{\partial L}{\partial q_i} - \ddot{q}_i\frac{\partial L}{\partial \dot{q}_i}\right) = 0, \tag{10.50}$$

where we used that L does not depend explicitly on t (so that no $\frac{\partial L}{\partial t}$ term is present) and the Euler-Lagrange equation (10.48). In many cases, the Hamiltonian will coincide with the energy (such as for the pendulum and the double pendulum), but there are also situations where they do not, in particular for time-dependent constraints (such as in the rotating pendulum discussed in the Exercises).

Remark 10.1. *The Hamiltonian can also be defined (in the same manner) for Lagrangians depending explicitly on time, but it will then in general not be conserved. In Hamiltonian Mechanics, the Hamiltonian is not interpreted as a function of q_i, $\dot{q}_i$ (and possibly t), but as a function of q_i, p_i (and possibly t), with generalized positions and momenta spanning* phase space. *On phase space, the Hamiltonian vector field (determined by the Hamiltonian) then describes the dynamics, in the same manner as the flow vector field $\mathbf{F}$ that we used in Chapter 4 to describe motion in the $(q, \dot{q})$ plane.*

A further case of conserved quantities occurs when the Lagrangian is independent of one of the generalized coordinates, say q_j. Such coordinates are called *cyclic*. Then

$$\frac{\partial L}{\partial q_j} = 0, \tag{10.51}$$

so by (10.48), the corresponding generalized momentum p_j is conserved. We study this in an example.

Example 10.2. *We consider the spherical pendulum, i.e., the pendulum which is not constrained to a plane. We describe the position of the mass m by angles θ, ϕ as*

$$\mathbf{r} = \ell\begin{pmatrix}\sin\theta\cos\phi\\ \sin\theta\sin\phi\\ -\cos\phi\end{pmatrix} \tag{10.52}$$

in coordinates such that $\mathbf{g} = -g\hat{\mathbf{z}}$ *(we have flipped the sign of* z *w.r.t. standard spherical coordinates in order to have* $\theta = 0$ *as the equilibrium). The corresponding velocity is*

$$\mathbf{v} = \ell\dot{\theta}\begin{pmatrix}\cos\theta\cos\phi\\ \cos\theta\sin\phi\\ \sin\phi\end{pmatrix} + \ell\dot{\phi}\sin\theta\begin{pmatrix}-\sin\phi\\ \cos\phi\\ 0\end{pmatrix}, \tag{10.53}$$

so that

$$T = \frac{m}{2}\ell^2\left(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2\right). \tag{10.54}$$

Hence,

$$L = \frac{m}{2}\ell^2\left(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2\right) + mg\ell\cos\theta. \tag{10.55}$$

As L *does not depend explicitly on* t*, the Hamiltonian (which coincides with the usual notion of energy in the present case) is conserved. In order to arrive at the Euler-Lagrange equations, we compute*

$$\frac{\partial L}{\partial\theta} = m\ell^2\cos\theta\sin\theta\dot{\phi}^2 - mg\ell\sin\theta \qquad \frac{\partial L}{\partial\phi} = 0 \tag{10.56}$$

$$\frac{\partial L}{\partial\dot{\theta}} = m\ell^2\dot{\theta} \qquad \frac{\partial L}{\partial\dot{\phi}} = m\ell^2\sin^2\theta\dot{\phi}. \tag{10.57}$$

We see that ϕ *is a cyclic coordinate, so the corresponding generalized momentum* $p_\phi = m\ell^2\sin^2\theta\dot{\phi}$ *is conserved. This is of course nothing but the* z*-component of the angular momentum. The Euler-Lagrange equation w.r.t. the other coordinate is now*

$$m\ell^2\ddot{\theta} = m\ell^2\cos\theta\sin\theta\dot{\phi}^2 - mg\ell\sin\theta = \frac{p_\phi^2}{m\ell^2}\frac{\cos\theta}{\sin^3\theta} - mg\ell\sin\theta. \tag{10.58}$$

Equilibria of motion occur at the zeros of the right hand side. As $\frac{\cos\theta}{\sin^3\theta}$ *decreases monotonically from* $+\infty$ *at* $\theta = 0$ *to* $-\infty$ *at* $\theta = \pi$*, passing through zero at* $\theta = \frac{\pi}{2}$*, and* $\sin\theta$ *is positive, monotonically increasing from* 0 *at* $\theta = 0$ *to* 1 *at* $\theta = \frac{\pi}{2}$ *and then monotonically decreasing to* 0 *at* $\theta = \pi$*, there is (for* $p_\phi \neq 0$*) exactly one zero of the right hand side, situated between* 0 *and* $\frac{\pi}{2}$*. In this range, the function is monotonically decreasing (as both summands are), so, according to our discussion in Section 4.2, the equilibrium is stable.*